



ACM RecSys 2026

The 20th ACM Recommender Systems Conference will take place in Minneapolis, Minnesota, USA, from Sep. 27 - Oct. 2, 2026.

Tutorial on **Recommender Systems and Sustainability: Bridging Algorithmic Design and Societal Impact**



<https://sustainablerecsys.github.io/SustainabilityTutorial26/>



Tutorial Organizers



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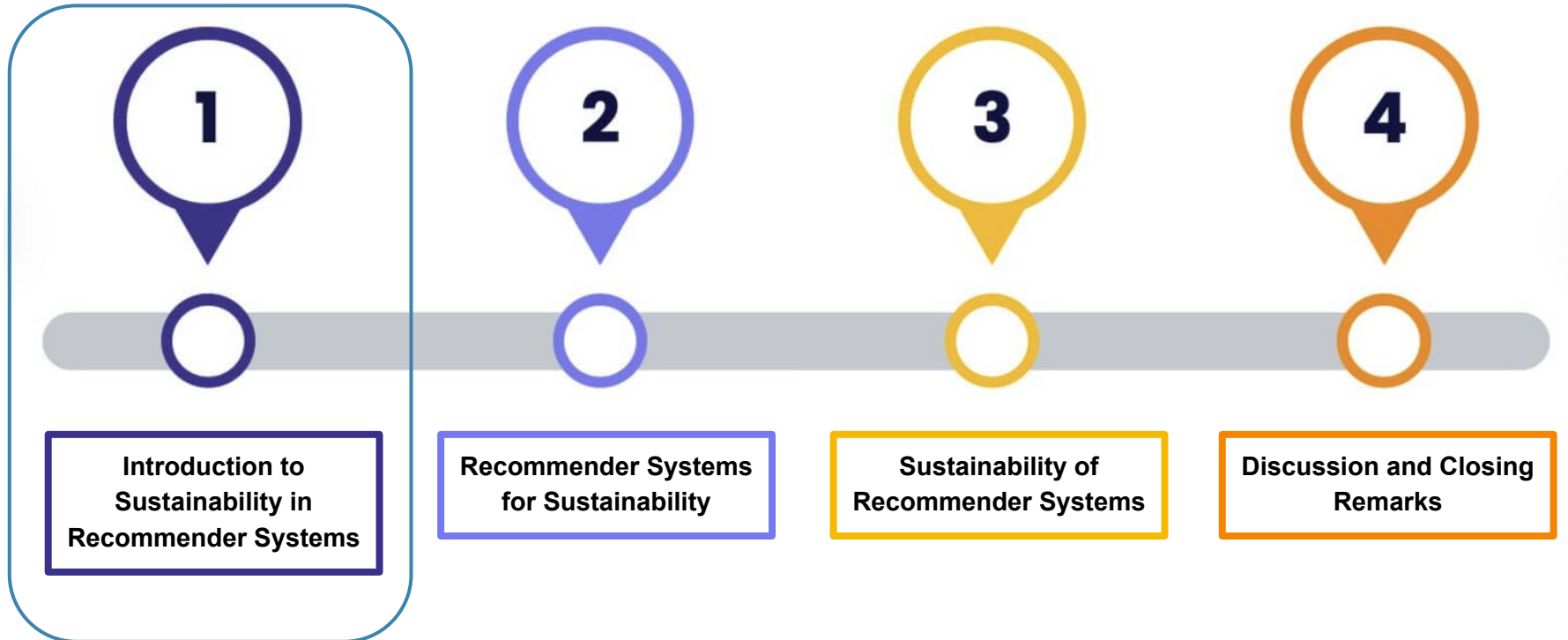
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Tutorial Outline



Motivation

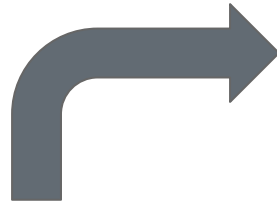


What is sustainable development?

**“Development that meets the needs of the present
without compromising the ability of future generations
to meet their own needs”**

*the globally accepted definition adopted in 1987 at the World Commission on
Environment and Development*

Motivation



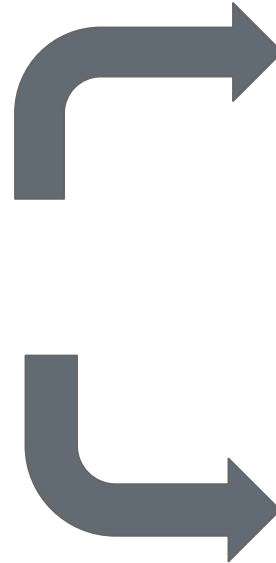
The 2030 agenda for sustainable development

It contains the **17 Sustainable Development Goals (SDGs)**, which are an urgent call for action by all countries - developed and developing - in a global partnership.

<https://sdgs.un.org/2030agenda>



Motivation



**AGENDA
2030**



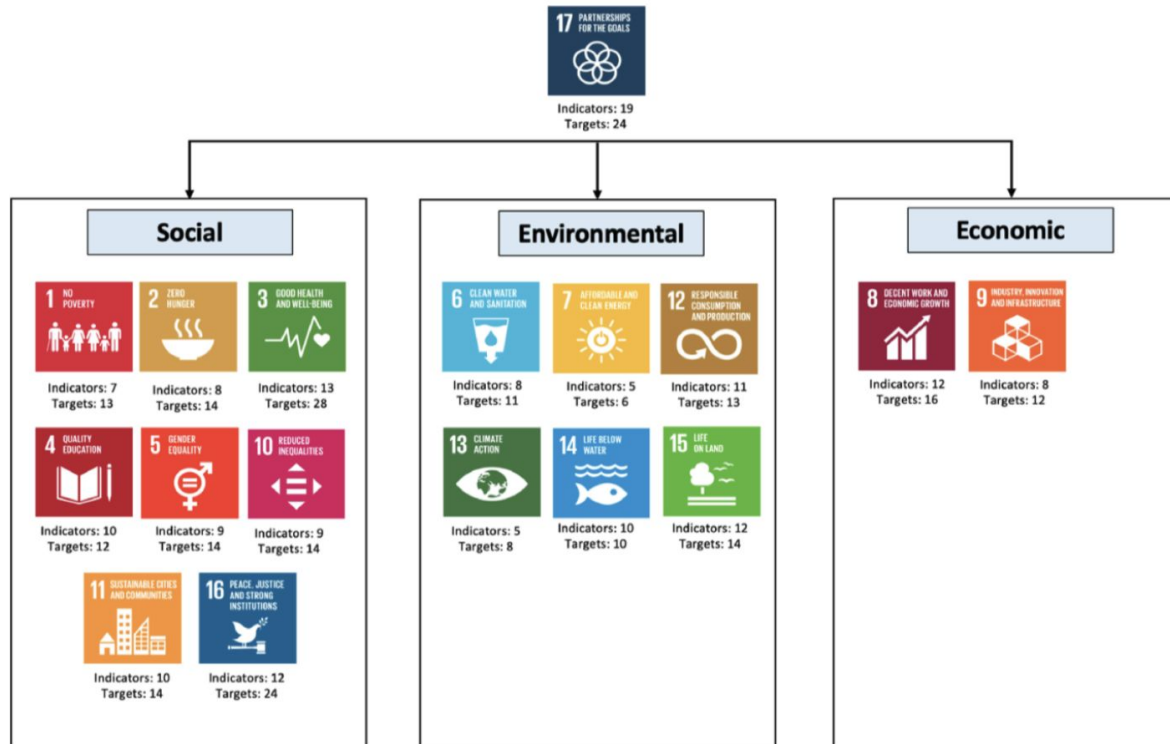
Motivation

SUSTAINABLE DEVELOPMENT GOALS



<https://sdgs.un.org/goals>

Motivation



M. Regona, et al. **Artificial intelligence and sustainable development goals: Systematic literature review of the construction industry**. SCS, page 105499, 2024.

Three Pillars of Sustainable Development



It is misleading to consider the pillars in isolation, as each Sustainable Development Goal **often impacts multiple pillars simultaneously**.

Motivation

Shifting from a siloed approach to an integrated approach for achieving SDGs is important.



A siloed approach cutting off the intrinsic interlinkages



An integrated approach through SDG interlinkages

AI and Sustainability

Motivation: AI and Sustainability



Van Wynsberghe, A. "Sustainable AI: AI for sustainability and the sustainability of AI." *AI and Ethics* 1.3 (2021): 213-218

Using AI to achieve SDGs:

We need to use #aiforgood, and harness its potential for sustainable development

Use SDGs to develop AI:

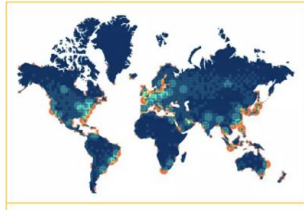
We need to use the driving principles of the SDGs when developing and deploying AI

Motivation: AI for Sustainability



Motivation: AI for Sustainability

AI4SDGs Cooperation Network



Satellite images analytics and
Agro-meteorological monitoring



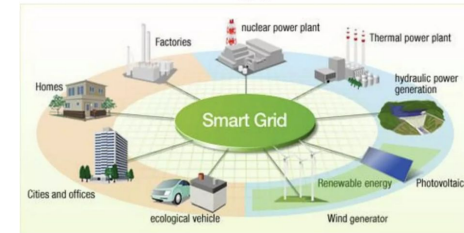
Personalized Learning



Predictive Healthcare
(analyzes historical data to prevent future target events)



• Smart renewable energy grids



Motivation: AI for Sustainability

<https://ai-for-sdgs.academy/>

Browse by UN Sustainable Development Goals



Recent Projects

Synergizing language learning:
SmallTalk AI In industry 4.0 and
Education 4.0

Goal 4 Quality Education

Online shopping consumer perception
analysis and future network security
service technology using logistic
regression model

Goal 8 Decent Work and Economic Growth

Rural micro-credit model design and
credit risk assessment via improved
LSTM algorithm

Goal 1 No Poverty

Goal 8 Decent Work and Economic Growth

A consumer emotion analysis system
based on support vector regression
model

Goal 8 Decent Work and Economic Growth

Machine learning based IoT system for
secure traffic management and
accident detection in smart cities

Find a project

Try searching for project title, abstract and comment that you are interested in.

Search by title, abstract or comment

Search

Submit a project

Share your project to the world and get it evaluated.

Submit



Rating on Sustainability

Each project is rated based on its potential impact on sustainable goals, using our sustainability index rating scheme.



Negativity Evaluation

Projects who have potential negative impacts on SDGs are rated and marked.



Comment on Sustainability

Every project comes with a detailed comment on its potential influence on SDGs, listing the reasons why it gets the rating.



Search

Search for any project title, content or SDGs that you are interested in.



Share Your Project

Submit your project information to get it listed and rated.

Motivation: AI for Sustainability

AI4SDGs Cooperation Network



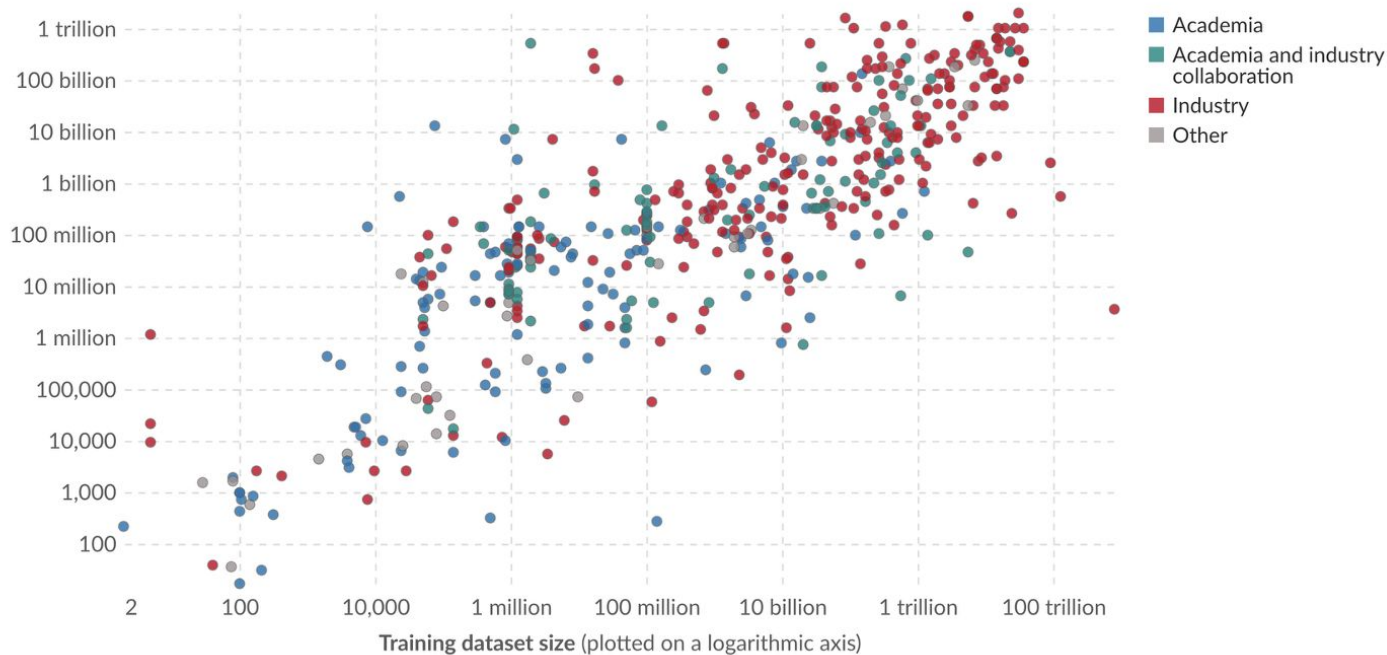
Is AI always an accelerator for the Sustainable Development Goals?

Motivation: Sustainability of AI



Motivation: Sustainability of AI

Number of parameters (plotted on a logarithmic axis)



Data source: Epoch AI (2026)

OurWorldinData.org/artificial-intelligence | CC BY

Motivation: Sustainability of AI

AI IMPACT RISK



**Bias and voids in
training data**



**Unauthorized use
of art, code, text, or
data**



**Power concentrated
in a few companies**



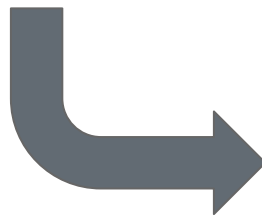
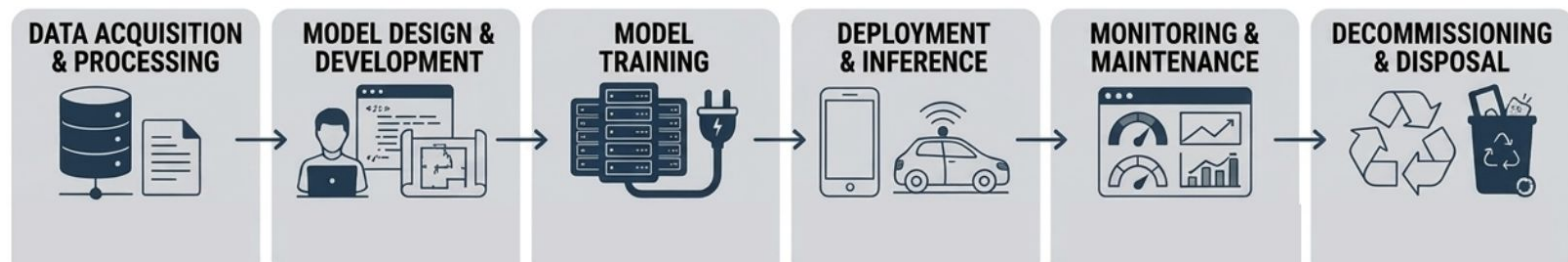
Loss of human jobs



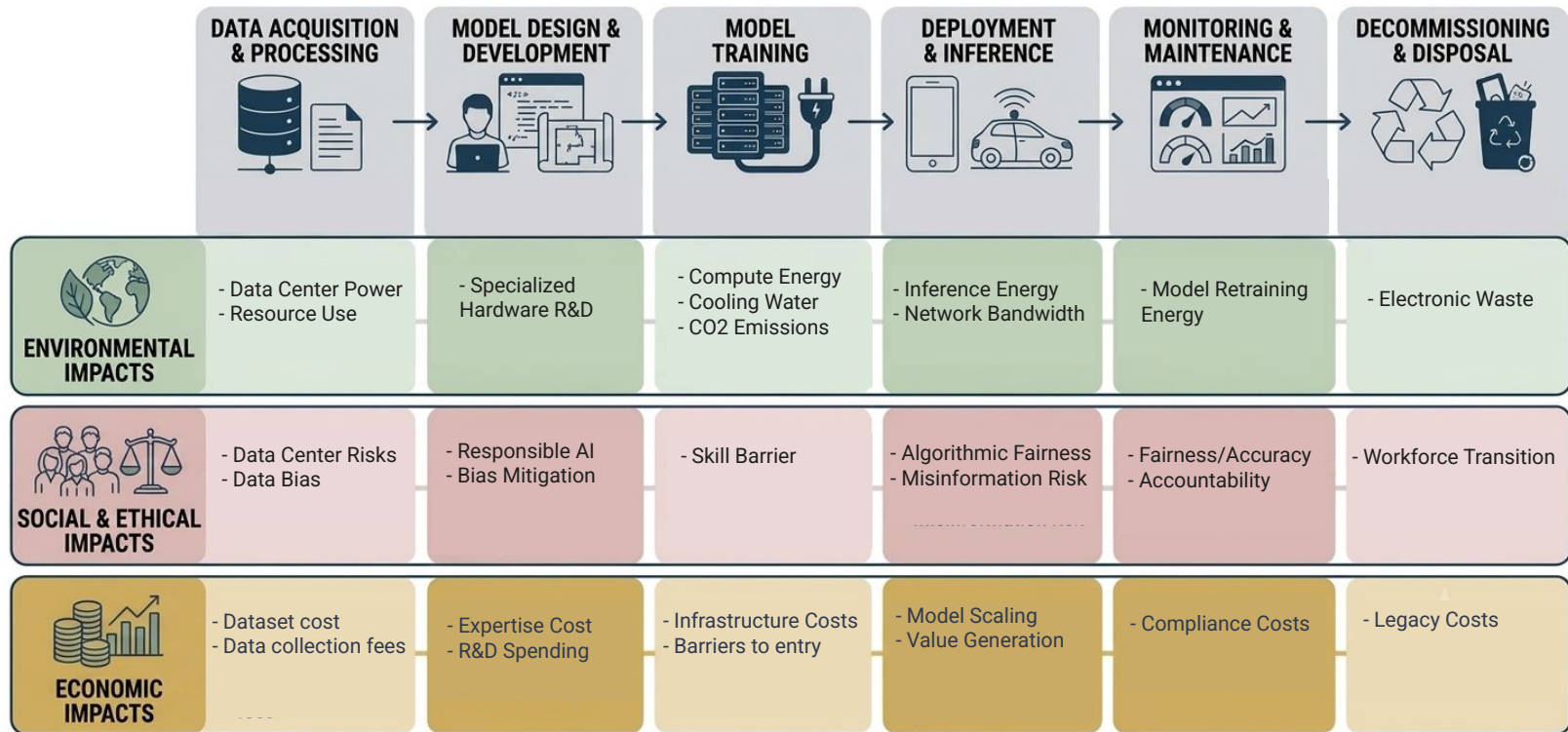
**Carbon and water
use**

Motivation: Sustainability of AI

AI lifecycle



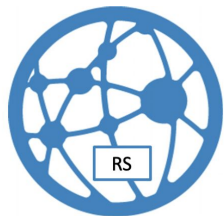
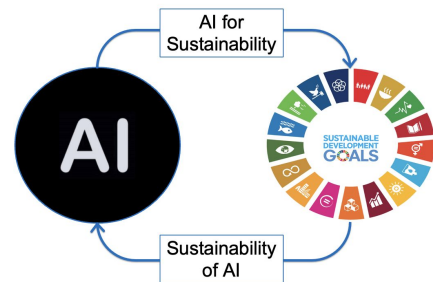
Motivation: Sustainability of AI



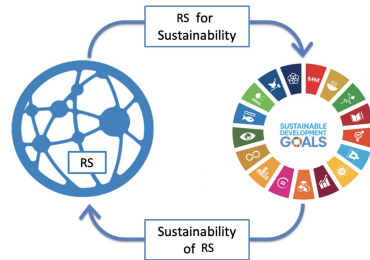
Our Vision



We should **balance**
the two perspectives



We focus on
Recommender Systems



RS and Sustainability

Our Vision: RS and Sustainability



Which can be the connection with Sustainability and
Recommender systems (RS)?

Why Recommender Systems?

Recommender Systems (RS) are AI technologies that:

- ◆ handle the information overload by providing **personalized item lists**
- ◆ support the everyday **decision-making process**

Recommender systems have always been designed to influence choices.

Now, **sustainability goals** indicate how to **guide these choices**.

Why Recommender Systems?

RS can be very effective in **actively promoting sustainable behaviors** and societal good

RS for tailored healthy food recommendations



Tourism RS for Sustainable City Trips



RS for Personalised Learning Experiences



RS for Promoting Virtuous Energy Consumption Behaviors



ECONOMIC



ENVIRONMENTAL



SOCIAL

Our Vision for RS



Our Vision for RS



Our Vision for RS



Computer Science Review 60 (2026) 100912



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Recommender systems and sustainability: a dual perspective

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Research Methodology

RS for Sustainability

Application of Recommender Systems to promote Sustainable Development Goals (SDGs), focusing on environmental, social, and economic pillars.

Sustainability of RS

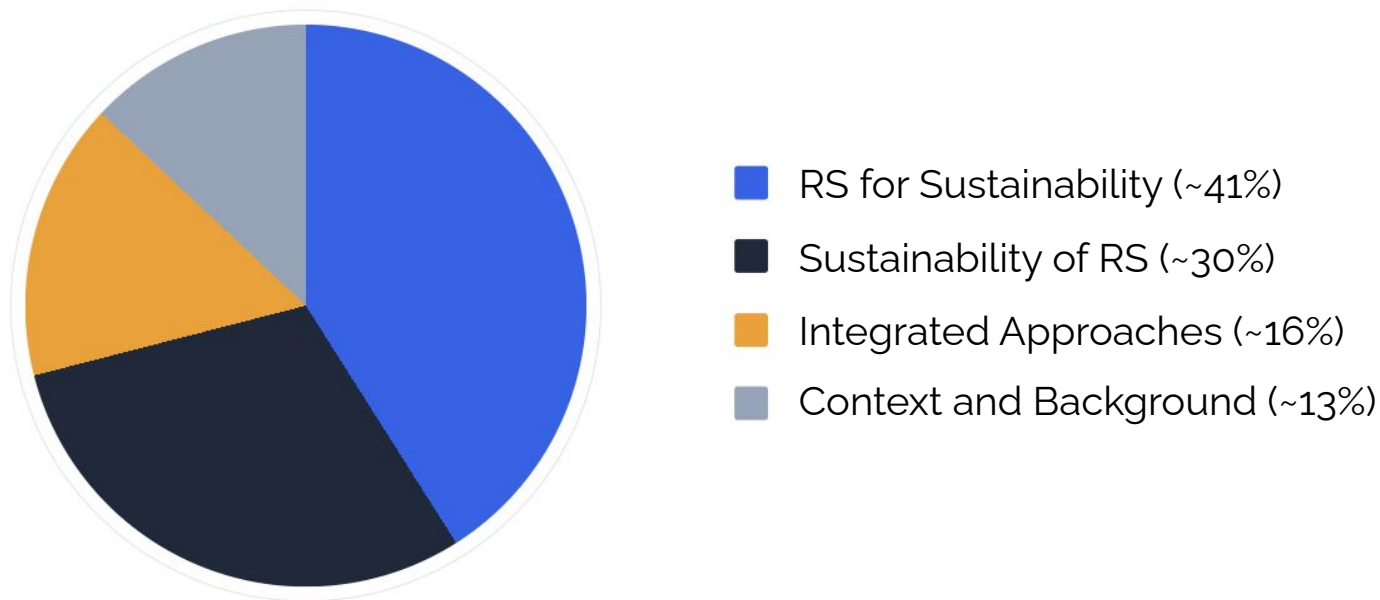
Designing intrinsically sustainable algorithms, prioritizing fairness, transparency, energy consumption, and economic efficiency.

Protocol inspired by PRISMA: Systematic search conducted in April 2025 via Google Scholar, Scopus, DBLP, and ACM Digital Library.

Selection Process

-  **Collection:** 250 initial records identified through targeted queries (e.g., "Green AI", "Fairness", "Carbon footprint").
-  **Screening:** Removal of 98 duplicates and subsequent manual analysis based on inclusion/exclusion criteria.
-  **Exclusion:** 43 works discarded as non-peer-reviewed or not focused on RS and sustainability.
-  **Outcome:** 109 studies selected for final analysis, documented to ensure reproducibility.

Literature Distribution (n=109)



Studies were categorized by sustainability pillar and specific technical topics.

What are we doing in this research direction?

- **Three editions of the International Workshop on** «Recommender Systems for Sustainability and Social Good», co-located with RecSys **2024**, **2025** and **2026**

- Contribution to the current literature:
 - ◆ **published volumes** on «Recommender Systems for Sustainability and Social Good» (Springer, CCIS Series)
 - ◆ **published (and under review) papers** on different techniques to reduce the environmental impact of RS, and to balance the social and economic impact of RS.

RecSoGood Workshop

Third International Workshop on Recommender Systems for Sustainability and Social Good co-located with RecSys 2026



The ACM Conference Series on
Recommender Systems



RecSoGood Volumes

Ludovico Boratto ·
Allegra De Filippo ·
Elisabeth Lex ·
Francesco Ricci (Eds.)

Communications in Computer and Information Science 2470

Recommender Systems for Sustainability and Social Good

First International Workshop, RecSoGood 2024
Bari, Italy, October 18, 2024
Proceedings



Ludovico Boratto · Allegra De Filippo ·
Elisabeth Lex · Noemi Mauro ·
Francesca Maridina Mallocci ·
Francesco Ricci (Eds.)

Communications in Computer and Information Science 2802

Recommender Systems for Sustainability and Social Good

Second International Workshop, RecSoGood 2025
Prague, Czech Republic, September 26, 2025
Proceedings



Our current works

1. A. De Filippo, G. Spillo, L. Boratto et al. Recommender systems and sustainability: **a dual perspective**. In **Computer Science Review**, 2026.
2. G. Spillo, A. De Filippo, et al. Balancing carbon footprint and algorithm performance in recommender systems: **A comprehensive benchmark**. In **Sustainable Computing: Informatics and Systems**, 2025.
3. G. Spillo, A. De Filippo, et al. Comparing **data reduction strategies** for energy-efficient **green recommender systems**. In **Journal of Intelligent Information Systems**, 2025.
4. G. Spillo, A. De Filippo, et al. Training Green and Sustainable Recommendation Models: Introducing **Carbon Footprint Data into Early Stopping Criteria**. In ACM **UMAP**, 2025.
5. G. Ramos, L. Boratto, et al. Temporal User-Agnostic Ranking: Detecting Preference Evolution while Preserving **Ethical Principles**. In ACM **SIGIR**, 2026.
6. E. Gomez, D. Contreras, L. Boratto, et al. Ensuring provider **fairness in recommender systems** across coarse- and fine-grained groups. In **International Journal of Human-Computer Studies**, 2026.



ENVIRONMENTAL



SOCIAL



ECONOMIC

Outline

What will we learn today?



1. RS for Sustainability

How can recommender systems be designed to not just reflect user preferences, but **to actively promote sustainable behaviors and societal good**?

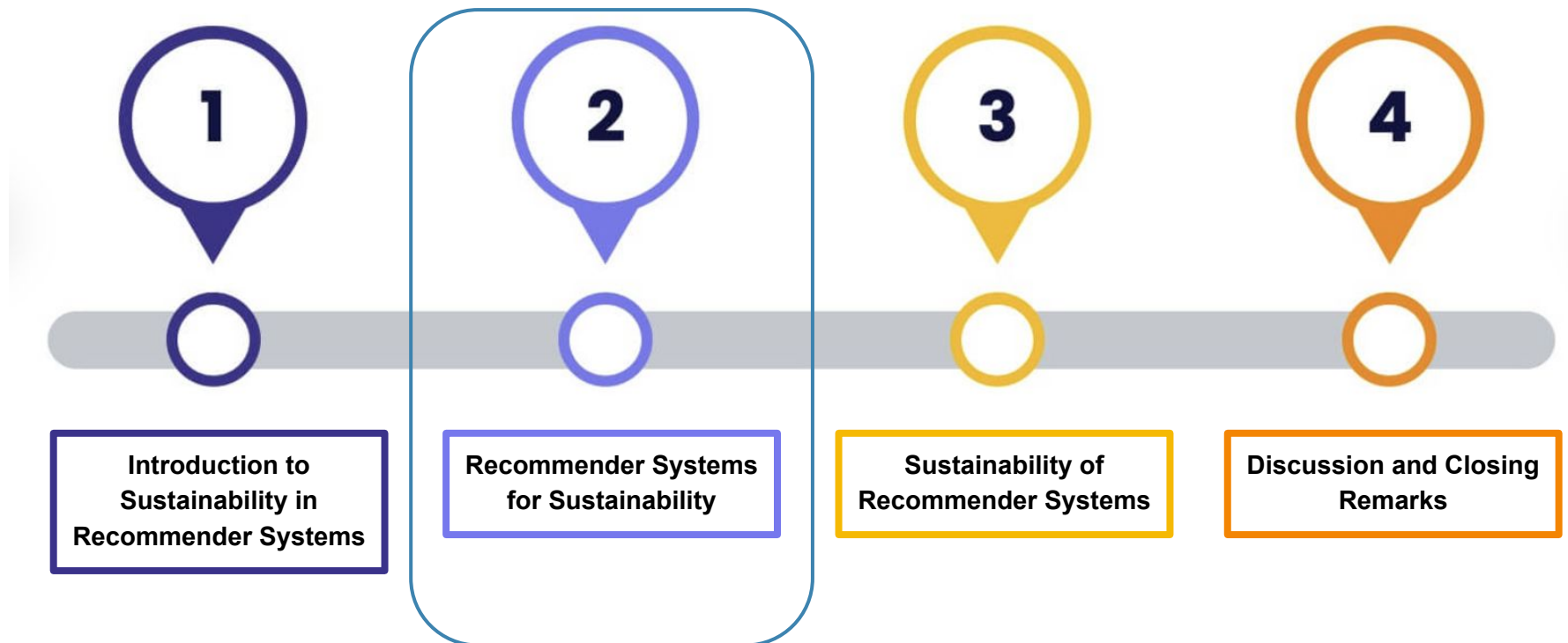
2. Sustainability of RS

What are the hidden **environmental and ethical costs** of deploying large-scale recommender systems, and how can we redesign them to **align with sustainability principles**?

3. The Dual Perspective

Can we envision a future where recommender systems **both enable sustainable choices and are themselves sustainable**, without compromising performance or user experience?

Tutorial Outline



1. RS for Sustainability



How can recommender systems be designed to not just reflect user preferences, but **to actively promote sustainable behaviors and societal good?**

RS for Sustainability

Refers to systems designed to **support SDGs through recommendation**

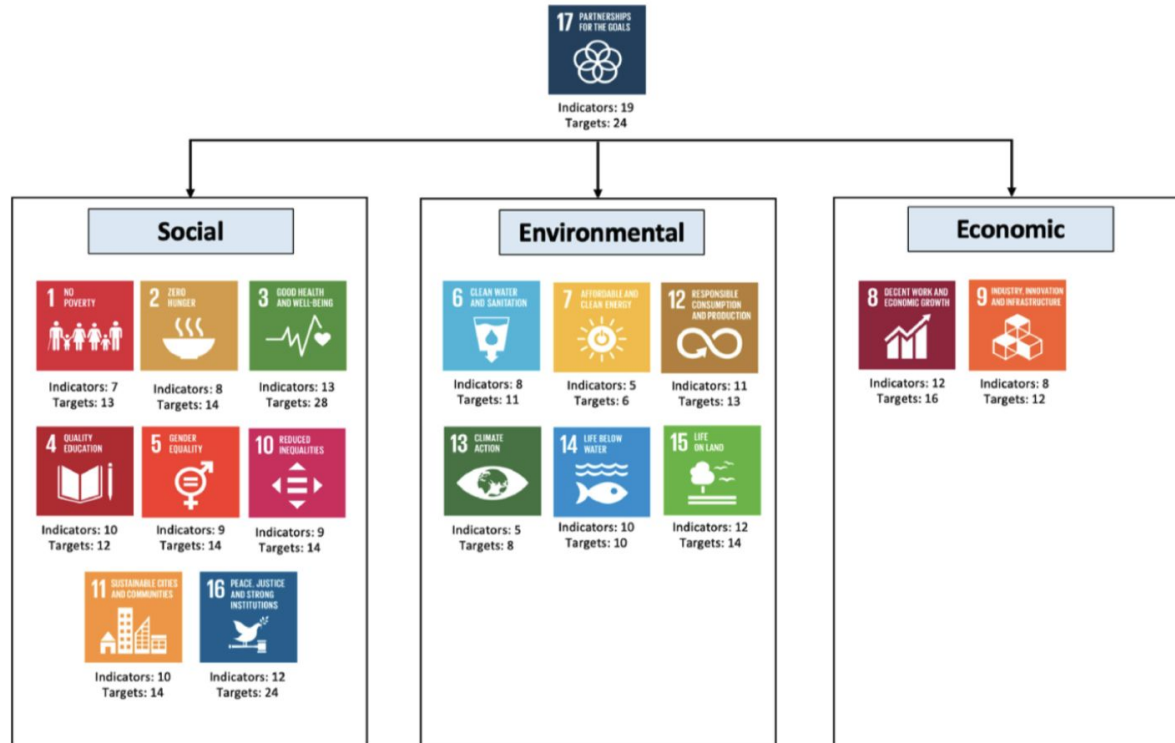
Recommendations lead to **interventions**, what we predict has an **impact**



Alignment with the broader vision of “AI for Sustainability”

Pillar map

SDGs and the Three Pillars



M. Regona, et al. **Artificial intelligence and sustainable development goals: Systematic literature review of the construction industry**. SCS, page 105499, 2024.

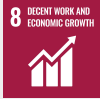
Social pillar

01	Equity and inclusivity		
02	Well-being		
03	Education and Empowerment		
04	Poverty and Hunger		
05	Urban Inclusion		
06	Justice and Institutions		

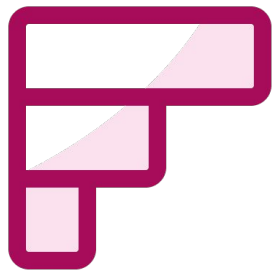
Environmental pillar

01	Resource preservation			
02	Ecological footprint reduction			
03	Climate change mitigation			
04	Marine ecosystem protection			
05	Terrestrial biodiversity			

Economic pillar

01	Economic growth and employment		
02	Industrial innovation		
03	Efficiency in production/consumption		

M. Regona, et al. **Artificial intelligence and sustainable development goals: Systematic literature review of the construction industry**. SCS, page 105499, 2024.



Are you targeting one or more pillars in your current research?



Social pillar

RS for Social Sustainability



Social pillar targets

- Equity
- Well-being
- Access to resources



Decision support by RS

- Poverty
- Health
- Education

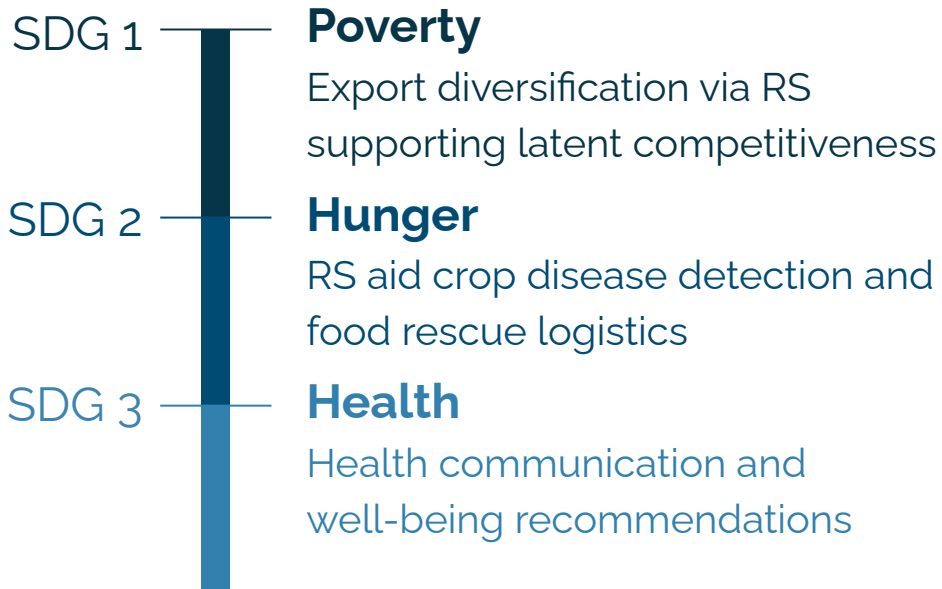


Impact

- Gender equality
- Inclusive cities

De Filippo, et al. "**Recommender systems and sustainability: a dual perspective.**" *Computer Science Review* 60 (2026): 100912.

Examples of RS for Social Sustainability



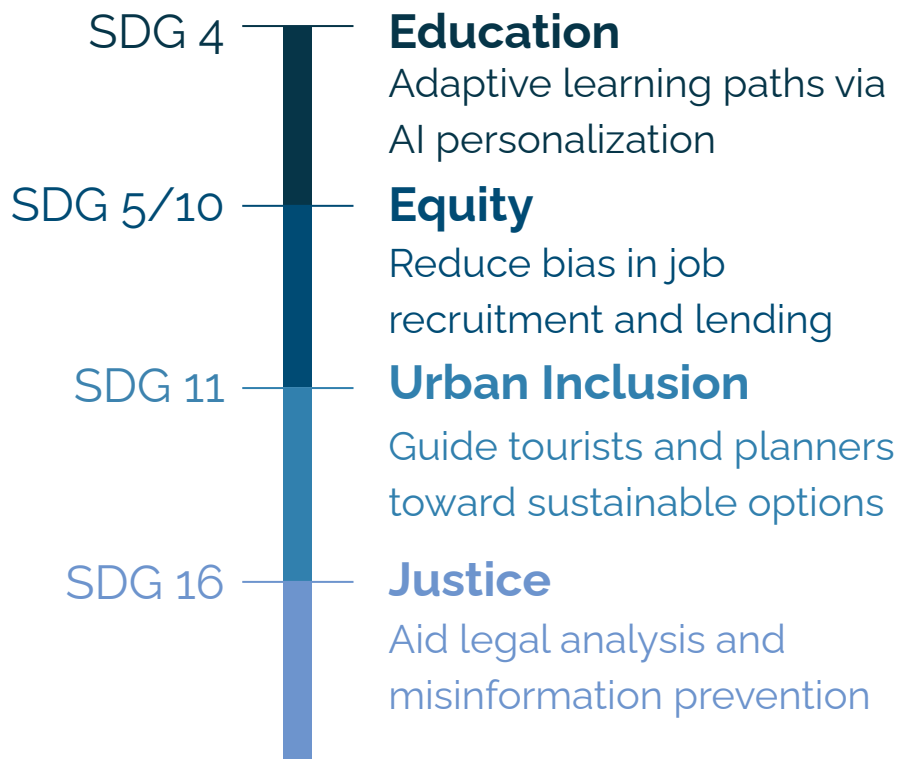
N. Che. *Intelligent export diversification: An export recommendation system with machine learning*. IMF Working Paper, 2020.

A. Ahmad, et al. *A survey on using deep learning techniques for plant disease diagnosis and recommendations for development of appropriate tools*. Smart Agricultural Technology, 3:100083, 2023

Z. Shi, et al. *Predicting and presenting task difficulty for crowdsourcing food rescue platforms*. In ACM Web Conf., pages 4686–4696, 2024

J.N. Cappella, et al. *Critical considerations for using cultural targeting and tailoring in health communication interventions*. Health Communication, pages 1–12, 2024

Examples of RS for Social Sustainability



A.Y. Huang, et al. *Effects of artificial intelligence-enabled personalized recommendations on learners' learning engagement, motivation, and outcomes in a flipped classroom*. Computers & Education, 2023

C. Meng, et al. *Dfrp: A dual-track feedback recommendation system for educational resources*. In IJCAI, 2024

T. Bogers, et al. *Fourth workshop on recommender systems for human resources (recsys in hr 2024)*. In ACM RecSys, 2024

K. Sivamayilvelan, et al. *Flexible recommendation for optimizing the debt collection process based on customer risk using deep reinforcement learning*. Expert Systems with Applications, 2024

A. Banerjee. *Fairness and sustainability in multistakeholder tourism recommender systems*. In ACM UMAP, 2023

P. Merinov. *Sustainability-oriented recommender systems*. In ACM UMAP, 2023

J. Dhanani, et al. *Legal document recommendation system: A cluster based pairwise similarity computation*. Journal of Intelligent & Fuzzy Systems, 2021

D. Sallami, et al. *Trust-based recommender system for fake news mitigation*. In ACM UMAP, 2023

SDG 1: Poverty as a decision-support driver

Social pillar: access to resources + decision support



People → financial choices
access, agency, constraints



Countries → export opportunities
capabilities, diversification,
development



RS task

Rank feasible actions

Deep dive: Export diversification as top-N recommendation

Input

Country × product matrix
RCA = revealed comparative advantage

Prediction

Predict missing RCA →
rank top-N export products

Goal

Sustainability link:
discover capability-fit diversification paths

01

agri varieties

02

light manufacturing

03

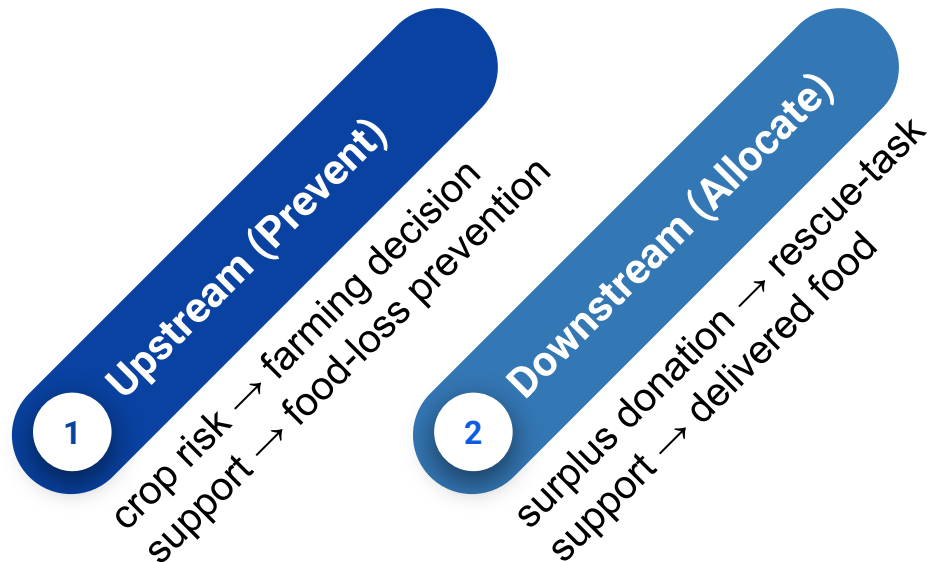
machinery/equipment

SDG 1: Take-home message

SDG 1 teaches us that RS for Sustainability can expand **opportunity sets under scarcity**: the user-item abstraction still works, but the objective **shifts from preference satisfaction to capability-aware decision support**

SDG 2: Zero hunger and food rescue

Food security needs prevention and allocation



Recommendation can support **food security operations**, and go **beyond individual food choices**

Ahmad, et al.. **A survey on using deep learning techniques for plant disease diagnosis and recommendations for development of appropriate tools**. Smart Agricultural Technology, 2023

Omara, et al. **A field-based recommender system for crop disease detection using machine learning**. Frontiers in Artificial Intelligence, 2023.

Bouni, et al. **Towards an efficient recommender systems in smart agriculture: A deep reinforcement learning approach**. Procedia Computer Science, 2022.

Shi, et al. **A recommender system for crowdsourcing food rescue platforms**. In ACM Web Conf., 2021.

Shi, et al. **Predicting and presenting task difficulty for crowdsourcing food rescue platforms**. In ACM Web Conf., 2024.

Deep dive: food rescue platforms



1 Learn with scarce labels

1,000 rescues labeled by experts

Past comments → BERT

Train on ratings, then difficulty



Pseudo-labels + expert labels

Train LightGBM on tabular features

Comments are used only for training.

FOR A NEW RESCUE

Features → LightGBM → easy?/hard?

2 Test the method

Offline ROC-AUC

Mean over 10 runs

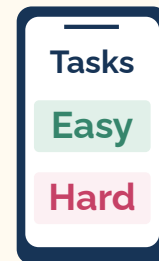
	Easy	Hard
Tabular only	0.543	0.495
No rating pretraining	0.709	0.563
Full method	0.710	0.685

Ratings help hard rescues;
easy gain is not significant.

3 Support choice

8 volunteers + 2 admins

Six storyboard designs



Show task difficulty
Keep all tasks visible
Explain predictions

Preferred by the volunteers

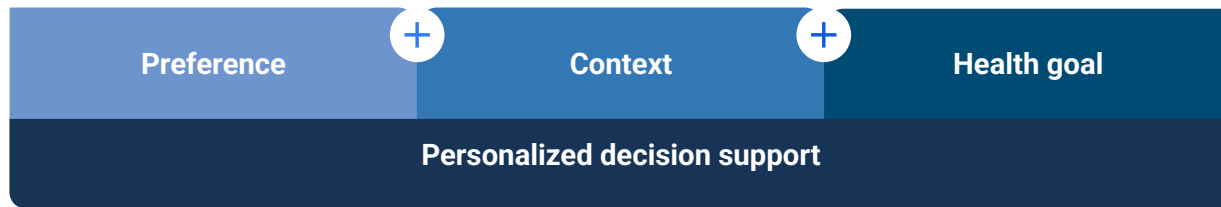
Retention and hunger-reduction gains were not measured

SDG 2: Take-home message

SDG 2 teaches that RS for Sustainability can become **food-security infrastructure**: recommendation supports both **prevention of food loss upstream** and **redistribution of surplus food downstream**, but SDG impact must be measured beyond platform accuracy

SDG 3: Health and well-being recommendations

Health RS guide behavior, going beyond content recommendation



Population level



tailored messages
awareness + prevention

Individual level



meals/activity/sleep
healthier routines

Impact needs **evidence**, not only engagement

Kim, et al.. *An experimental study of recommendation algorithms for tailored health communication*. Computational Communication Research, 2019.

Cappella, et al. *Critical considerations for using cultural targeting and tailoring in health communication interventions*. Health Communication, 2024.

Alcaraz-Herrera, et al. *Evorecsys: Evolutionary framework for health and well-being recommender systems*. UMUI, 2022.

Deep dive: Health and Well-being Recommendations

Approach

Profile, preferences, goal



CF-guided genetic search

Mean error: preference fit, nutrition,
activity fit, consistency/diversity



Meal + activity bundles

Tailored portions and duration

Algorithm test · 50 trials/variant

Lower fitness error vs standard GA

Evaluation

205 volunteers · 3 bundles each

4.15/5

Perceived
healthiness

3.64/5

Perceived
appeal

44 volunteers · 220 choices

5 blind comparisons per volunteer

56.4%

EvoRecSys (124)

43.6%

CF baseline (96)

Perceived quality ≠ demonstrated health improvement

SDG 3: Take-home message

SDG 3 teaches that, in health and well-being, recommendation becomes a **responsible behavior-change intervention** that must **balance preference fit, goals, constraints, safety, autonomy, and long-term impact**

SDG 4: Personalized education

Resource match



Which resource is relevant?

SDG 4 aim

Learner state



Prior knowledge,
goal, motivation



Adaptive path



Resource Activity Feedback



Capability



Knowledge, skills,
lifelong learning

Support learning progress, not just content access

Huang, et al. (2023). *Effects of artificial intelligence-enabled personalized recommendations on learners' learning engagement, motivation, and outcomes in a flipped classroom*. Computers & Education.

Deep dive: Learning as a consequence of gaps' targeting

Systems programming. Same flipped course and videos. With RS: 43 / Without: 59

Approach

Self-test (SPRT)

Concept mastery

Viewing logs

Time per video

1 Concepts not fully mastered

2 Shortest viewing time first



After-class review-video list

Assess learning gains and who benefits from them.

Equity and long-term retention not evaluated

Evaluation

Whole cohort

Higher video engagement;
learning-gain difference not significant.

Moderate motivation · n = 48

Mean gain (posttest – pretest); $p < .05$

With RS, n=20

18.20

No RS, n=28

4.14

SDG 4: Take-home message

SDG 4 teaches us that recommendation can be **capability-building**: not only predicting what a learner may click, but **sequencing resources, activities, and feedback** so learners can progress toward **knowledge, skills, and lifelong learning**.

SDGs 5/10: Equity-oriented social applications

Access domains



Jobs



Credit



Organ/blood
matching

RS decision support



Rank, match, suggest

Equity questions

- | Who is seen?
- | Who is excluded?
- | Who bears risk?



Historical bias can become unequal access

Deep dive: vehicle-loan debt collection

Vehicle-loan collection, recommendations to field agents



≠ Better collection does not prove fairer treatment. Equity not evaluated.

SDGs 5/10: Take-home message

SDGs 5/10 teaches RecSys that some **recommendations are access gatekeepers**: quality is not separable from who is **surfaced**, who is **excluded**, who receives **opportunity**, and who bears the **risk of errors**

SDGs 11/16: Cities, tourism, institutions, misinformation

Shared spaces



Crowding



Public-interest decision support



Trips · planning



**Inclusive
cities**

Shared trust



Information
overload



Legal cases · news



Access, integrity

Human oversight

Goals, not guarantees

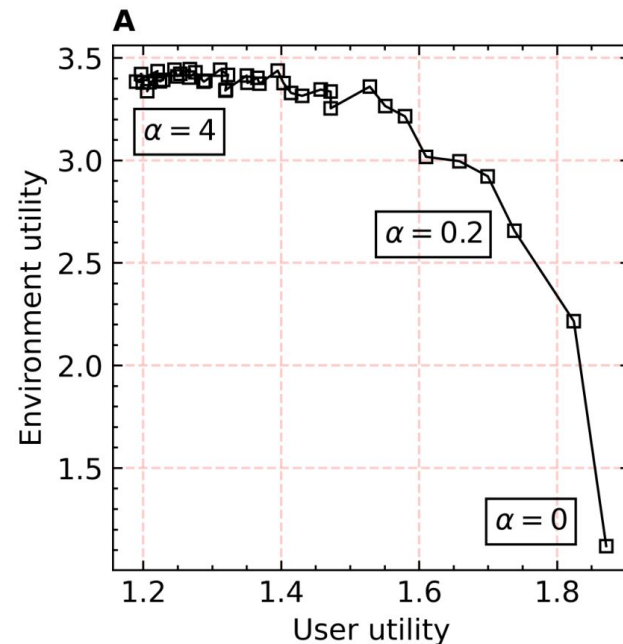
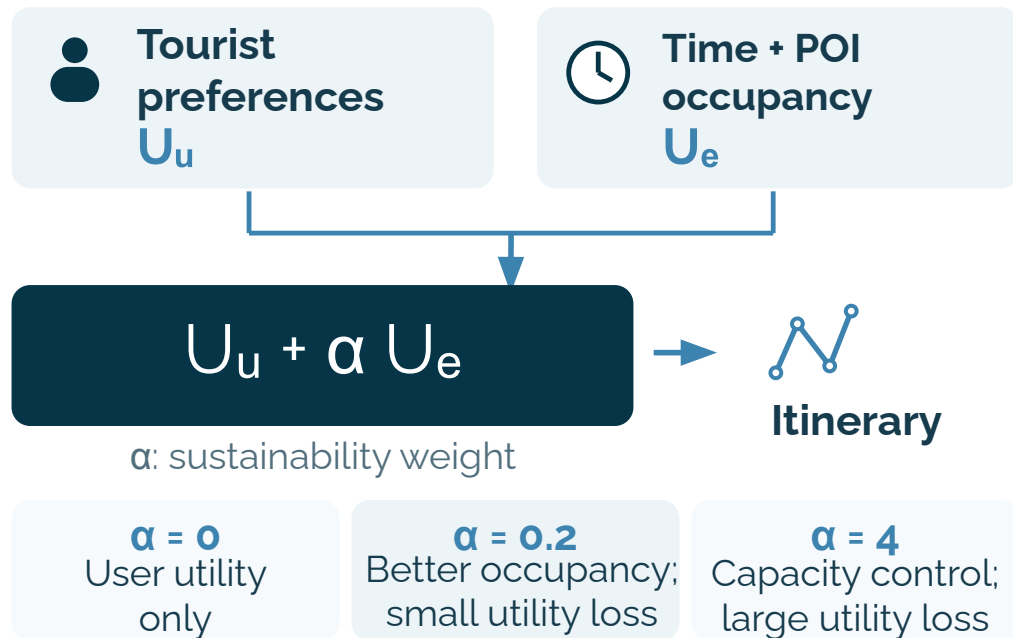
Individual utility ↔ collective impact ↔ accountability

Banerjee. 2023. **Fairness and Sustainability in Multistakeholder Tourism Recommender Systems**. UMAP '23.

Dhanani, et al. 2021. **Legal document recommendation system: A cluster based pairwise similarity computation**. Journal of Intelligent & Fuzzy Systems.

Dorsaf Sallami, et al. 2023. **Trust-based Recommender System for Fake News Mitigation**. UMAP '23 Adjunct.

Deep dive: A good trip for whom?



Simulation, not deployment. Occupancy utility is not measured CO₂e.

Deep dive: Similarity is not credibility

1 ESTIMATE



Sharing histories

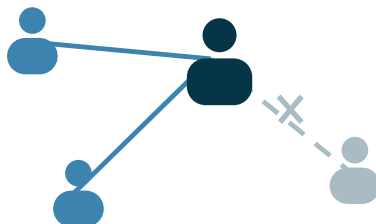
Base reliability

$$R = \frac{g + 1}{g + f + 2}$$

g: genuine-labelled shares
f: fake-labelled shares

2 FILTER NEIGHBOURS

Similar neighbours



Updated $R > 0.5$

Then aggregate

3 RANK & EVALUATE



Ranked news

FANAR

TF@K ↓

Fake-labelled share
of top-K news

Lower on FNEW
than compared baselines

GOVERNANCE REQUIREMENT

Offline result; classifier-assigned labels.

Scores $\neq$ truth. Human oversight and contestability remain necessary

SDGs 11/16: Take-home message

SDGs 11/16 expands the **unit of accountability**: a recommendation should be assessed not only for the **person receiving** it, but also for the **communities, institutions,** and **information environments** affected by it

Take-home messages: social pillar

Three takeaways: RS can support **access to resources**; many applications involve **vulnerable populations**; evaluation must go **beyond accuracy to real-world social outcomes**

Environmental pillar

RS for Environmental Sustainability



Environmental pillar
targets

- Preservation
- Sustainable resource use



Decision support by RS

- Energy and water savings
- Reduce waste
- Support biodiversity

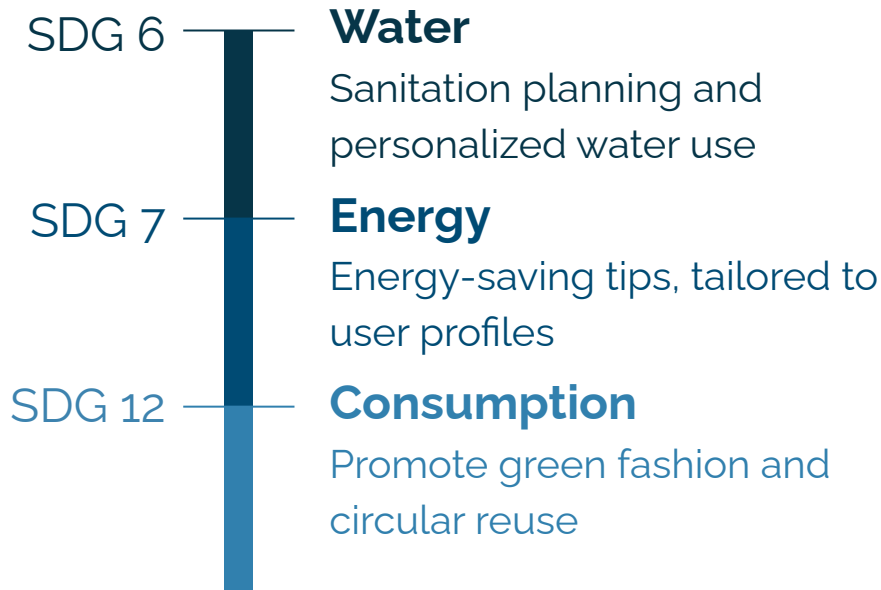


Impact

- Responsible habits
- Local environmental action

De Filippo, et al. "**Recommender systems and sustainability: a dual perspective.**" *Computer Science Review* 60 (2026): 100912.

Examples of RS for Environmental Sustainability



F. Magalhaes Filho, et al. ***Sustainable sanitation management tool for decision making in isolated areas in brazil.*** IJRPH, 16(7):1118, 2019

D. Arsene, et al. ***Decision support strategies for household water consumption behaviors based on advanced recommender systems.*** water 2023, 15, 2550, 2023

V. Riabchuk, et al. ***Utility-based context-aware multi-agent recommendation system for energy efficiency in residential buildings.*** Information Fusion, 112:102559, 2024

S.Yuksel, et al. ***An Integrated Expert Recommender system approach to environmental service priorities in renewable energy.*** Env. Research Communications, 6(9):095001, 2024

A. Cossatin, et al. ***Promoting green fashion consumption through digital nudges in recommender systems.*** IEEE Access, 2024

G. van Capelleveen, et al. ***Toward building recommender systems for the circular economy: Exploring the perils of the european waste catalogue.*** Journal of env. man., 277:111430, 2021

Examples of RS for Environmental Sustainability

SDG 13	Climate Reduce emissions via local energy clustering and VR-based awareness tools
SDG 14	Marine ecosystems RS for aquaculture and illegal fishing alerts
SDG 15	Terrestrial biodiversity RS-based anti-poaching systems

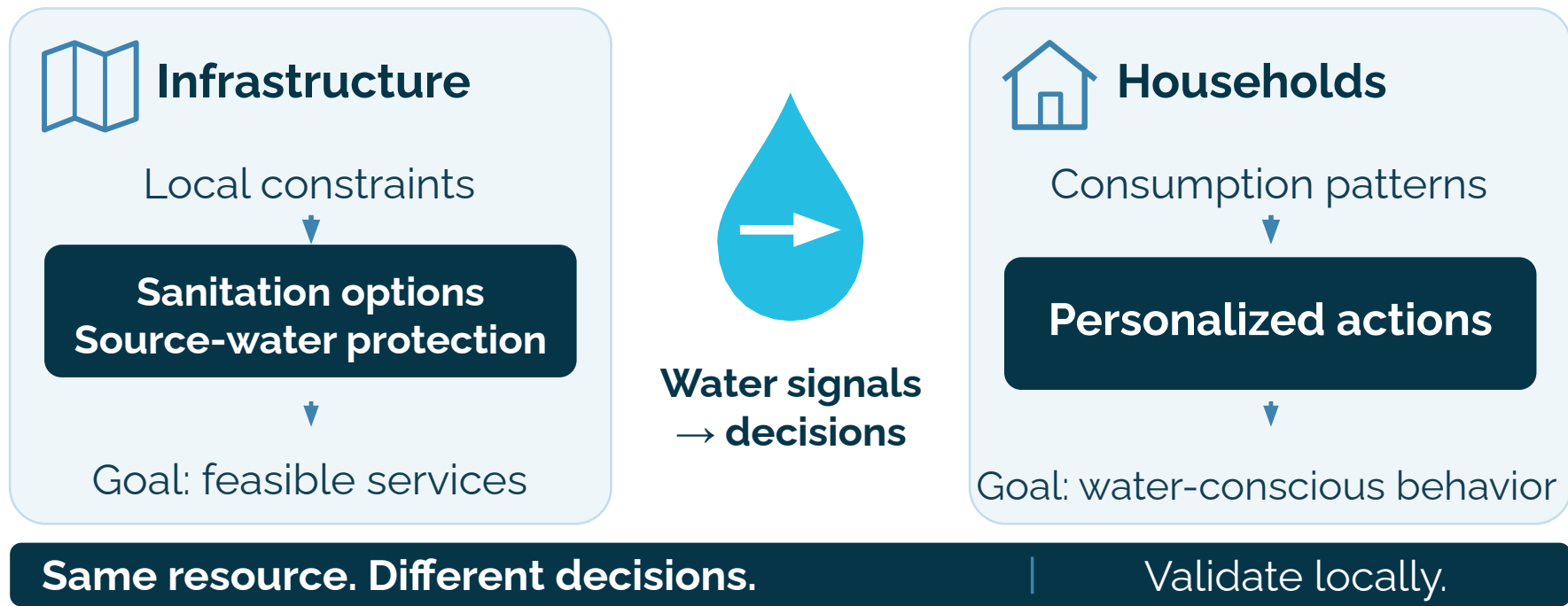
S. Zhang, et al. *Personalized recommendation algorithm based on electricity users' usage behavior portrait*. In IET, number 21, pages 89–98. IET, 2024

A. Clocchiatti, et al. *Vr4green-walk through the (visual) effects of climate change*. In SIGCHI, pages 1–3, 2023

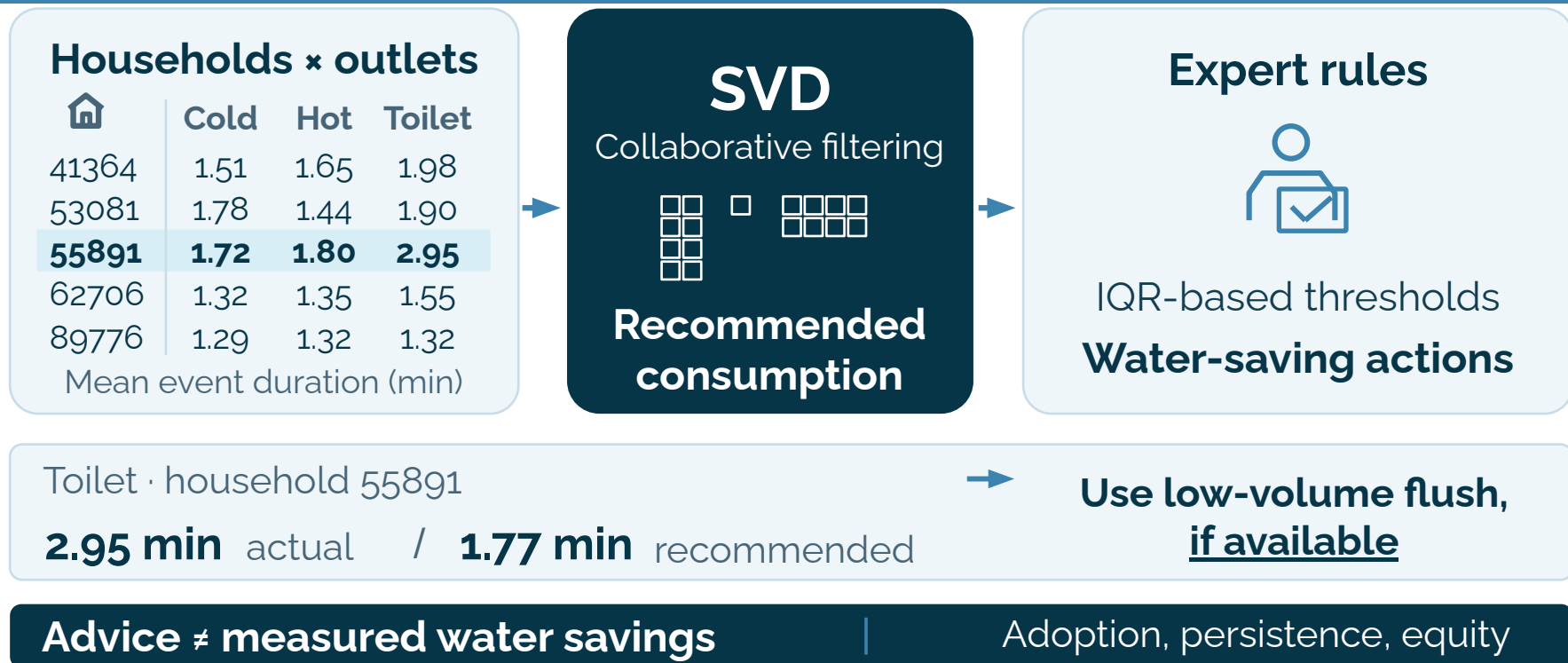
M. Praba, et al. *Smart fish farming recommendation system using k-means algorithm*. In Confluence, pages 333–338. IEEE, 2023

P. Mitra, et al. *Info-wild: Knowledge extraction and management for wildlife conservation*. In CIKM, pages 5281–5284, 2023

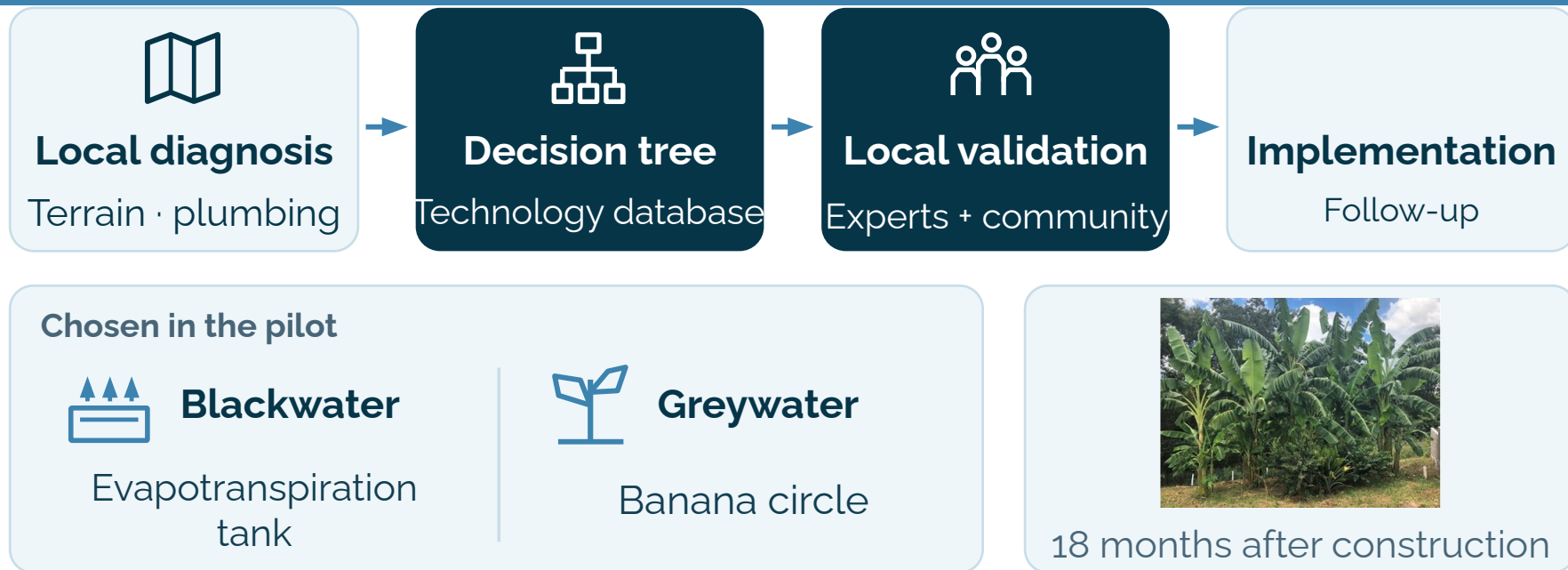
SDG 6: Water and sanitation



Deep dive: integrating an action layer in consumption models



Deep dive: Infrastructure recommendations with local agreement

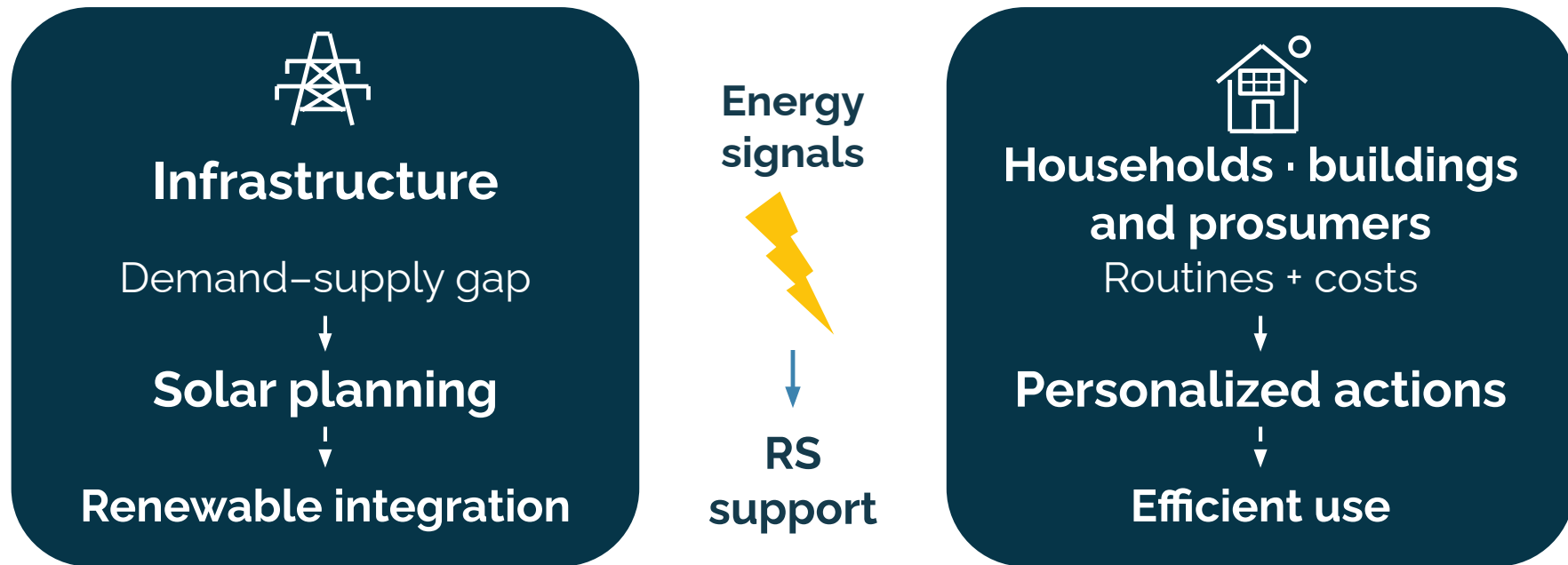


Local implementation ≠ universal suitability

SDG 6: Take-home message

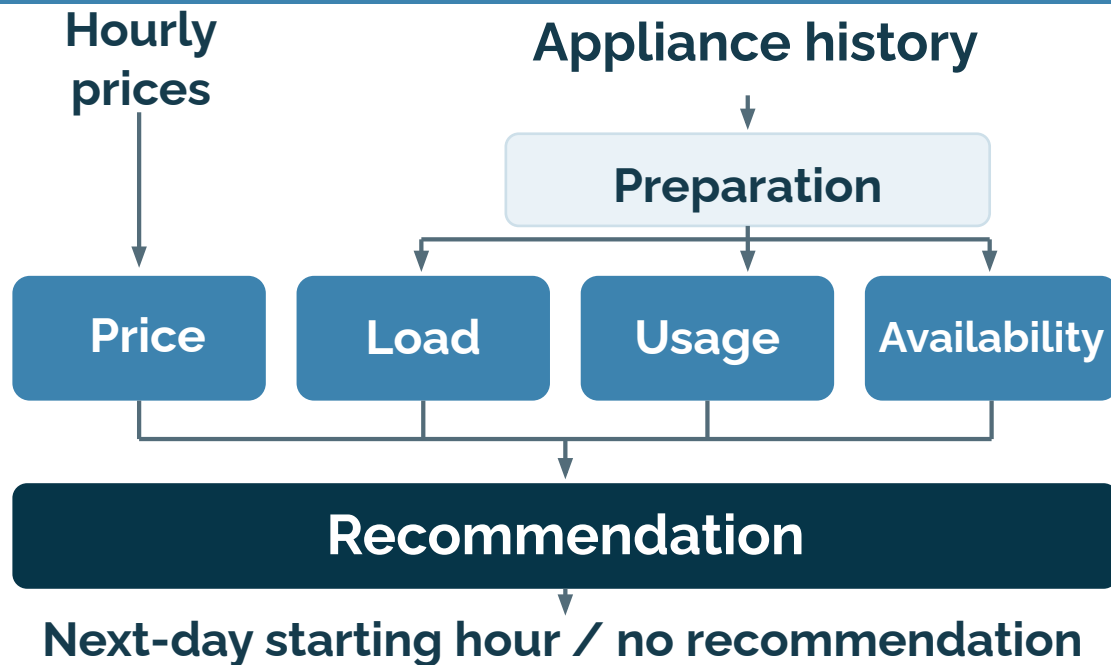
SDG 6 shows that recommender systems can turn **resource-use patterns** and **local infrastructure conditions into actionable decisions**, but those decisions must be **feasible, locally validated, and evaluated through actual water and sanitation outcomes**

SDG 7: Energy efficiency as personalized action



Same system with different decision scales

Deep dive: Recommend when, not just what



Filter

$$P(\text{use}) > \tau_s$$

$$P(\text{available}) > \tau_u$$

**Minimize
whole-run cost**

$$\Sigma (\text{price} \times \text{load})$$

Potential cost savings $\neq$ measured environmental impact

Deep dive: Evaluate both
actions and predictions

Energy recommendation $\neq$ static ranking



Time

When



Context

Occupancy
weather · tariffs



Constraints

Comfort · safety
availability
affordability



Impact

kWh · CO_{2e}
cost ·
persistence

Load shifting $\neq$ impact reduction

Beyond accuracy and acceptance

Deep dive: A cheap hour is not enough

Candidate hour: 08:00 for all three devices

Tumble dryer



Usage too unlikely

No recommendation

Washing machine



Usage condition met

08:00 issued

Dishwasher



Usage too unlikely

No recommendation

Usage probability gates whether a recommendation is issued

Deep dive: Potential savings vary across households

Potential relative cost savings



3–21%
 potential relative
 cost savings

Wholesale-price
 proxy

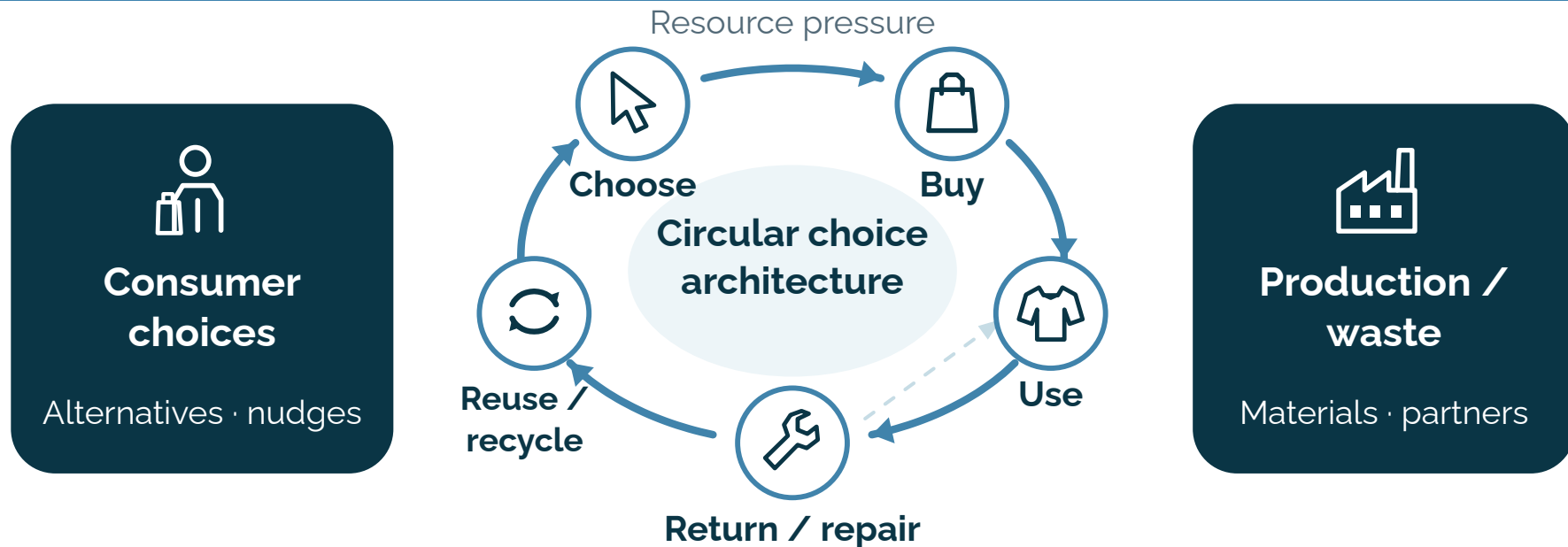
Not measured:
 adoption or emissions

Cost-saving opportunities, not demonstrated SDG 7 impact

SDG 7: Take-home message

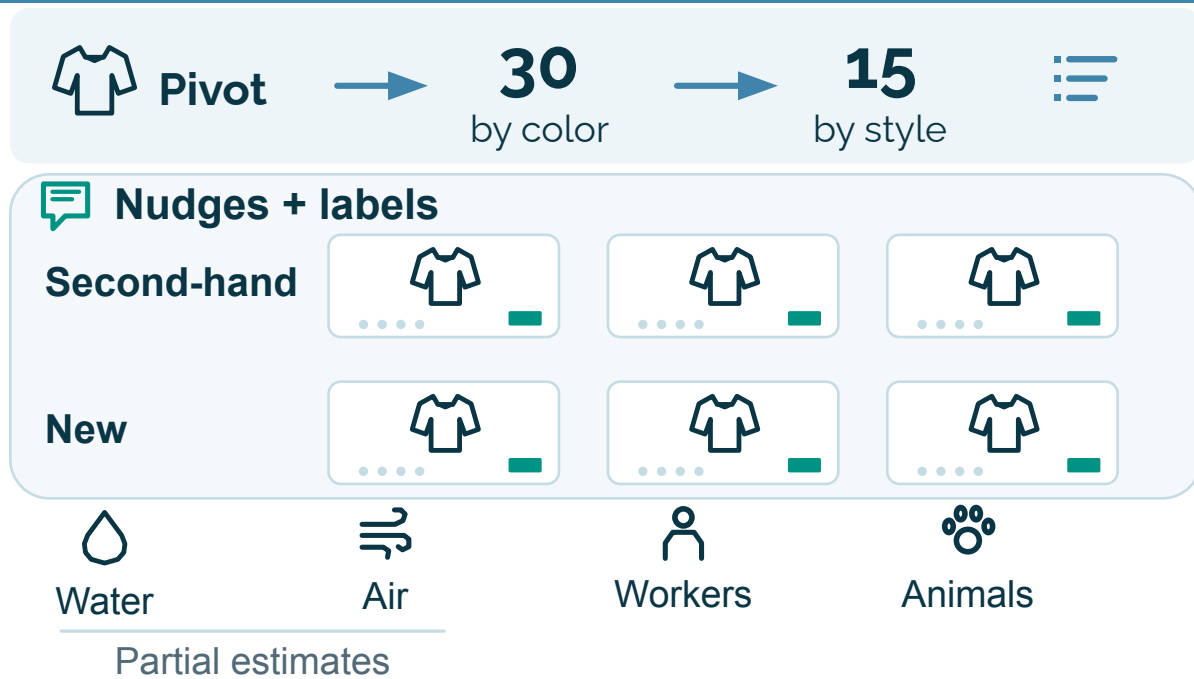
SDG 7 reframes recommendation as **choosing an energy action that fits the situation and its constraints**, and **demonstrating its consequences** beyond prediction accuracy or recommendation acceptance

SDG 12: Responsible consumption and circular economy



Keep materials in circulation. Measure resource use, not sales

Deep dive: Shaping choices via the interface

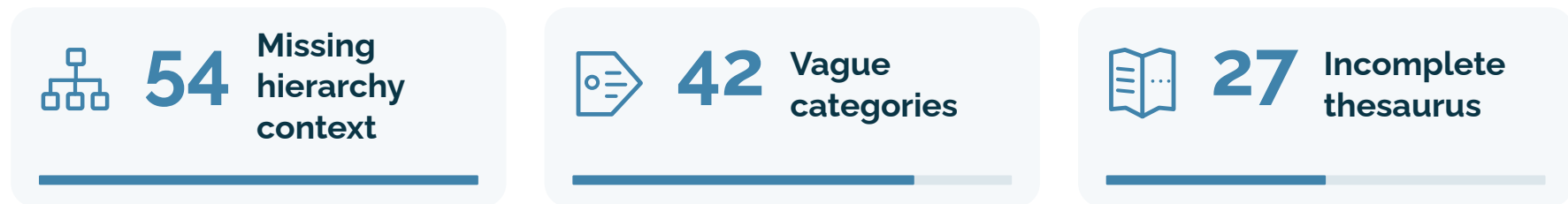


Selection $\neq$ purchase $\neq$ impact

Deep dive: Integrating representation in circularity



Annotation failures



Correct tag $\neq$ feasible match $\neq$ waste diverted

SDG 12: Take-home message

SDG 12 reframes **recommendation as circular choice architecture**: support decisions throughout a **product's or material's lifecycle**, and judge **success by resource use and waste**, not simply by whether another item was selected or sold

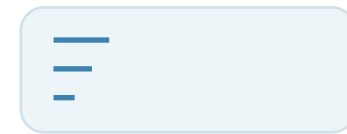
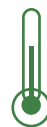
SDG 13: Climate action



Local choices



Climate consequences



Actionable
recommendations



Optimization

Energy • urban design

Mitigation + adaptation



Awareness / behavior

Explanations • feedback

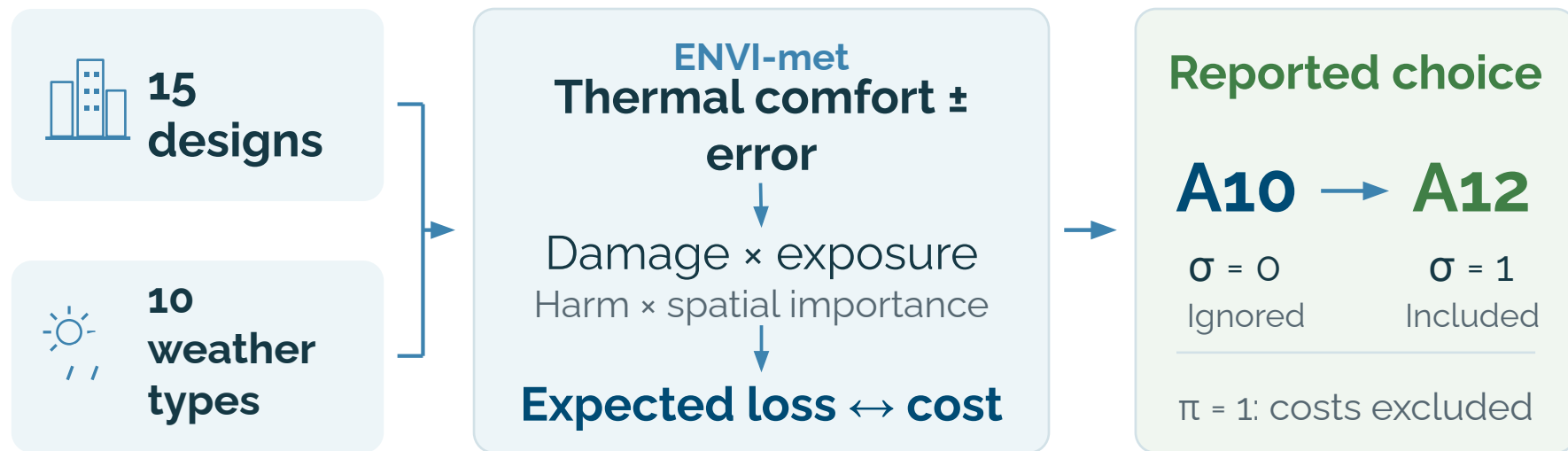
Understand → act

Make climate consequences visible enough to act

Nevat et al. (2020). *Recommendation system for climate informed urban design under model uncertainty*. Urban Climate.
 Sardanios et al. (2021). *The emergence of explainability of intelligent systems: Delivering explainable and personalized recommendations for energy efficiency*. International Journal of Intelligent Systems.
 Clocchiatti, A. et al. (2023). *Demo: VR4Green—Walk through the (Visual) Effects of Climate Change*. CHIItaly 2023.

Deep dive: How uncertainty changes the recommendation

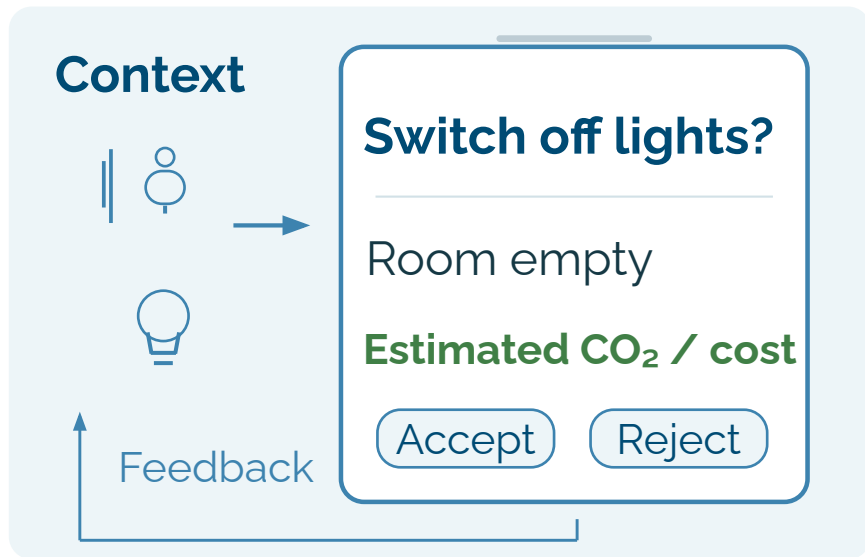
Urban planning, Singapore simulation



Adaptation support $\neq$ measured CO₂ reduction

Hypothetical costs; expert validation required.

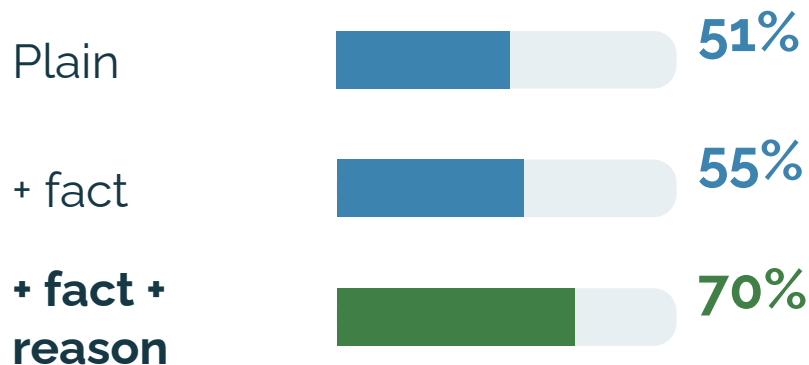
Deep dive: Acceptance and climate impact are different concepts



 VR4Green: awareness, not ranking

Mean acceptance

8 users • simulated office



Acceptance ≠ sustained change ≠ CO₂ reduction

SDG 13: Take-home message

SDG 13 teaches us to **connect local choices to climate consequences** (emissions and climate risk) while distinguishing a promising recommendation from demonstrated climate impact

SDG 14/15: Ecosystem protection



Marine ecosystems

Water conditions



Species candidates



Goal: preservation



Terrestrial ecosystems

Research agenda

Wildlife observations



Conservation knowledge



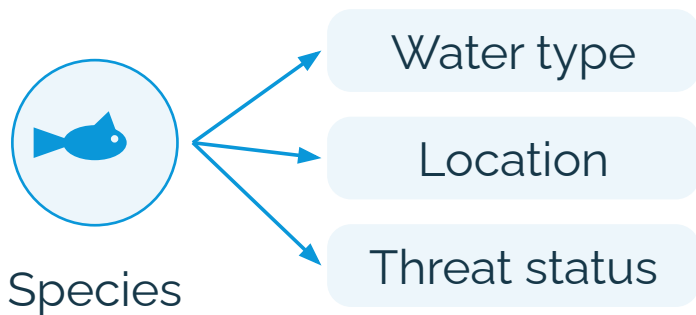
Goal: biodiversity

People decide, ecosystems are the intended beneficiaries

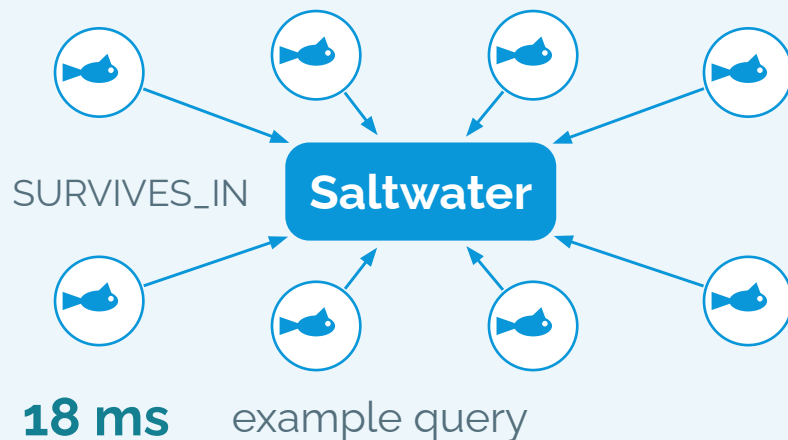
Deep dive: Match species to conditions

FishBase → Neo4j

5,874 nodes · 7,096 relationships



Cypher → candidates

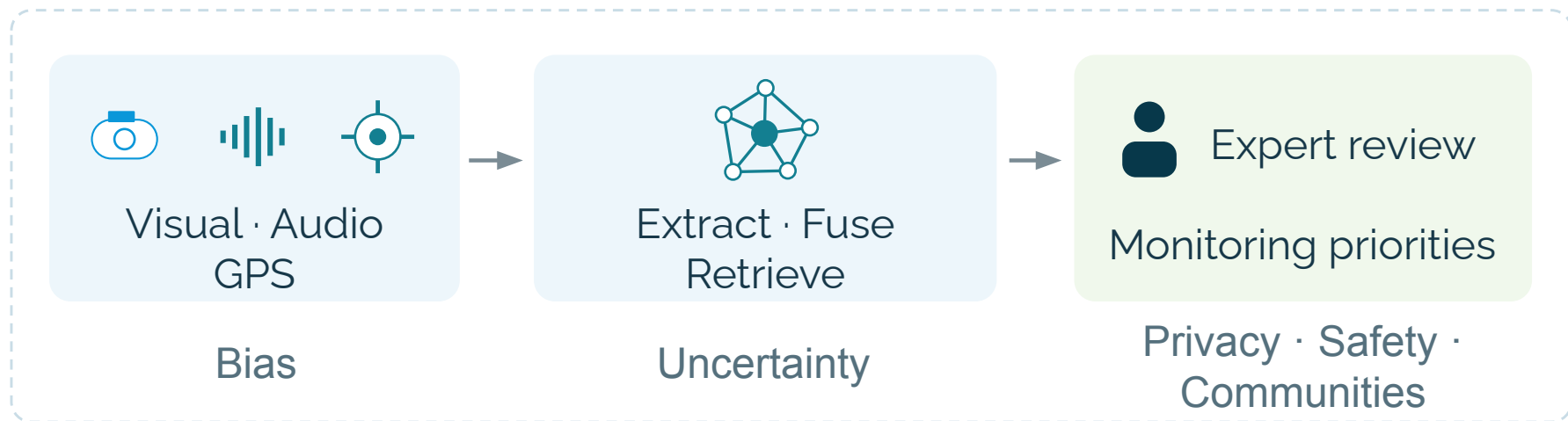


Condition matching ≠ ecological validation

Deep dive: Knowledge is not yet conservation

Workshop agenda, not an evaluated system

Illustrative workflow



Field validation before biodiversity claims

SDGs 14/15: Take-home message

SDGs 14/15 reframe **recommendation as conservation decision support**: translating **ecological, geospatial, and expert evidence** into **human-reviewed choices** whose success must ultimately be assessed through **ecosystem outcomes**

Take-home messages: environmental pillar

Three takeaways: **sustainability signals are often external to user preference**; **impact metrics are domain-specific**; many systems **require sensor/infrastructure data**

Economic pillar

RS for Economic Sustainability



Economic pillar targets

- Work
- Innovation
- Sustainable production



Decision support by RS

- Identify jobs
- Match skills
- Foster industry innovation



Impact

- Sustainable consumption choices

De Filippo, et al. "**Recommender systems and sustainability: a dual perspective.**" *Computer Science Review* 60 (2026): 100912.

Examples of RS for Economic Sustainability

SDG 8

Jobs and Innovation

RS support open innovation and workforce collaboration

SDG 9

Industry

RS help industries reduce impact and close skill gaps

J. Konstan, et al. *Recwork: Workshop on recommender systems for the future of work*. In ACM RecSys, pages 675–677, 2022.

E.L. Zickler, et al. *A recommender system to close skill gaps and drive organisations' success*. In IBICA, pages 806–815. Springer, 2022

SDG 8: Jobs, work, and open innovation

Opportunity gaps



Jobs



Skills



Innovation



RS coordination



People ↔ Jobs

Skills ↔ Opportunities

Ideas ↔ Partners



SDG aims



Decent work



Employability



Resilience

Goal: provide better opportunities

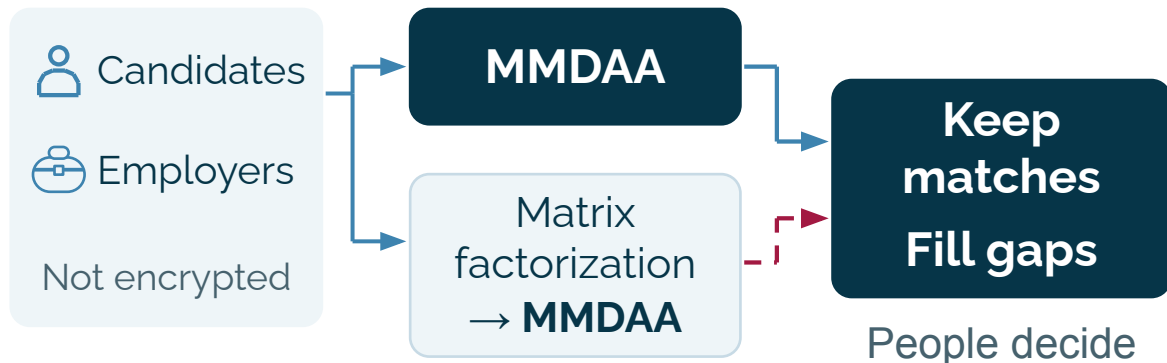
Gutiérrez, et al. (2019). *Explaining and Exploring Job Recommendations: a User-driven Approach for Interacting with Knowledge-based Job Recommender Systems*. Proceedings of the 13th ACM Conference on Recommender Systems.

Konstan, et al. (2022). *RecWork: Workshop on Recommender Systems for the Future of Work*. Proceedings of the 16th ACM Conference on Recommender Systems.

Ji, et al. (2025). *Recommender systems as co-design tools: empowering human-AI hybrid intelligence in complex design networks*. Journal of Engineering Design.

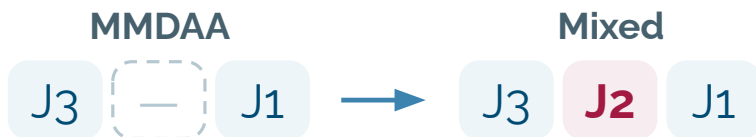
Deep dive: Work matching

Partial rankings Multi-match deferred acceptance



Example

Candidate 2



Real preferences

75 **24**
 candidates employers

Candidate results

Mixed vs MMDAA

Lower displacement
 in later rounds

**Zero withholding
 earlier**

Matching stability ≠ employment stability

Deep dive: Recommend the next collaboration

Activity + goal



GraphSAGE link prediction



Suggested links



10 practitioners

Perceived usefulness

Collaboration not tracked

Predicted links $\neq$ sustainable innovation

SDG 8: Take-home message

SDG 8 reframes recommendation as **opportunity coordination**: connecting people, skills, jobs, ideas, and organizations in ways that can **support decent work and sustainable value creation**

SDG 9: Sustainable industry and infrastructure

Knowledge gaps



Processes
Materials • energy



Capabilities
Skills • expertise



Information
Reports • labels



Goals



Resource
efficiency



Innovation
capacity



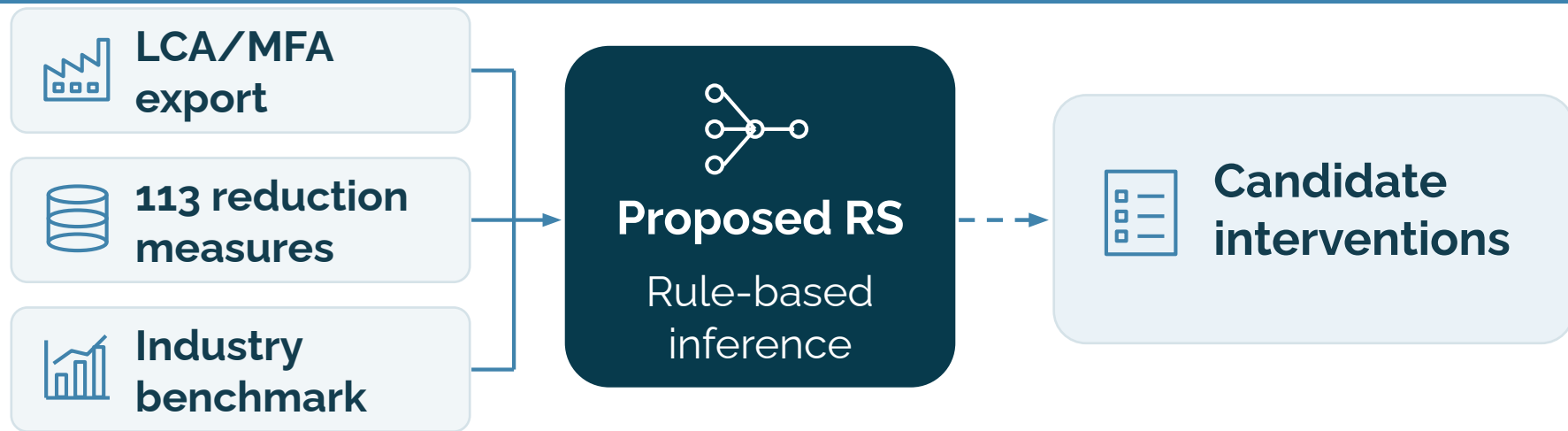
Informed
decisions

Fayed and Wohlgemuth. 2023. ***Design of a recommender system to improve the environmental impact of companies based on their material and energy balances.*** EnvirolInfo 2023.

Hillebrand, et al. 2023. ***sustain.AI: a Recommender System to analyze Sustainability Reports.*** ICAIL 2023.

Geninatti Cossatin, et al. 2024. ***Promoting Green Fashion Consumption Through Digital Nudges in Recommender Systems.*** IEEE Access

Deep dive: The profile is a process

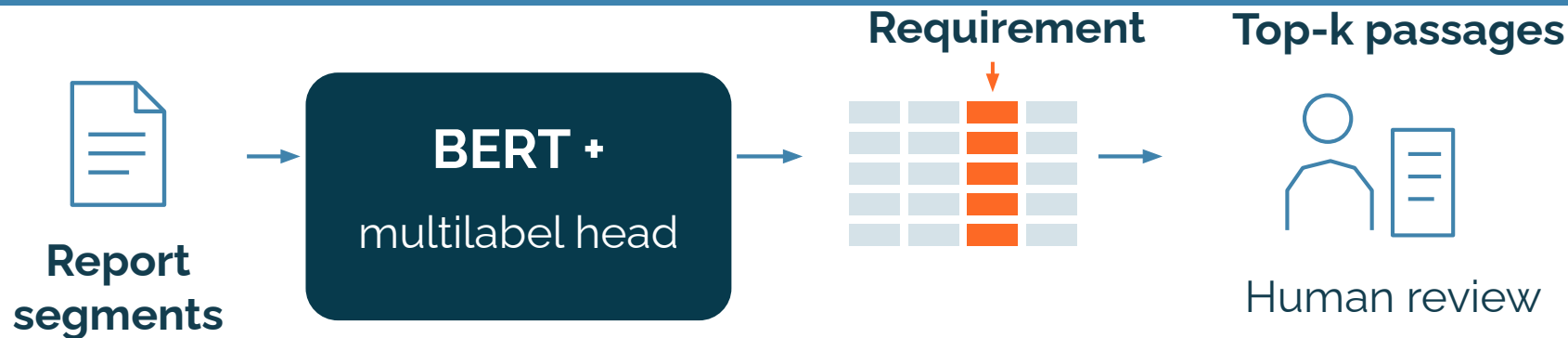


 **Examined data** | **Insufficient information for most measures**

Required: expert validation → adoption → measured impact

Fayed and Wohlgemuth. 2023. *Design of a recommender system to improve the environmental impact of companies based on their material and energy balances*. EnviroInfo 2023.

Deep dive: find passages



9%
relevant segments

28.4
Standard sampling

GRI test MAP@3 (%)

35.9
Weighted random sampling

Better retrieval $\neq$ verified sustainability.

SDG 9: Take-home message

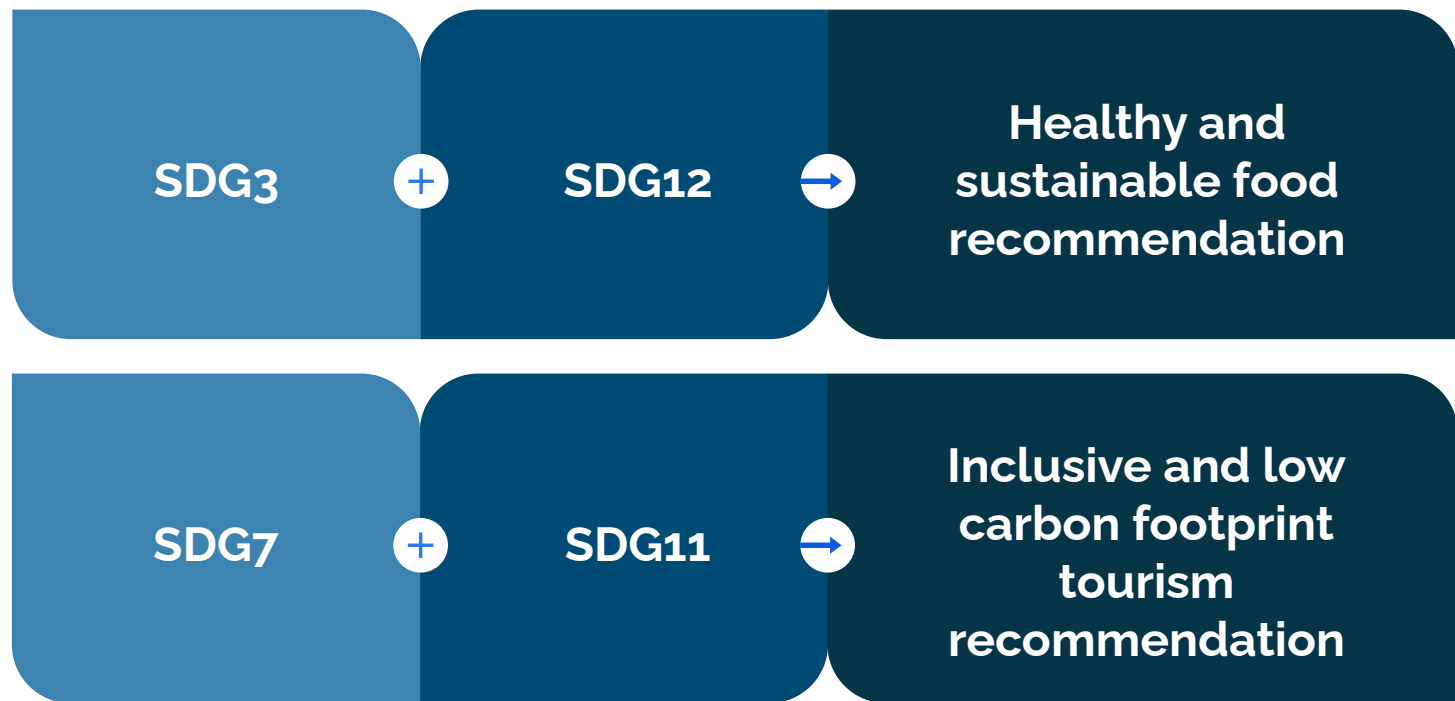
SDG 9 reframes recommendation as **industrial knowledge transfer**: connecting **processes**, **capabilities**, and **sustainability information** to **decisions** whose **feasibility** and **external impact** must be validated

Take-home messages: economic pillar

Three takeaways: **stakeholders are often organizations**, not just users; recommendations can **mediate collaboration**; economic impact must be balanced with **environmental/social goals**

Toward cross-pillar RS

Toward Cross-Pillar RS



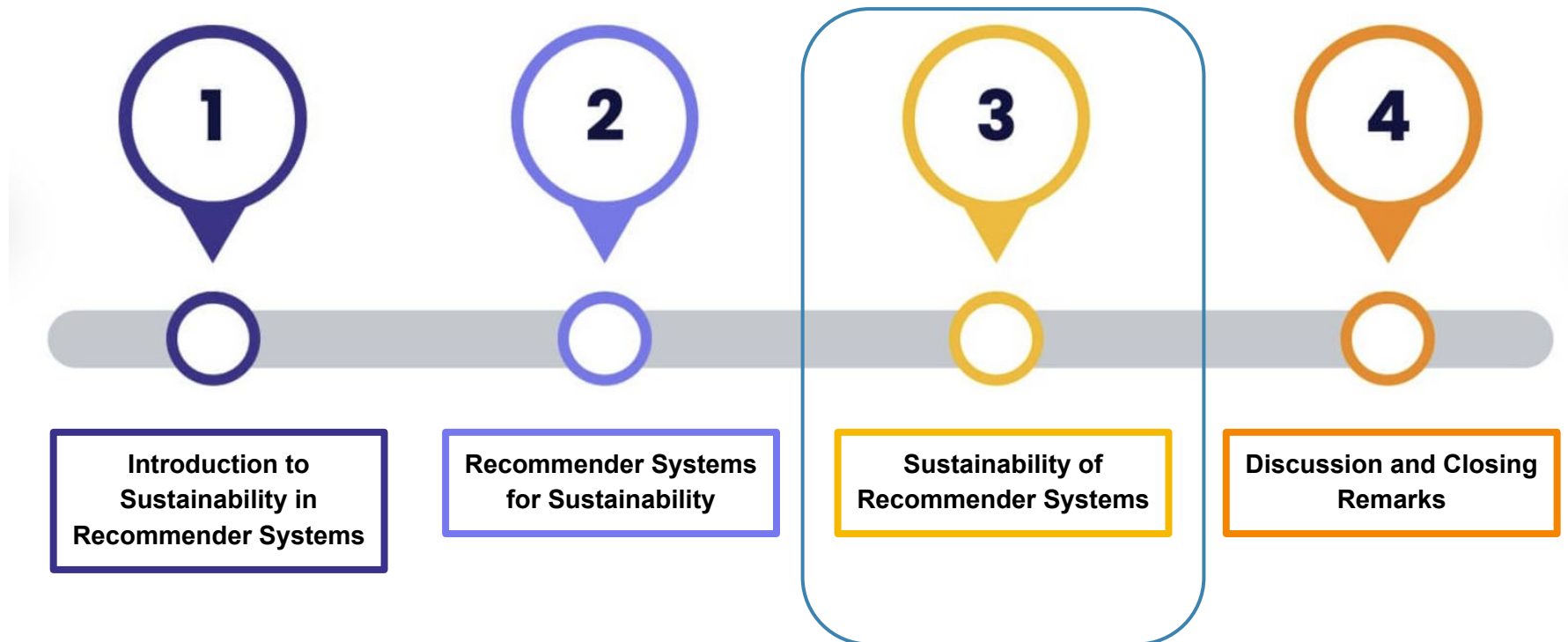
Open Issues and Limitations



1	Lack of real-world deployment
2	How do RS impact SDGs?
3	Need stronger interdisciplinary validation
4	Better evaluation frameworks

De Filippo, et al. "**Recommender systems and sustainability: a dual perspective.**"
Computer Science Review 60 (2026): 100912.

2. Sustainability of RS



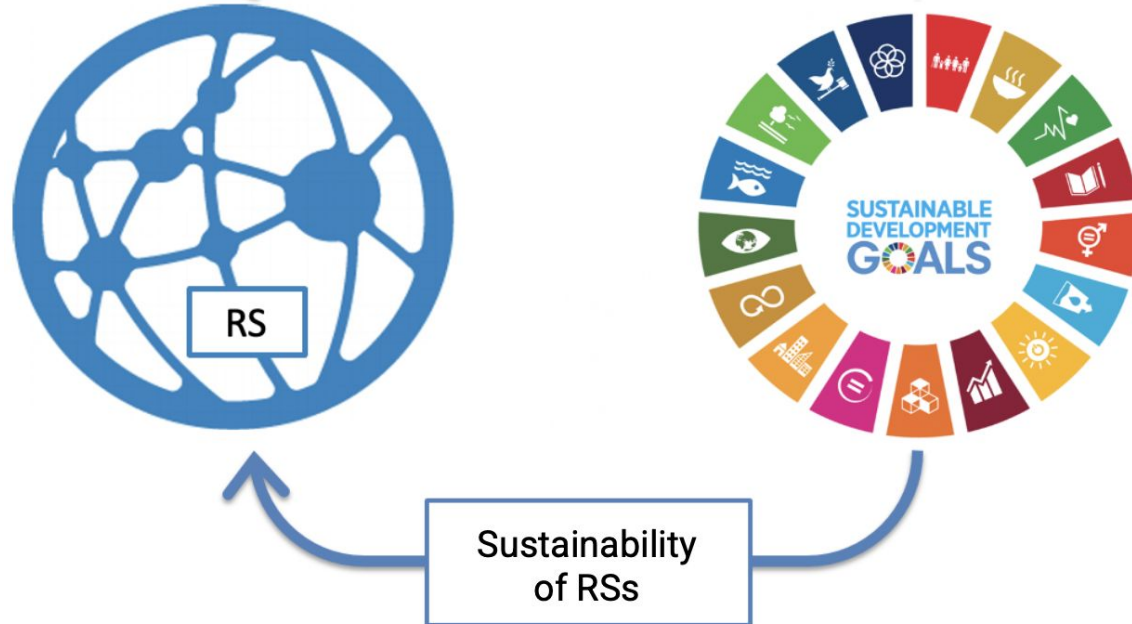
2. Sustainability of RS



2. Sustainability of RS

What are the hidden **environmental and ethical costs** of deploying large-scale recommender systems, and how can we redesign them to **align with sustainability principles**?

2. Sustainability of RS



2. Sustainability of RS

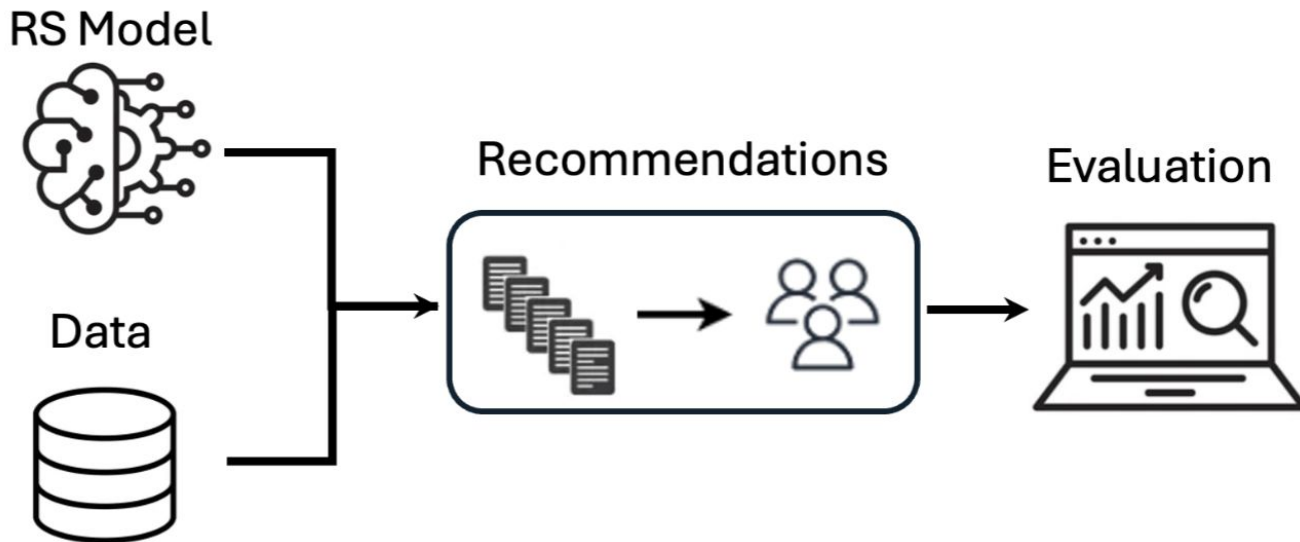
Can SDGs support the design and the development of Recommender Systems?



2. Sustainability of RS

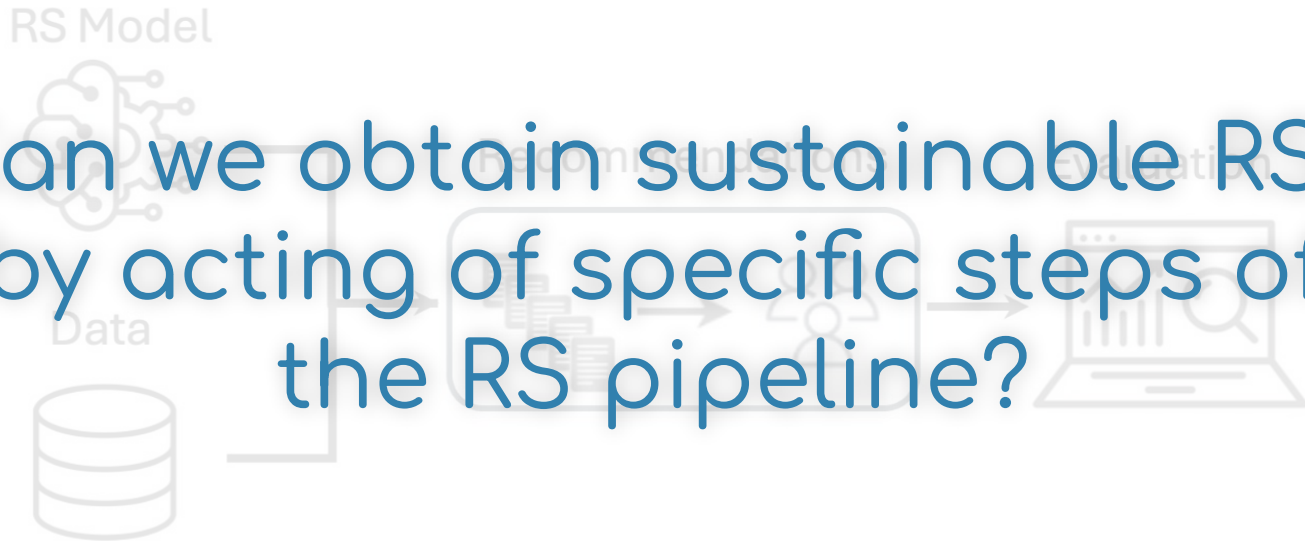
- Designing and developing of **RSs should follow the SDGs**
(*Van Wynsbergh*)
- We focus **not only on accuracy**, but also on **beyond-accuracy**
(*Kaminskas et al.*)
 - diversity
 - fairness
 - transparency
 - environmental impact

2. Sustainability of RS



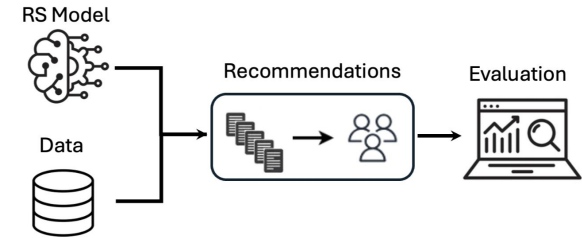
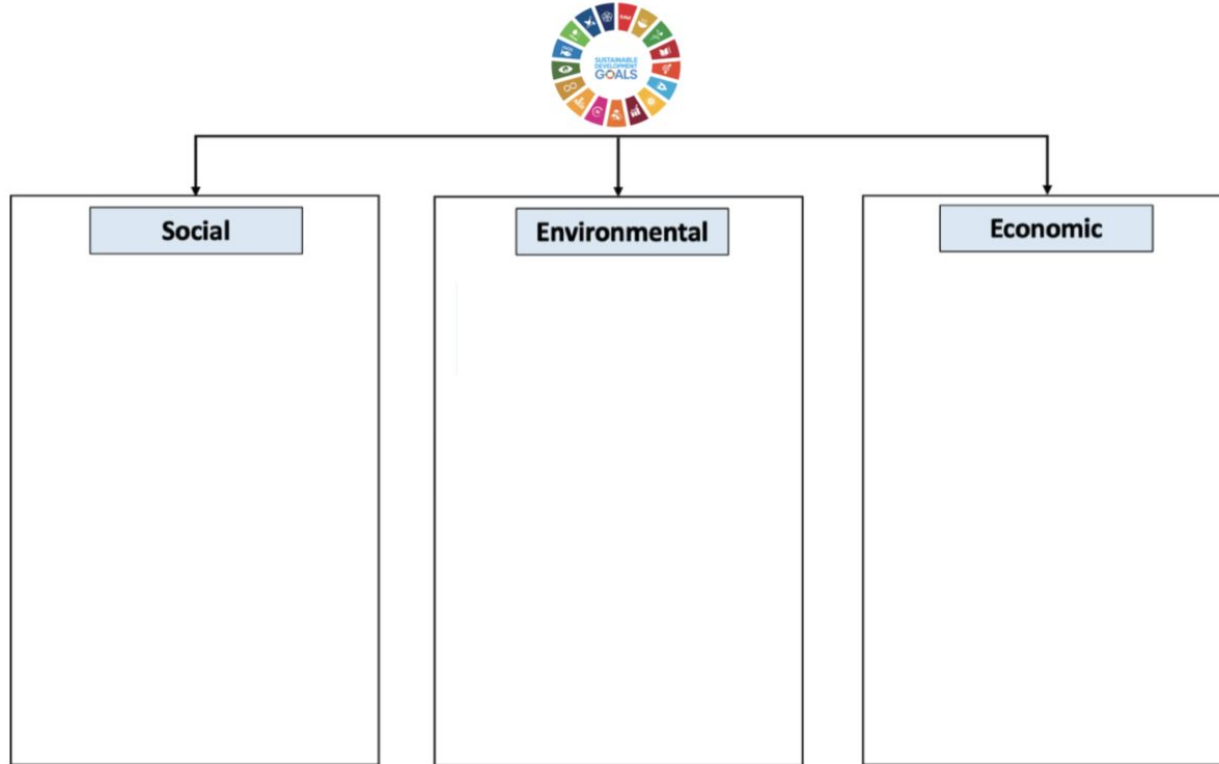
2. Sustainability of RS

Can we obtain sustainable RSs
by acting of specific steps of
the RS pipeline?

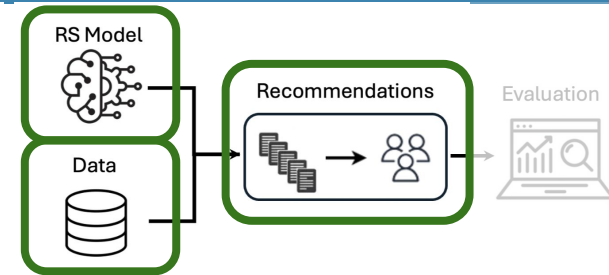
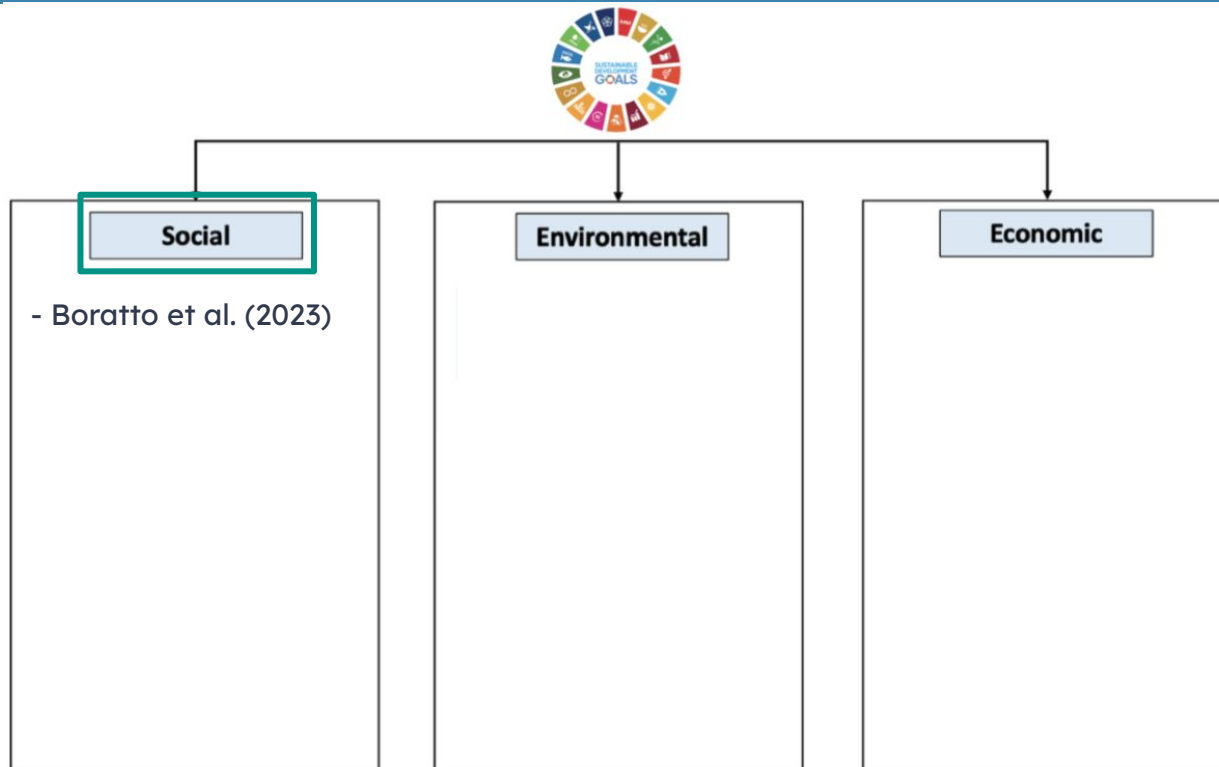


The background features a faint, light-blue diagram of the RS (Recommendation System) pipeline. It starts with a 'Data' icon (a cylinder) on the left, which feeds into a box labeled 'RS Model'. From the 'RS Model' box, an arrow points to a box labeled 'Recommendation', which then points to a box labeled 'Evaluation'. The 'Evaluation' box contains a laptop icon. The entire diagram is rendered in a very light, semi-transparent style.

2. Sustainability of RS



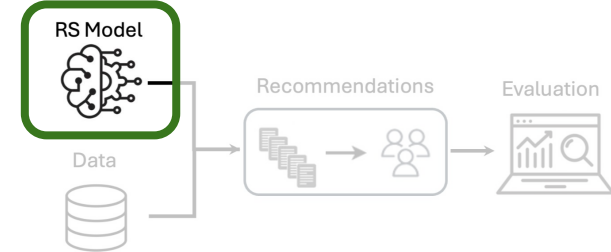
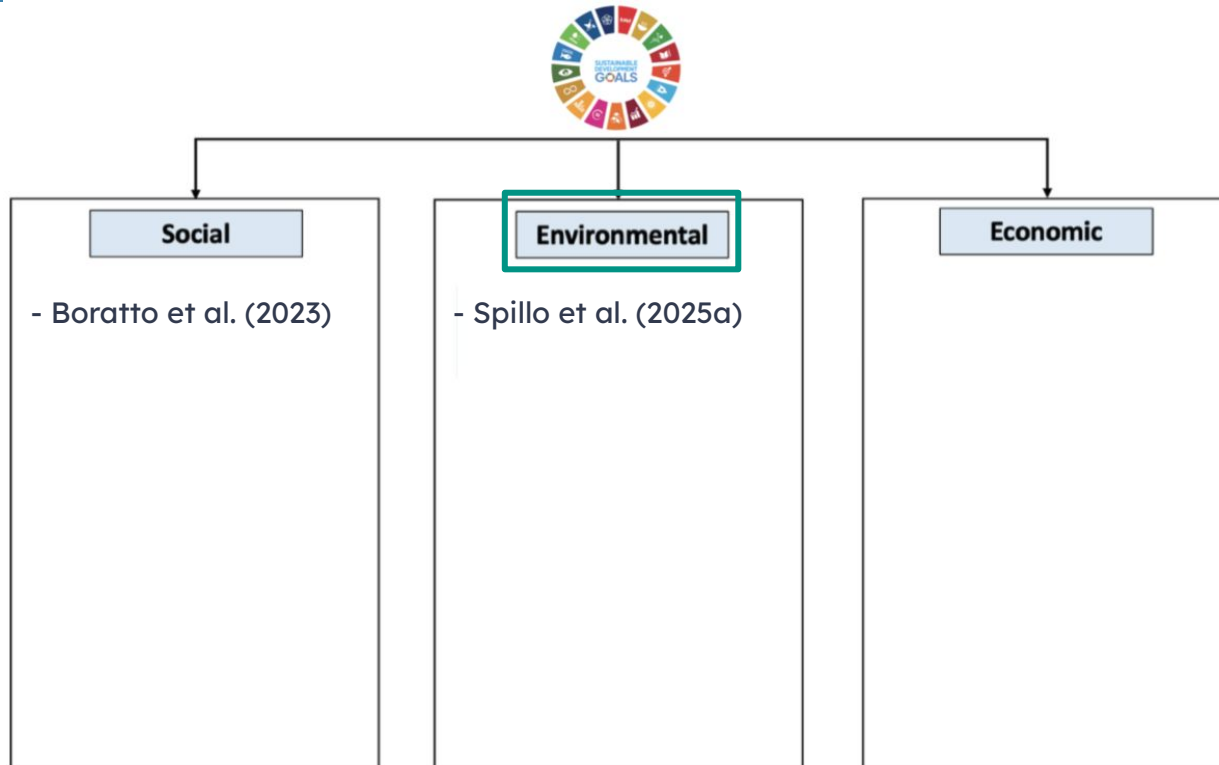
2. Sustainability of RS



Boratto et al. (2023): define the concept of **applicability** of mitigation strategies in the field of consumer unfairness.

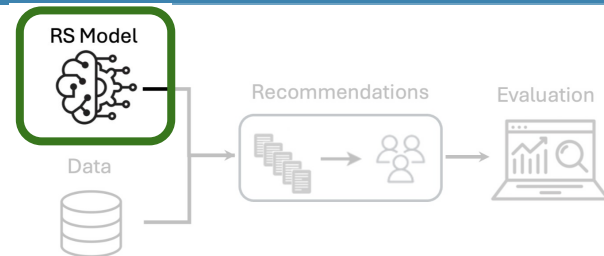
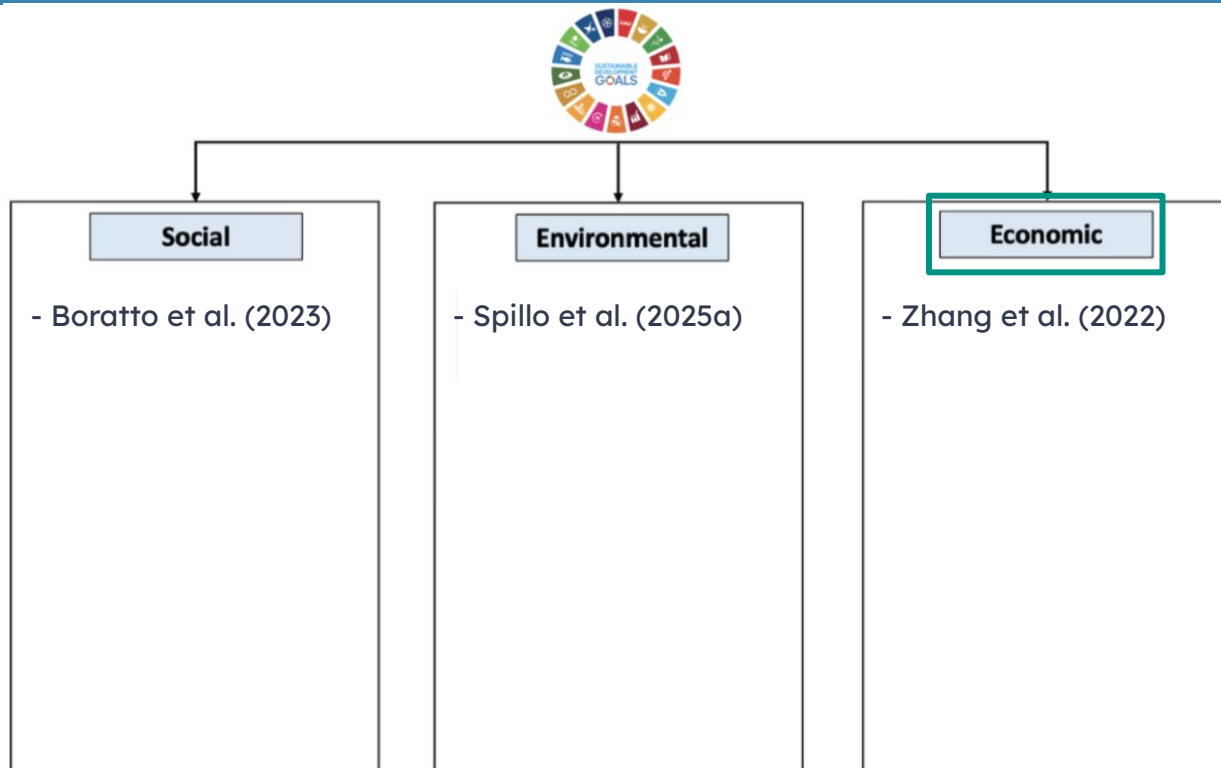
Applicability is a mitigation procedure's adaptability across diverse RS without significant alterations: affecting the model design, pre-processing the datasets, post-processing the recommendations

2. Sustainability of RS



Spillo et al. (2025a) investigates how the emission data can be exploited during the training of a RS models to **early-stop** when the improvements in the validation metric are not justified by the increase of the total emitted CO₂e

2. Sustainability of RS



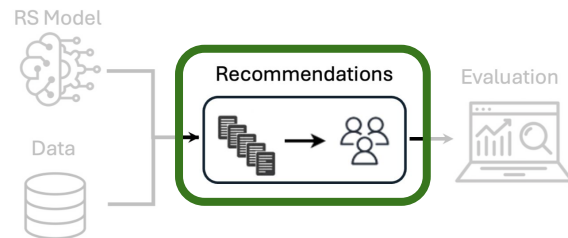
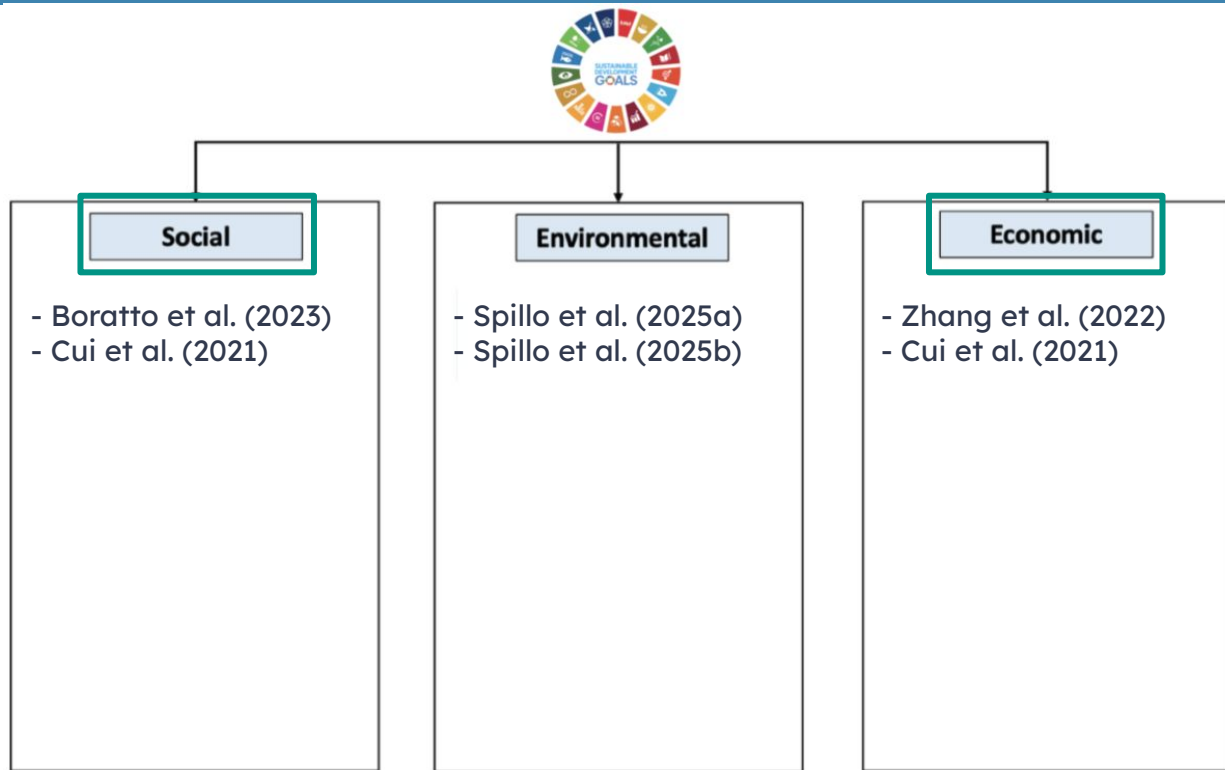
Zhang et al. (2022) introduces a RS that models how **user preferences** are related to **item prices** in session-based e-commerce scenarios.

2. Sustainability of RS



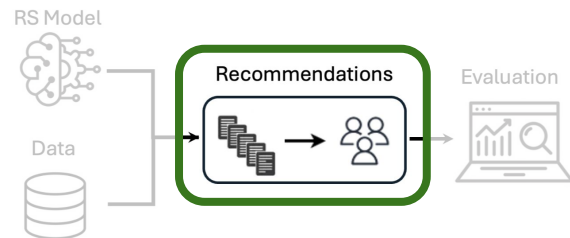
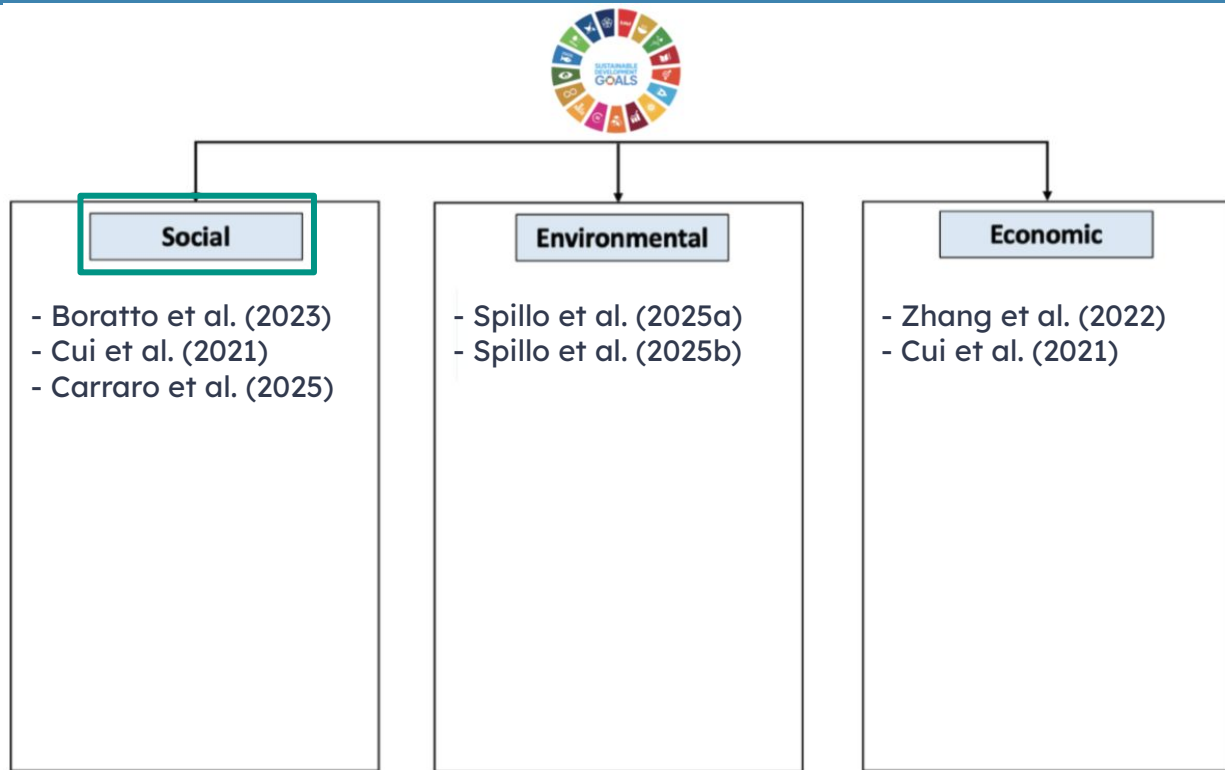
In Spillo et al. (2025b), the authors propose quantitatively investigate how **reducing the quantity of training data** affects the overall energy efficiency of the models. **Side-effects on diversity and average popularity** are observed, and different data reduction strategies are explored.

2. Sustainability of RS



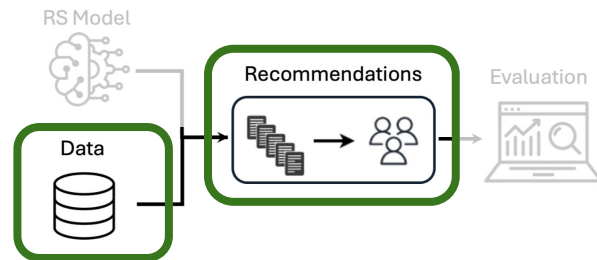
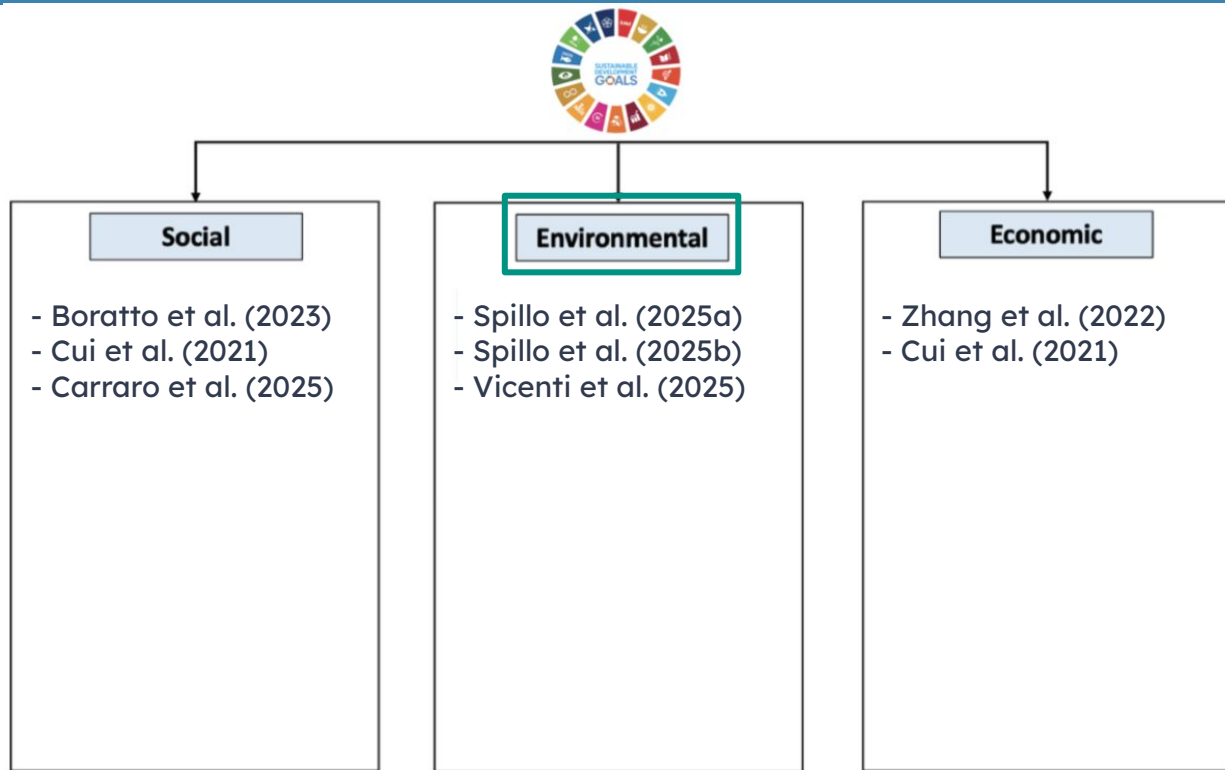
In Cui et al. (2021), the authors propose a **post-hoc adjustment strategy** that balances recommendation **utility** and **group fairness** by maximizing a weighted sum of both metrics. This is obtained by reordering recommended items across sensitive groups. The method is entirely non-parametric and performs well even with sparse interaction data.

2. Sustainability of RS



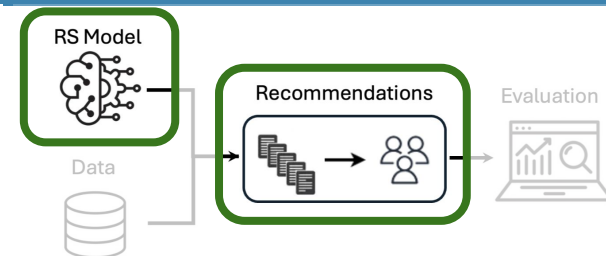
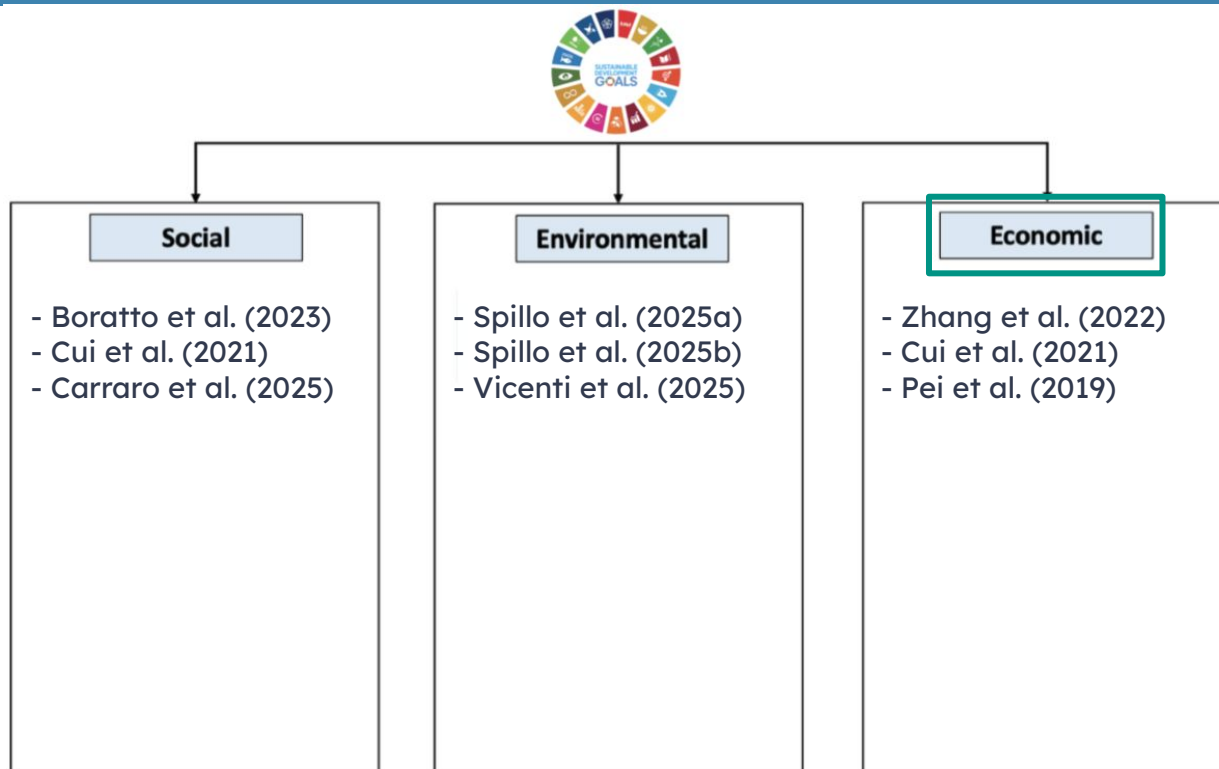
In Carraro et al. (2025), LLMs are investigated as **post-hoc rerankers** to increase item **diversity**. Several strategies are investigated, and some include item side information (e.g., book plot). Results show that LLMs can interpret the concept of diversity, but lower-ranked relevant items and generate invalid recommendations.

2. Sustainability of RS



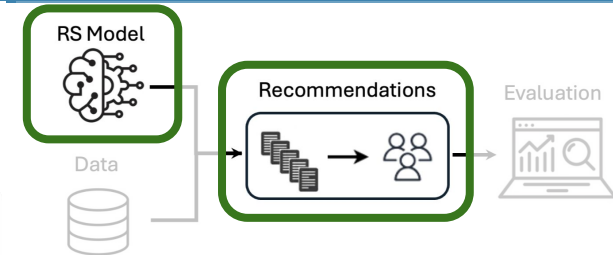
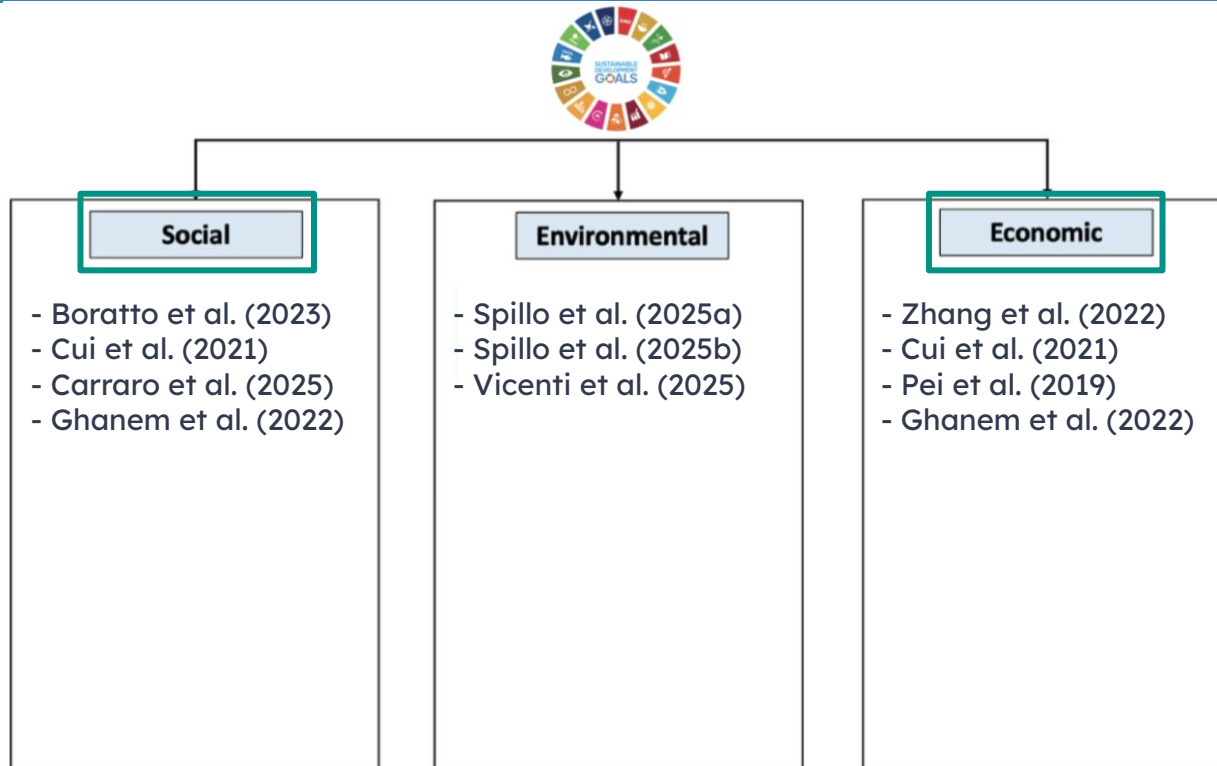
In Vicenti et al. (2025), the authors propose a zero-shot prompting strategy to **estimate the Product Carbon Footprint** of the whole life-cycle of items in popular e-commerce domains. Then, the authors propose a methodology to **rerank the recommendations** of any RS model **balancing accuracy and sustainability** of such items.

2. Sustainability of RS



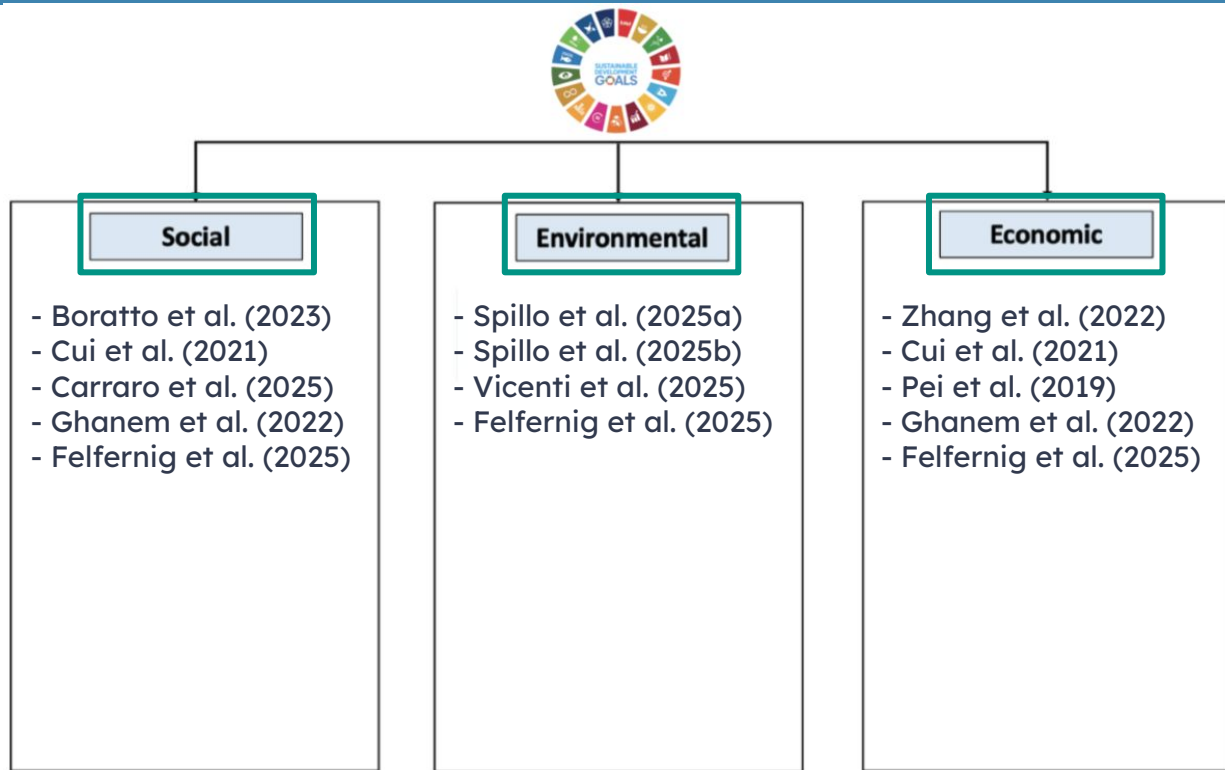
In Pei et al. (2019), the authors propose a **value-aware RS**. The RS generates a recommendation list, and each user action on this list is associated to a profit. Then, a reinforcement learning module is used to **maximize the profit** whose reward is the aggregated monetized user actions.

2. Sustainability of RS



In Ghanem et al. (2019), the authors investigate the **balance between consumer-trust and long-term profitability**. Giving more weight to **user satisfactions** than profitability leads the highest cumulative **long-term business success** (20% more than pure consumer satisfactions and pure profitability).

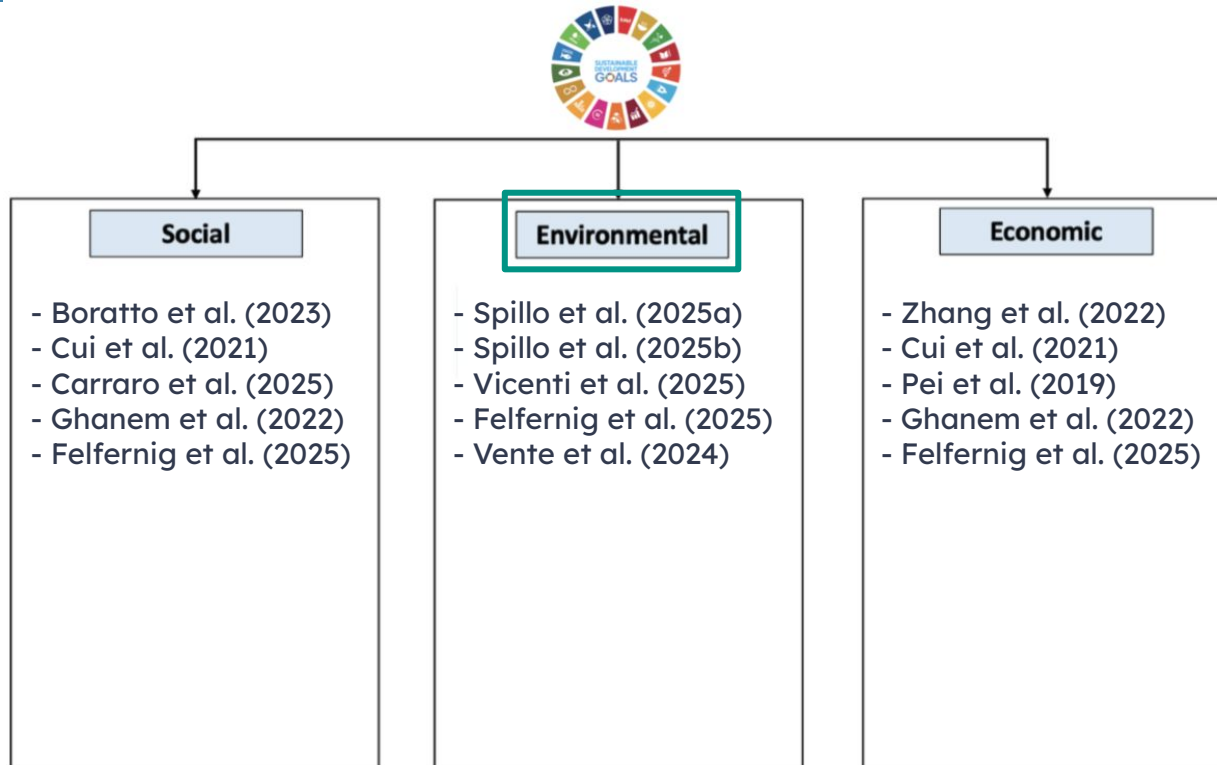
2. Sustainability of RS



Felfernig et al. (2025) introduce new evaluation **metrics** for **social, environmental, and economic sustainability** in RS.

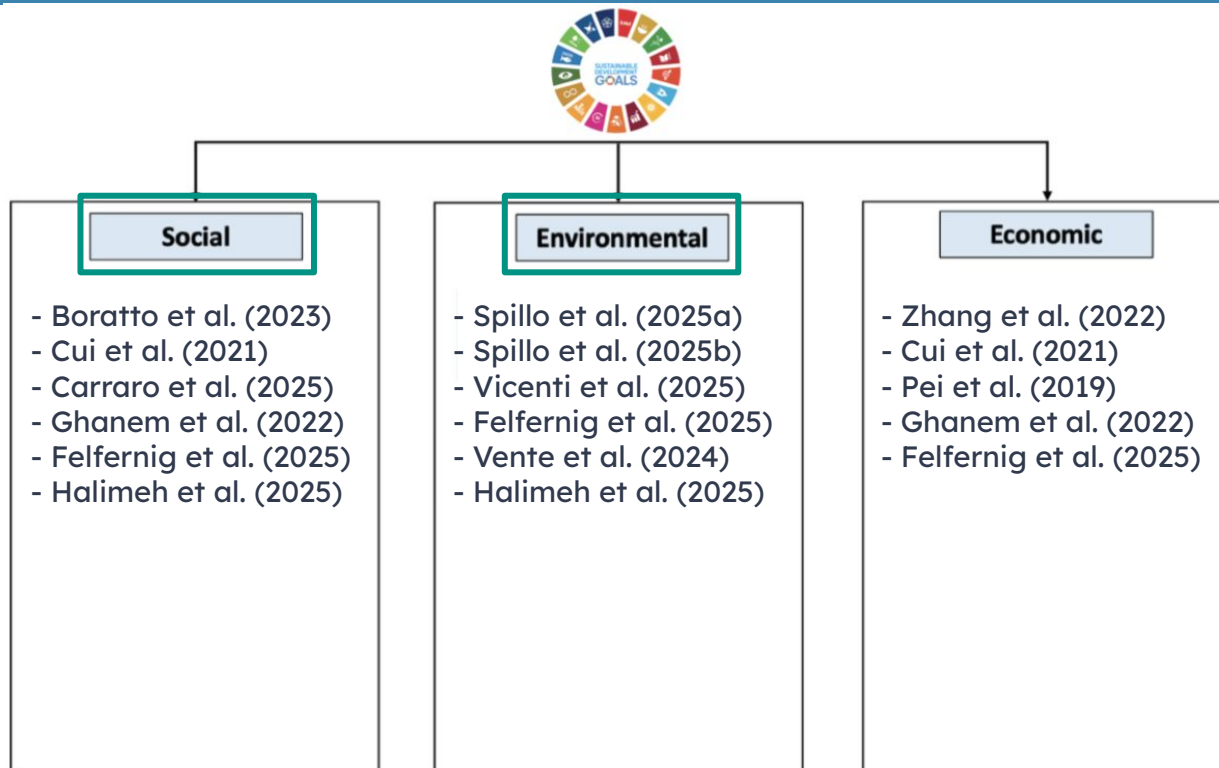
- Greenes of items, sustainable item rate, energy consumption, etc.
- Fairness, diversity, serendipity, accessibility, inclusivity, etc.
- Local business, long-term consumer satisfaction, producer fairness exposure, etc.

2. Sustainability of RS



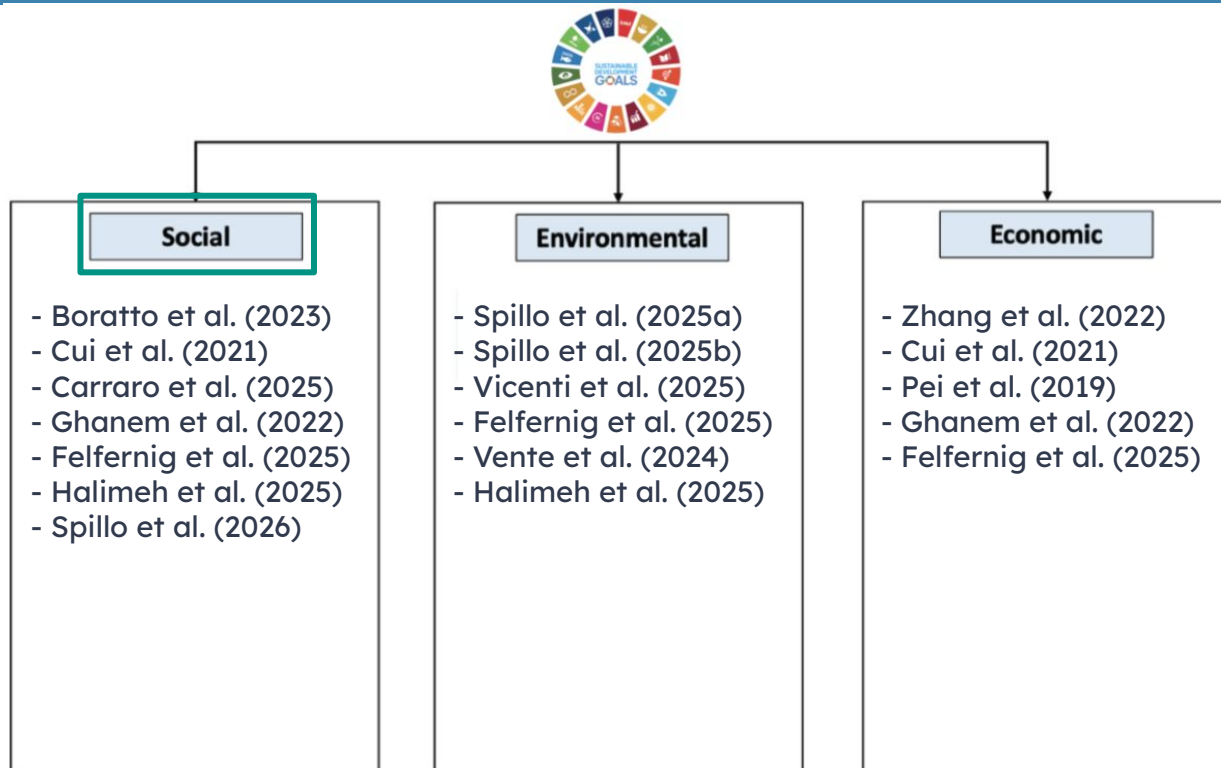
Vente et al. (2024) provide a full **benchmark of different RS** on several datasets, different implementations, and hardware to provide a comprehensive overview of the **carbon footprint of RS**.

2. Sustainability of RS



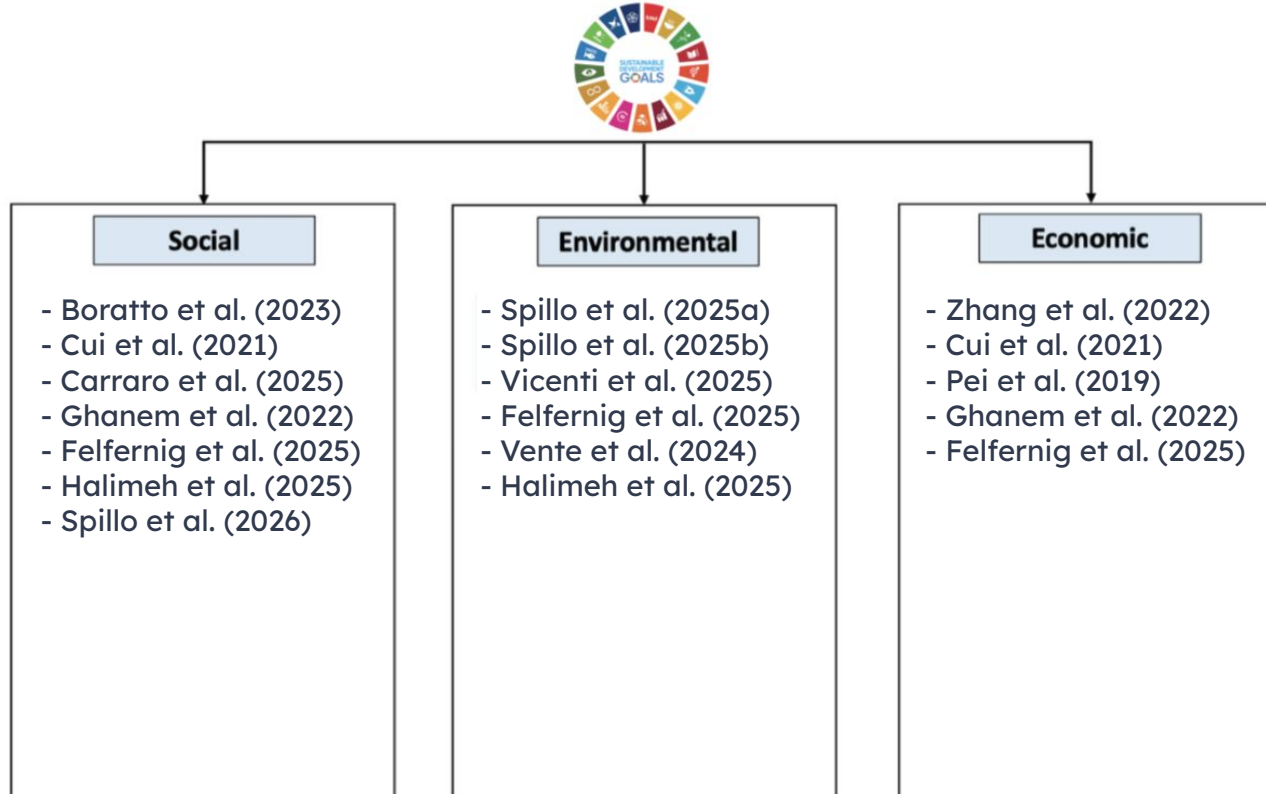
Halimeh et al. (2025) evaluate, through a user-study, how explanation enriched with **nudges related to environmental and social sustainability** can influence the decision-making process of the users. 4 decision information nudging strategies are explored.

2. Sustainability of RS

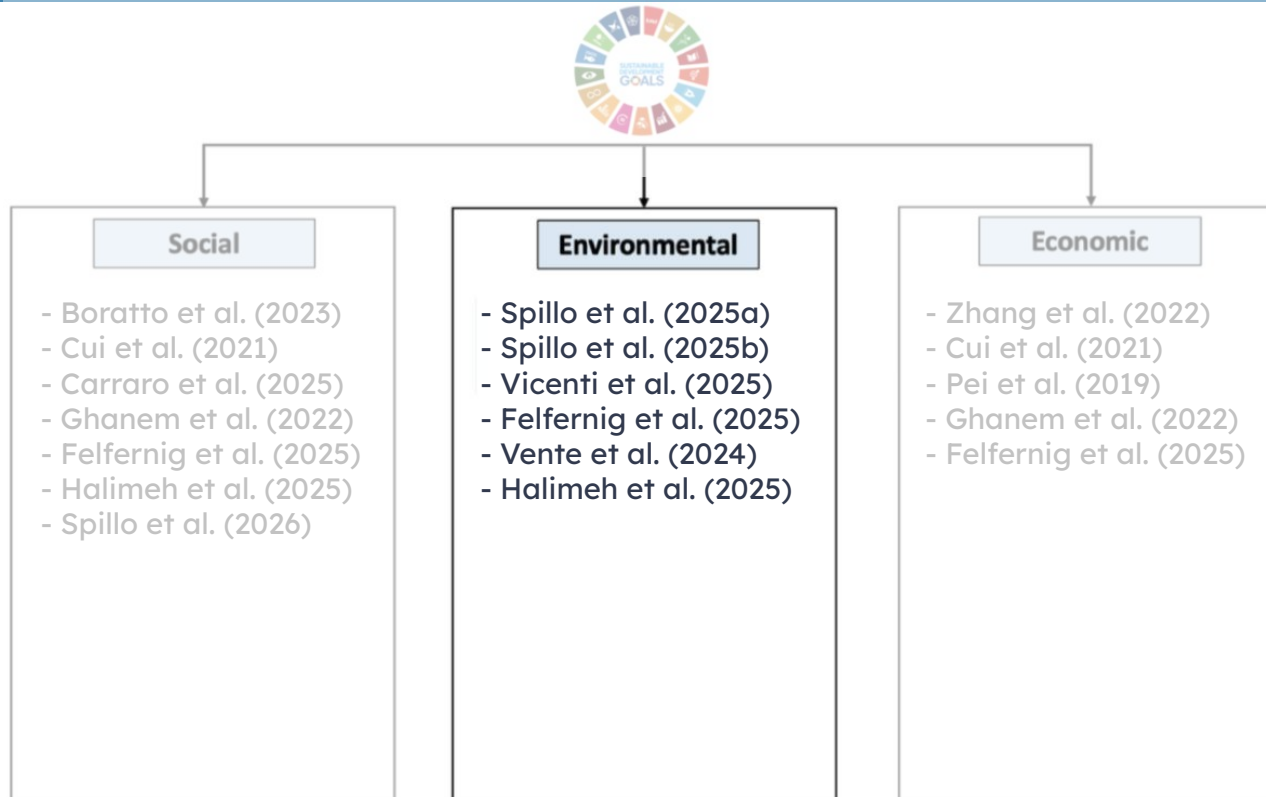


Spillo et al. (2026) provide a first benchmark of the **accuracy-fairness trade-off in multimodal RS**, assessing it when fairness-oriented models are equipped with multimodal features.

2. Sustainability of RS



2. Sustainability of RS



2. Sustainability of RS

Hands-on session notebook



<https://shorturl.at/U5QMt>

2. Sustainability of RS

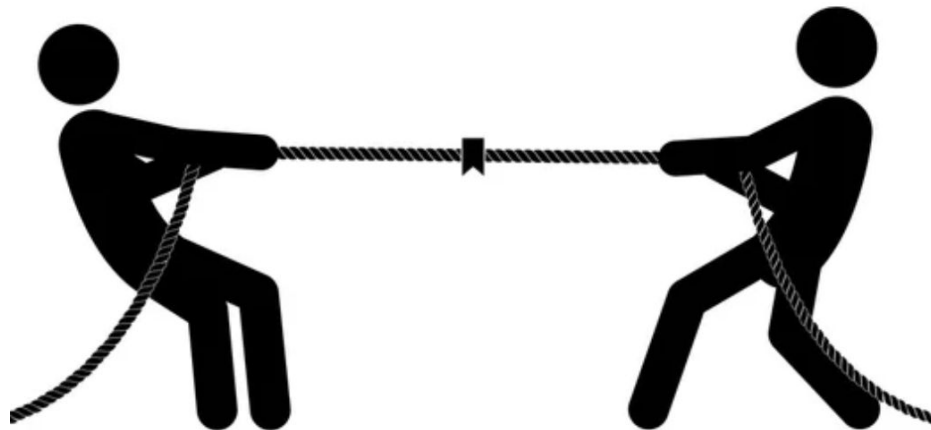
- Overview on the analysis of the environmental sustainability of RecSys
- Carbon footprint:
 - Emissions of *CO₂-equivalent* (CO₂-eq)
 - Related to training step

2. Sustainability of RS

The notebook is composed of the following sections:

1. **Trade-off analysis** between recommendation performance and CO₂-eq emissions of state-of-the-art models
2. **Data reduction strategies** to improve the trade-off

2. Sustainability of RS



Environment

Do more **complex** recommendation **algorithms** show performances that justify their **higher carbon footprint**?

2. Sustainability of RS

Main tools for this notebook:

- **RecBole** to train and evaluate RS models
- **CodeCarbon** to measure the carbon footprint



2. Sustainability of RS

A measure used to express the total **warming impact** of various greenhouse gases in **terms of carbon dioxide (CO₂)**



Energy source mix used for the computation (i.e., coal, gas, nuclear, etc.)



$$\text{Emissions}(\text{CO}_2\text{-eq}) = \text{CarbonIntensity} * \text{EnergyConsumption}$$

Energy consumption required by the computation
(read from the hardware during the training)



```
37 # Start emissions tracking
38 tracker = EmissionsTracker()
39 tracker.start()
```



Before training, we start the CO2-eq tracking

```
40
41 # Train the model
42 epochs = 3
43 device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
44 model.to(device)
45
46 for epoch in range(epochs):
47     for images, labels in trainloader:
48         images, labels = images.to(device), labels.to(device)
49
50         optimizer.zero_grad()
51         outputs = model(images)
52         loss = loss_fn(outputs, labels)
53         loss.backward()
54         optimizer.step()
55
56     print(f"Epoch {epoch+1}/{epochs}, Loss: {loss.item():.4f}")
57
58 # Stop emissions tracking
59 emissions_kg = tracker.stop()
```

```
37 # Start emissions tracking
38 tracker = EmissionsTracker()
39 tracker.start()
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Before training, we start the CO2-eq tracking

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56     print(f"Epoch {epoch+1}/{epochs}, Loss: {loss.item():.4f}")
57
```

```
58 # Stop emissions tracking
59 emissions_kg = tracker.stop()
```



When the training is complete, we stop the tracking and read the CO2-eq value

timestamp	2026-07-01T08:37:29
project_name	codecarbon
run_id	6afcbff3-7d95-47db-855e-5b765fc2eea1
experiment_id	5b0fa12a-3dd7-45bb-9766-cc326314d9f1
duration	40.144025
emissions	0.000146
emissions_rate	0.000004

cpu_power	2.958935
gpu_power	25.229313
ram_power	10.0
cpu_energy	0.000032
gpu_energy	0.000279
ram_energy	0.000107
energy_consumed	0.000418
water_consumed	0.0

```

100%|██████████| 9.91M/9.91M [00:00<00:00, 18.3MB/s]
100%|██████████| 28.9k/28.9k [00:00<00:00, 451kB/s]
100%|██████████| 1.65M/1.65M [00:00<00:00, 4.11MB/s]
100%|██████████| 4.54k/4.54k [00:00<00:00, 3.51MB/s]
[codecarbon WARNING @ 08:36:47] Multiple instances of codecarbon are allowed to run at the same time.
[codecarbon INFO @ 08:36:47] [setup] RAM Tracking...
[codecarbon INFO @ 08:36:47] [setup] CPU Tracking...

```

```

Training completed.
Total emissions (kg): 0.14615079752406243

```



Total emissions

For this notebook, we consider:

- Dataset:
 - MovieLens-1M
- RS models:
 - **ItemKNN, LightGCN, NGCF, DGCF**
- Metrics:
 - Precision, Recall, NDCG, AvgPop, Gini Index, Emissions

```
# dataset is 'ml1m'
# rec_model is 'BPR'
checkpoint_dir = f'checkpoints/{dataset}/{rec_model}/'
epochs = 50
emb_size = 32
kg_emb_size = 32
metrics = ['Precision', 'Recall', 'ndcg',
           'AveragePopularity', 'GiniIndex']

os.makedirs(checkpoint_dir, exist_ok=True)

parameter_dict = {
    'epochs': epochs,
    'embedding_size': emb_size,
    'kg_embedding_size': kg_emb_size,
    'checkpoint_dir': checkpoint_dir,
    'benchmark_filename': ['train', 'valid', 'test'],
    'repeatable': True,
    'metrics': metrics
}
```


Same as before:

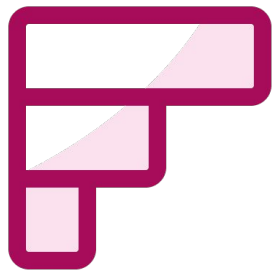
1. We instantiate the emission tracker object
2. We start the RecBole training
3. We stop the tracking and save the emissions
4. We print the emissions

```
# init an emission tracking object, then start tracking
proj_name = f'{rec_model}_{dataset}'
tracker = EmissionsTracker(project_name=proj_name)
tracker.start()

# train the recommendation model
run_recbole(model=rec_model, dataset=dataset,
            config_dict=parameter_dict)

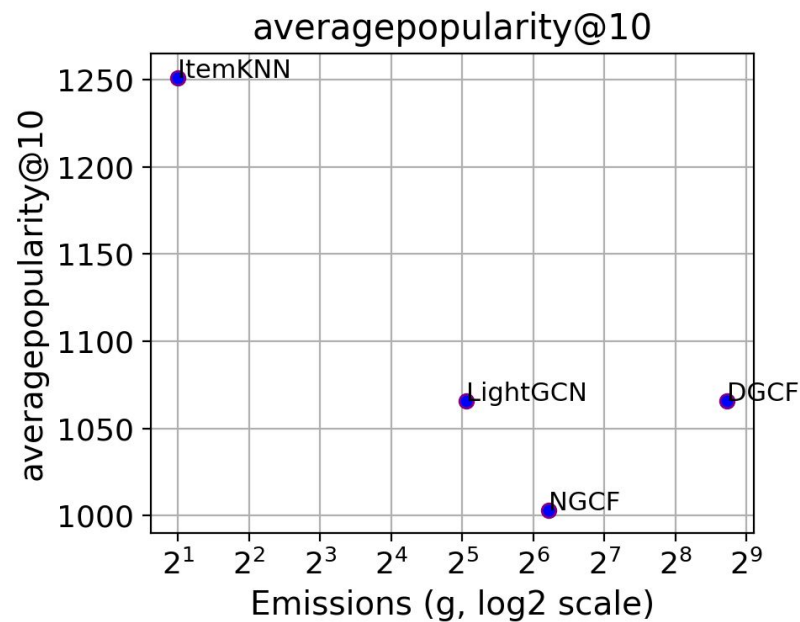
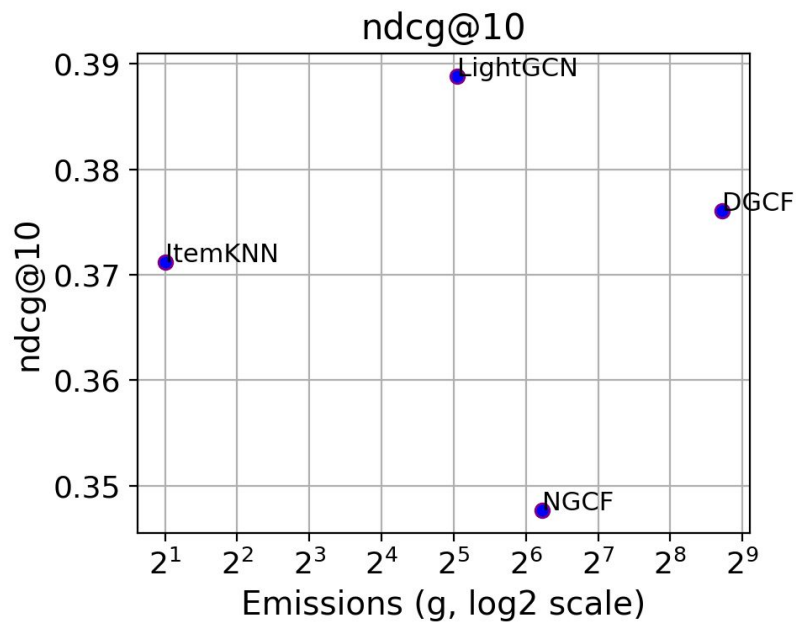
# stop the tracking and save the emissions as a float
emissions = tracker.stop()

print(f"Emissions for {rec_model}: {emissions}")
```



Which model do you think shows the best trade-off?





Let's change the place of computation.

This data is related to the average carbon intensity, that is about 475g/kWh.
Scotland has a carbon intensity of 161 g/kWh (Source: UK Government).

To perform the change, we divide each emission value for the world average carbon intensity, and multiply the resulting value (that is, the energy consumed) times the carbon intensity of Scotland.

```
1 # first, we make a copy of the result data
2 import copy
3
4 results_scotland = copy.deepcopy(results)
5 for model in results_scotland:
6     results_scotland[model]['carbon_metrics']['emissions'] = results_scotland[model]['carbon_metrics']['emissions'] / 475 * 161
```

	Model	Recall@10	ndcg@10	avgPopularity@10	giniIndex@10	Emissions (g)
0	Item-KNN	0.1499	0.3712	1251.5237	0.9239	1.985
1	LightGCN	0.1648	0.3888	1065.7047	0.8526	33.383
2	NGCF	0.1488	0.3477	1002.7306	0.8275	74.125
3	DGCF	0.1593	0.3760	1065.7581	0.8504	425.410

World average emissions

	Model	Recall@10	ndcg@10	avgPopularity@10	giniIndex@10	Emissions (g)
0	Item-KNN	0.1499	0.3712	1251.5237	0.9239	0.672811
1	LightGCN	0.1648	0.3888	1065.7047	0.8526	11.315080
2	NGCF	0.1488	0.3477	1002.7306	0.8275	25.124474
3	DGCF	0.1593	0.3760	1065.7581	0.8504	144.191600

Scotland emissions

2. Sustainability of RS



Spillo, G., De Filippo, A., Musto, C., Milano, M., & Semeraro, G. (2025). **Balancing carbon footprint and algorithm performance in recommender systems: A comprehensive benchmark.** Sustainable Computing: Informatics and Systems, 101286.



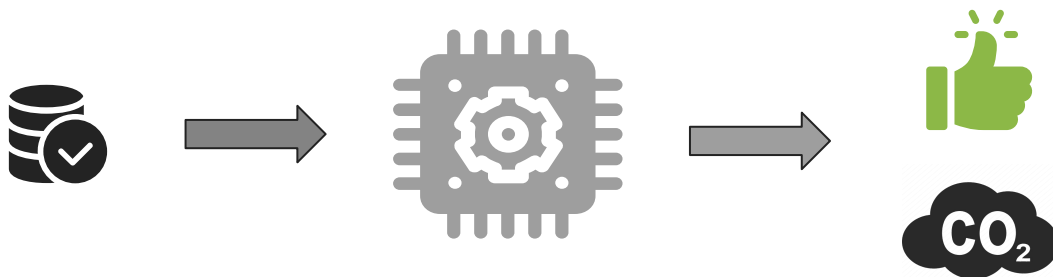
Spillo, G., De Filippo, A., Musto, C., Milano, M., & Semeraro, G. (2023, September). **Towards sustainability-aware recommender systems: analyzing the trade-off between algorithms performance and carbon footprint.** In Proceedings of the 17th ACM Conference on Recommender Systems (pp. 856-862).

2. Sustainability of RS

After benchmarking,
can we improve the
accuracy-emissions trade-off?

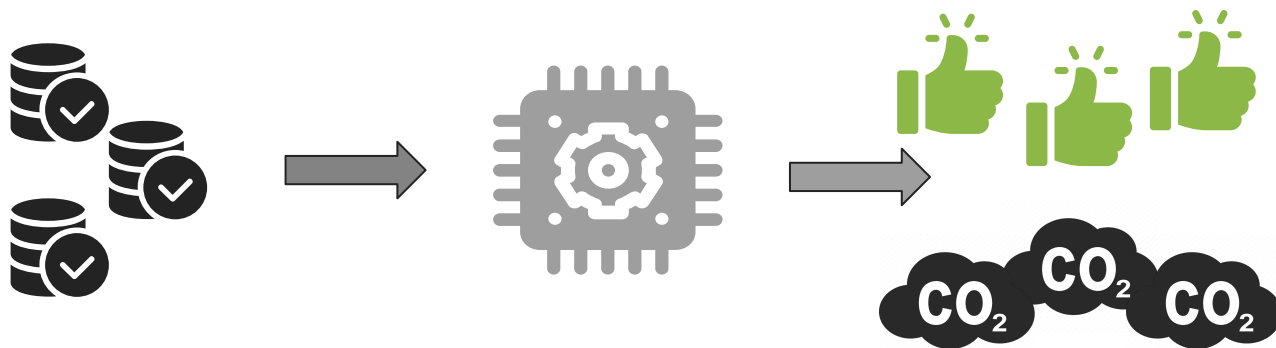
2. Sustainability of RS

- RSs typically require **lots of training data**
- Data-intensive technology: the more the data
 - The better the performances



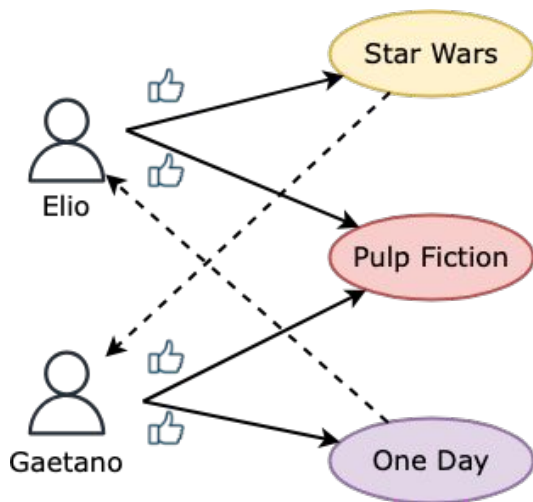
2. Sustainability of RS

- RSs typically require **lots of training data**
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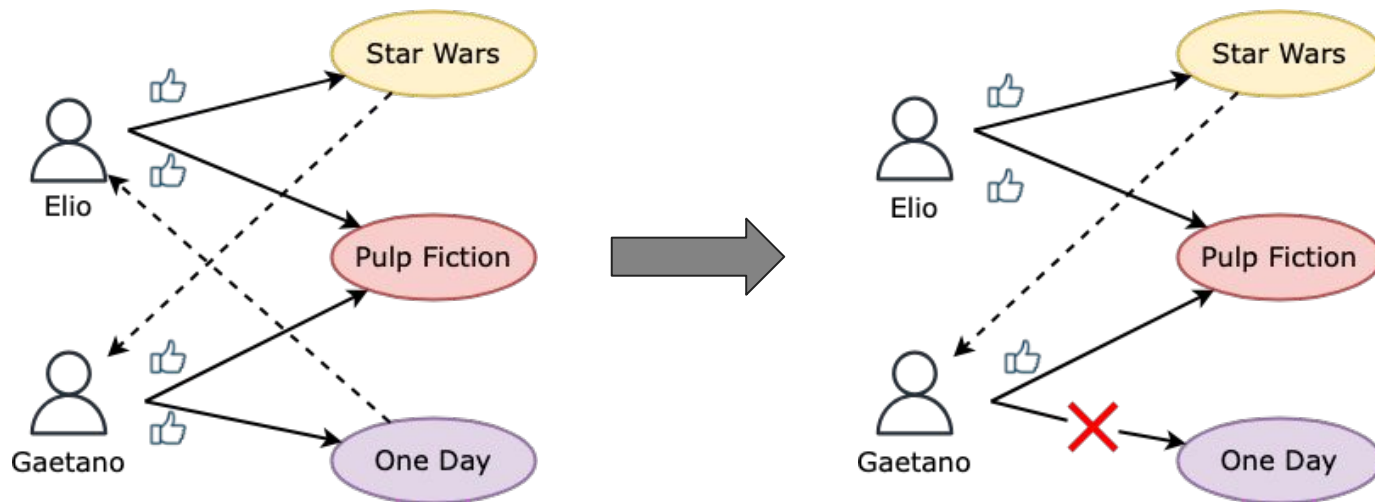
2. Sustainability of RS

- What happens if we train a RS using only a subset of the training data?
 - Do emissions decrease?
 - Do performances decrease?
 - How much?

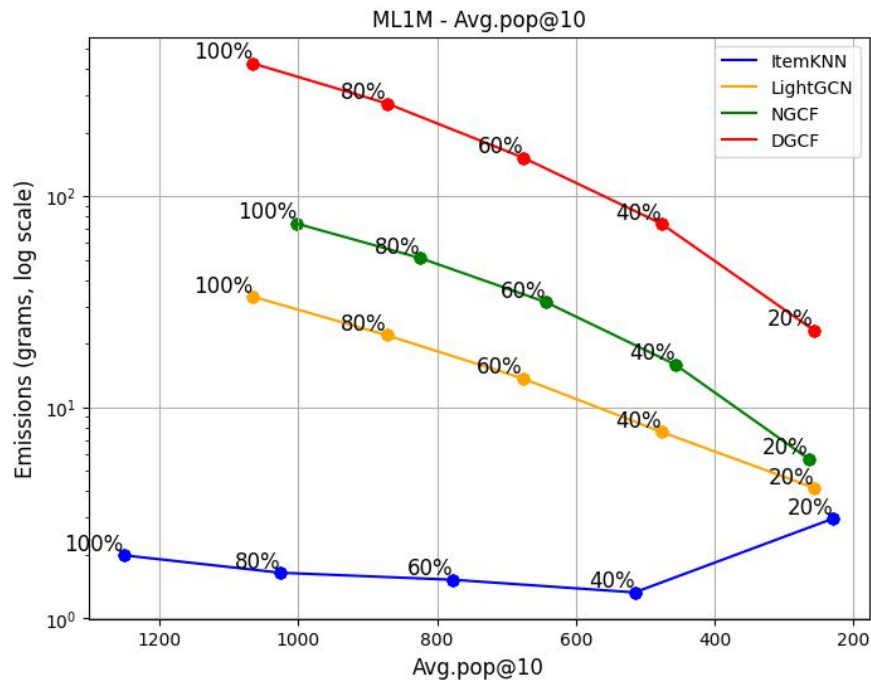
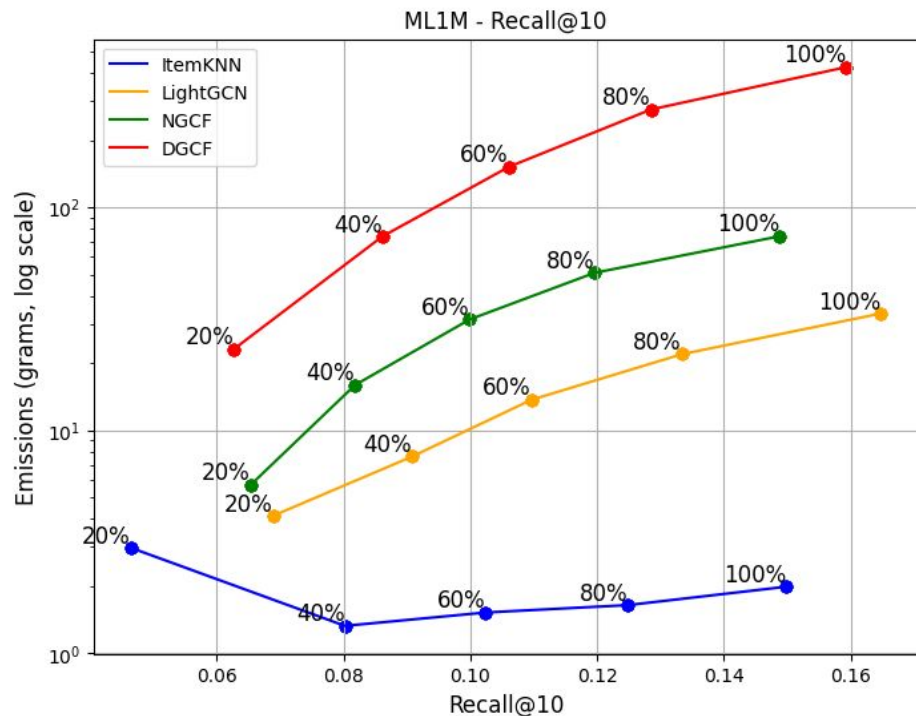


2. Sustainability of RS

- What happens if we train a RS using only a subset of the training data?
 - Do emissions decrease?
 - Do performances decrease?
 - How much?



- We randomly drop edges in the training data
- We train and evaluate the RS models at different quantity of data
 - While tracking CO2-eq





Spillo, G., De Filippo, A., Musto, C., Milano, M., & Semeraro, G. (2025). Comparing data reduction strategies for energy-efficient green recommender systems. *Journal of Intelligent Information Systems*, 63(6), 1837-1863.

2. Sustainability of RS

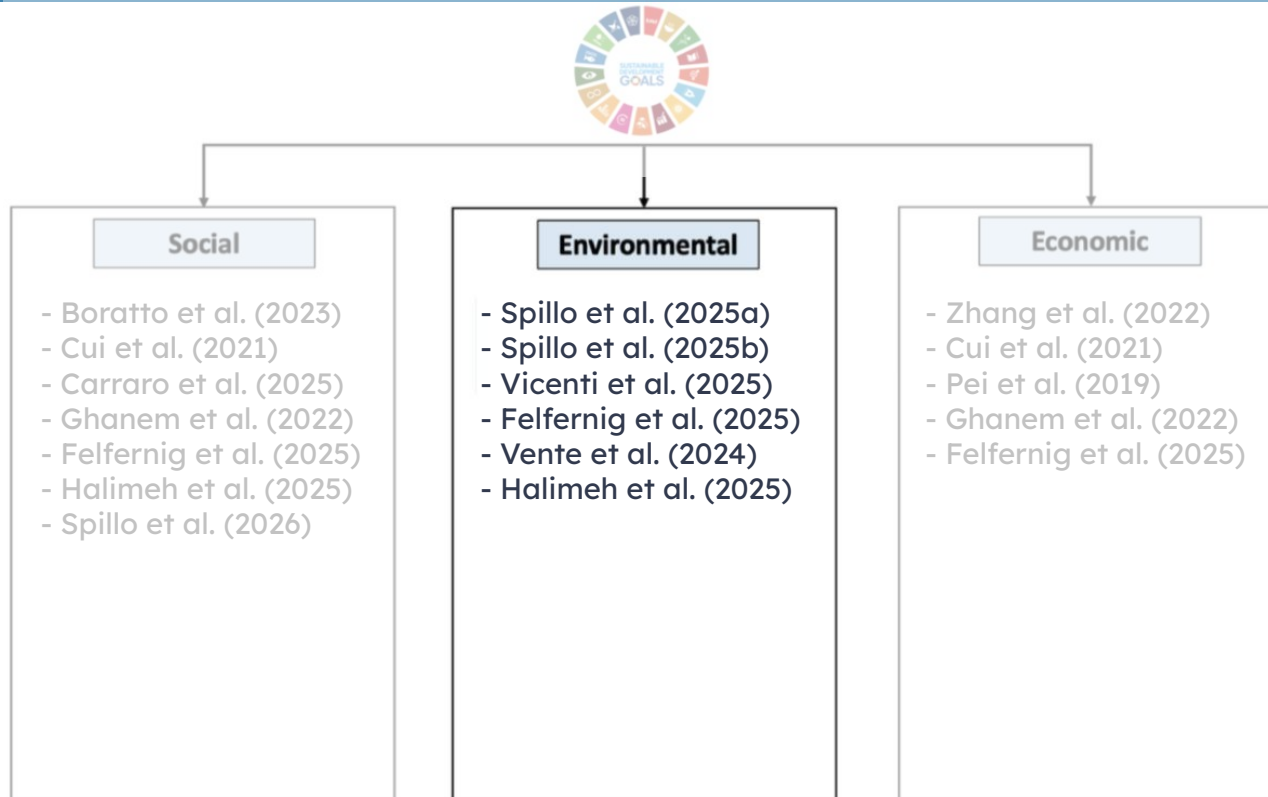


Spillo, G., De Filippo, A., Musto, C., Milano, M., & Semeraro, G. (2025). **Comparing data reduction strategies for energy-efficient green recommender systems**. Journal of Intelligent Information Systems, 63(6), 1837-1863.

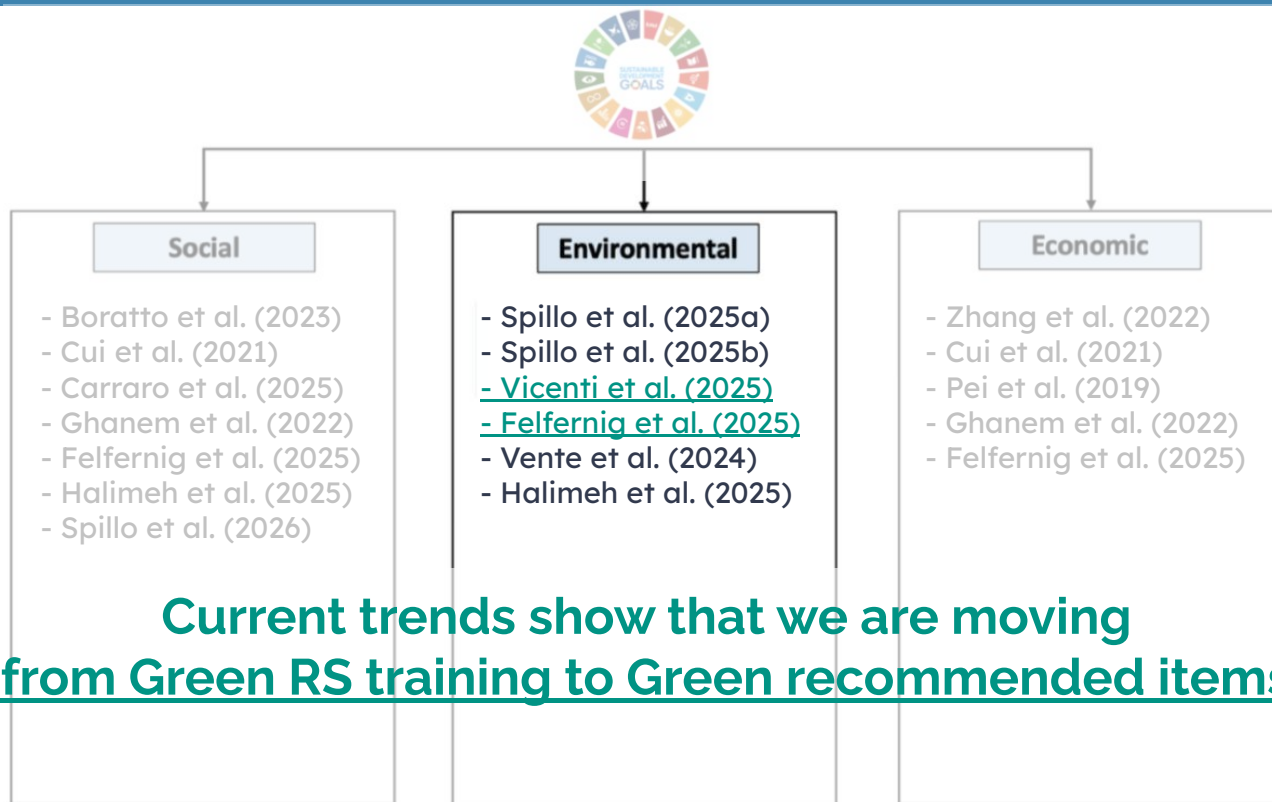


Spillo, G., De Filippo, A., Musto, C., Milano, M., & Semeraro, G. (2024, October). **Towards green recommender systems: Investigating the impact of data reduction on carbon footprint and algorithm performances**. In Proceedings of the 18th ACM Conference on Recommender Systems (pp. 866-871).

2. Sustainability of RS



2. Sustainability of RS



2. Sustainability of RS

Eco-AMAZON: Enriching E-commerce Datasets with Product Carbon Footprint for Sustainable Recommendations

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Michela Milano
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Giovanni Semeraro
University of Bari Aldo Moro, Italy
giovanni.semeraro@uniba.it

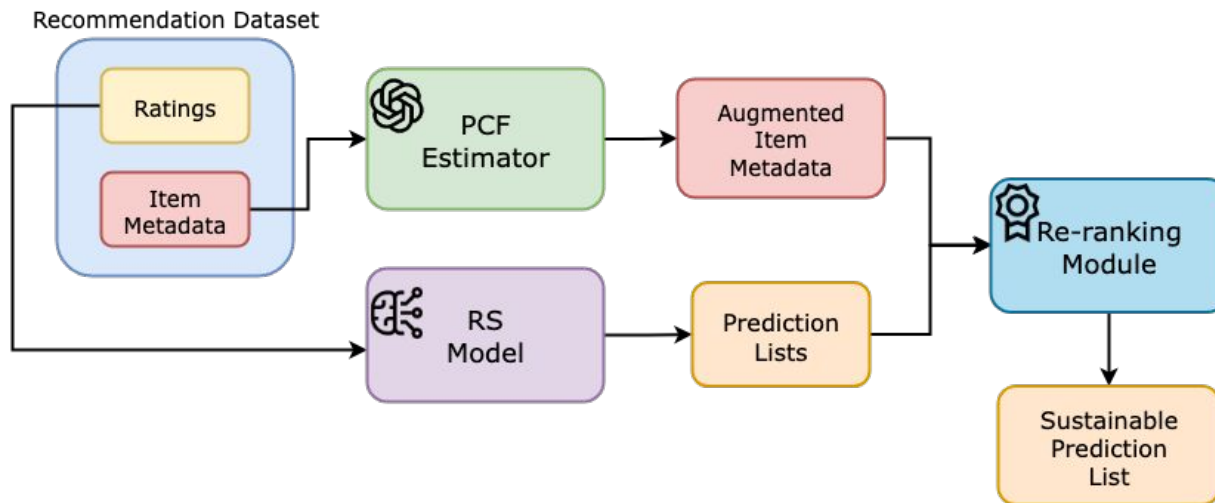
2. Sustainability of RS

- **Environmental impact of e-commerce is a growing concern**
 - But RS datasets lack item-level carbon footprint (PCF) data
- **Existing PCF estimation methods are impractical for RS research**
 - PCF sources (LCA, EPD platforms) are proprietary, narrow, or misaligned with RS benchmarks
- **LLMs offer a scalable opportunity to close this gap**
 - LLMs can infer PCF in a zero-shot setting, enabling a scalable, reusable resource for sustainable RS research

2. Sustainability of RS

- We designed a framework to **estimate the Product Carbon Footprint (PCF)** using only textual input and off-the-shelf LLMs
- We show an open methodology for enriching datasets with environmental metadata
- We show the applicability of these estimates in downstream tasks, such as sustainability-aware recommendation

2. Sustainability of RS



2. Sustainability of RS

- We designed a prompt that:
 - First, prioritizes **official data** provided by Environmental Product Declarations (EPD) reports - if any

You are an expert in life cycle analysis (LCA) and CO2e emission calculation for electronic products. You must estimate the CO2e emissions, based on the entire life cycle (cradle-to-grave), for the following electronic product.

Product data: {product_data}

INSTRUCTIONS:

1. FIRST, **check if there are any official carbon footprint reports** or environmental product declarations (EPD) from the manufacturer for this specific product. If found, use these official values as your primary source.

2. Sustainability of RS

- We designed a prompt that:
 - First, prioritizes official data provided by Environmental Product Declarations (EPD) reports - if any
 - If no data is available, then the model has to **estimate the PCF** based on widely adopted protocols, materials, manufacturing process, transportation, etc.

2. **If NO official manufacturer reports are available, then estimate emissions** following these protocols:

- **GHG Protocol Product Standard** for system boundaries and calculation methodology
- **ISO 14040/14044** for Life Cycle Assessment principles
- **PAS 2050 and ISO/TS 14067** for carbon footprint calculation guidelines.

3. For estimation, consider:

- **Main materials composition**
- **Manufacturing processes**
- **Transportation**
- Use phase energy consumption
- End-of-life disposal

2. Sustainability of RS

- We designed a prompt that:
 - First, prioritizes official data provided by Environmental Product Declarations (EPD) reports - if any
 - If no data is available, then the model has to estimate the PCF based on widely adopted protocols, materials, manufacturing process, transportation, etc.
 - The model **has to follow a solid methodology** and document the output

- 4. **Use the most recent emission factors** and scientific data available
- 5. **Document your sources** and assumptions in the explanation
- 6. Clearly state if you're using manufacturer data or estimation

2. Sustainability of RS



iPhone 16

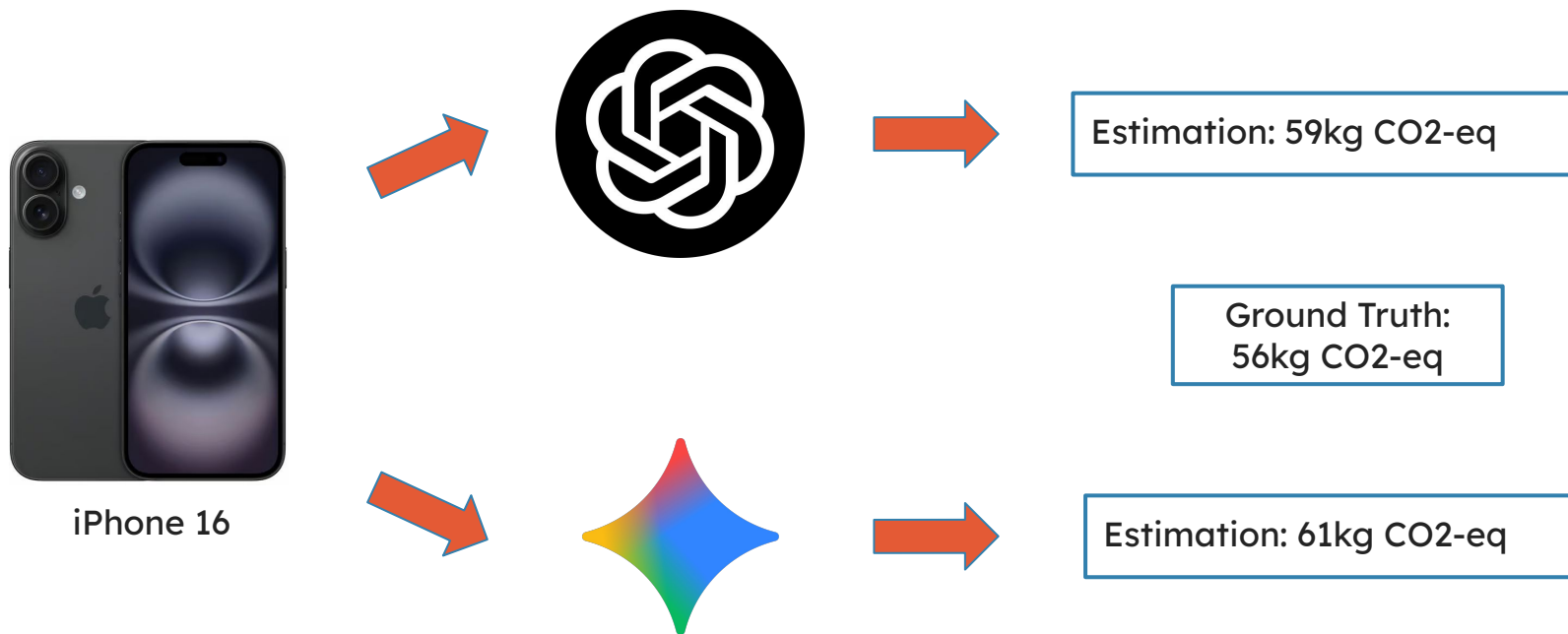
Ground Truth:
56kg CO₂-eq

2. Sustainability of RS



Ground Truth:
56kg CO₂-eq

2. Sustainability of RS



2. Sustainability of RS

- In parallel, we train RSs and save the prediction lists, along with each user-item recommendation score
- Finally, we use PCF and the recommendation scores to **re-rank the prediction lists**, so to obtain the **Sustainable-aware Score (SaS)**:
 - $\text{pred}(u,i)$ is the recommendation score for user u and item i
 - $\text{PCF}(i)$ is the value estimated by the LLM for item i
 - α is the parameter that balances accuracy and sustainability

$$\text{SaS}(u,i) = \alpha * \text{pred}(u,i) + (1-\alpha) * \text{PCF}(i)$$

2. Sustainability of RS

Item	GT	GPT	Gemini
<i>Electronics</i>			
Logitech Wireless Mouse	3.97	4.00	4.00
Logitech G PRO Wireless Mouse	7.84	5.25	6.35
Apple TV 3rd gen	46.00	46.75	92.50
Lenovo Chromebook C330 2-in-1	263.00	203.75	178.90
ThinkPad P1 Gen 5	312.00	347.50	639.25
<i>Clothing</i>			
The Long Sleeve T-Shirt	4.51	6.60	6.05
The T-Shirt	5.70	5.32	5.55
Kids' Multisport Shoes	6.15	6.00	7.30
Men's Wool Runner	10.36	8.17	9.50
501® Original Jeans	33.40	33.40	33.00
<i>Home&Kitchen</i>			
Steelcase DO Chair	82.20	135.25	78.39
KI Postura+ Chair	11.90	10.33	9.80
Benchmark OVO Table	66.90	182.50	78.07
Benchmark Coffee Table	17.10	69.75	25.48
Futura Line Toilet Paper	2040.00	1075.00	3861.25

2. Sustainability of RS

Category	GPT-o3-mini			Gemini-2.5-flash		
	MAE	Spearman	NDCG	MAE	Spearman	NDCG
Electronics	48.1	0.963	0.848	73.1	0.964	0.871
Home and Kitchen	102.91	0.919	0.978	153.22	0.939	0.988
Clothing	7.6	0.756	0.942	8.1	0.821	0.953

Table 1: PCF estimation performance. MAE scores are reported in kg. Spearman scores are normalized $\in [0, 1]$.

2. Sustainability of RS

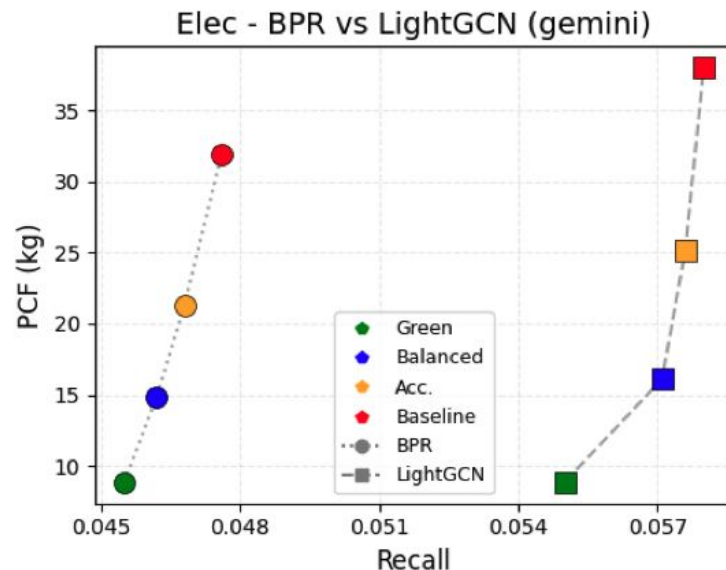
Category	GPT-o3-mini			Gemini-2.5-flash		
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Table 1: PCF estimation performance. MAE scores are reported in kg. Spearman scores are normalized $\in [0, 1]$.

LLMs can effectively predict higher PCF values for more impactful items

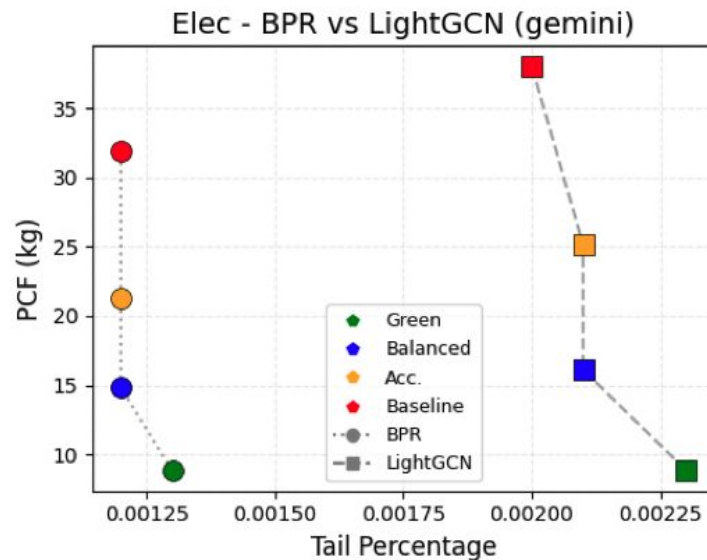
2. Sustainability of RS

- We observe a **slight accuracy reduction** when more weight is given to the green aspect
 - from 0.059 to 0.055 in LightGCN
- The average **PCF** of the recommendation list **decreases dramatically**
 - from 38kg to <10kg



2. Sustainability of RS

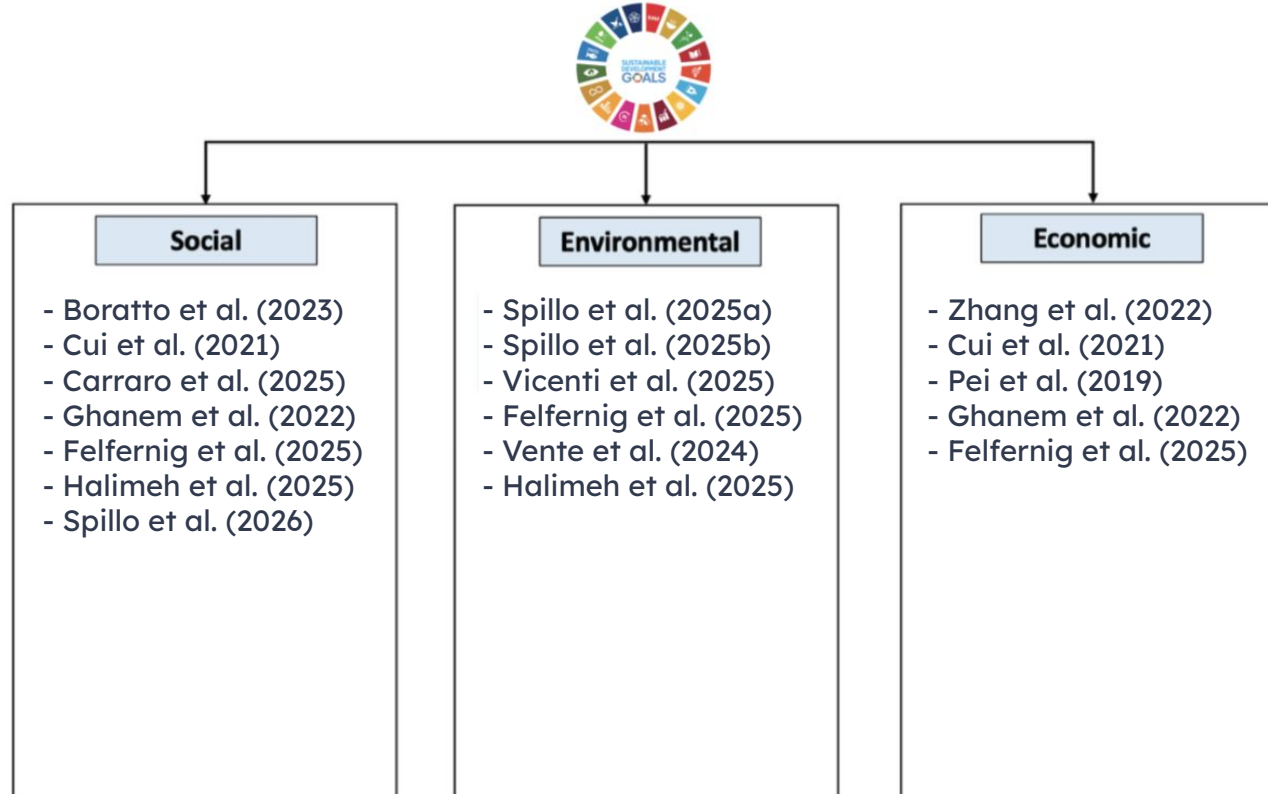
- In parallel, **more long-tail items are recommended**
 - Beyond-accuracy side-effect
- PCF-aware re-ranking can be used as a popularity mitigation strategy
 - further research is needed



2. Sustainability of RS

- **LLMs estimate PCF reliably for ranking tasks:** absolute error remains non-negligible, but rank correlation is strong (often >0.85), correctly ordering items by sustainability
- **Eco-Amazon lowers the barrier to sustainability-aware RS, offering PCF-enriched datasets,** a reproducible estimation pipeline, and GT values across three domains
- **PCF-based re-ranking effectively balances accuracy and sustainability:** higher α reduces footprint with limited performance loss, lower α recovers accuracy
- **Calibration remains an open challenge,** with variability across domains and models motivating future hybrid LLM + LCA pipelines.

2. Sustainability of RS



2. Sustainability of RS



**Research is going in the right direction,
but we are missing something**

2. Sustainability of RS

- Many works related to the fairness cover the **social pillar**
 - Even when they were not framed under the concept of sustainability
- Recent research focused mostly on **environmental pillar**
- Few works focused on **economic pillar**

2. Sustainability of RS

- Many works related to the fairness cover the **social pillar**
 - Even when they were not framed under the concept of sustainability
- Recent research focused mostly on **environmental pillar**
- Few works focused on **economic pillar**
- Multi-pillar approaches are mostly overlooked!
 - Multi-pillar findings are observed as a side-effects

2. Sustainability of RS

- **Sustainability silos**
 - Improvements pursued along one pillar
 - without regard for their collateral impact on the other pillars
- Thus, we have:
 - RS that are **fair** but environmentally **costly**
 - RS **efficient** but socially **inequitable**

2. Sustainability of RS

- **Sustainability silos**
 - Improvements pursued along one pillar
 - without regard for their collateral impact on the other pillars
- Thus, we have:
 - RS that are **fair** but environmentally **costly**
 - RS **efficient** but socially **inequitable**

Can we quantify this trade-off?



2. Sustainability of RS

Can We Have It All? Balancing Accuracy, Fairness, and Carbon Emissions in Sustainable Recommender Systems

Giuseppe Spillo^{a,*}, Allegra De Filippo^b, Cataldo Musto^a, Ludovico Boratto^c,
Michela Milano^{b,d} and Giovanni Semeraro^a

^aUniversity of Bari Aldo Moro, Via Orabona, 4, Bari, 70125, Italy

^bUniversity of Bologna, Via Zamboni, 33, Bologna, 40126, Italy

^cUniversity of Cagliari, Via Ospedale, 72, Cagliari, 09124, Italy

^dFondazione Bruno Kessler, Via Santa Croce, 77, Trento, 38122, Italy

2. Sustainability of RS

- We consider SOTA fairness-oriented RS
 - FOCF, PFCN, FairGo, NFCF
- Evaluate them together across
 - accuracy
 - fairness
 - carbon emissions

2. Sustainability of RS

- We consider SOTA fairness-oriented RS
 - FOCF, PFCN, FairGo, NFCF
- Evaluate them together across
 - accuracy
 - fairness → social pillar
 - carbon emissions → environmental pillar

2. Sustainability of RS

- **Accuracy**
 - NDCG
 - Recall
 - Hit
 - MRR
 - Precision
 - MAP

2. Sustainability of RS

- **Accuracy**
 - NDCG
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 - Hit
 - MRR
 - Precision
 - MAP
- **Beyond-accuracy**
 - Gini Index
 - Item Coverage
 - Shannon Entropy
 - Popularity Percentage

2. Sustainability of RS

- **Accuracy**

- NDCG
- Recall
- Hit
- MRR
- Precision
- MAP

- **Beyond-accuracy**

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- Popularity Percentage

- **Fairness**

- Value Unfairness
- Absolute Unfairness
- NonParity Unfairness
- Differential Fairness

2. Sustainability of RS

- **Accuracy**

- NDCG
- Recall
- Hit
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- **Beyond-accuracy**

- Gini Index
- Item Coverage
- Shannon Entropy
- Popularity Percentage

- **Fairness**

- Value Unfairness
- Absolute Unfairness
- NonParity Unfairness
- Differential Fairness



Gender sensitive attribute

Value Unfairness: is there a bias? Do the model's prediction errors differ systematically between groups?

2. Sustainability of RS

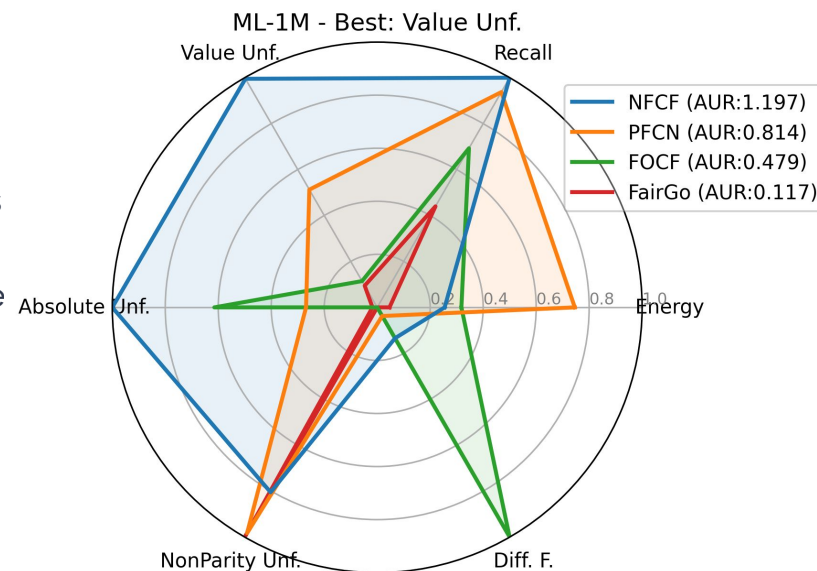
	ML-1M	LFM-1K
# users	6,040	271
# items	3,706	51,611
# interactions	1,000,209	200,608
Sparsity	95.53%	98.57%
Gender (M,F)	(4331, 1709)	(157, 114)

2. Sustainability of RS

Submitted to Journal

Trade-off Among Accuracy, Fairness, and Emissions (ML-1M)

- NFCF offers the best trade-off for Value and Absolute Unfairness, pairing low unfairness with strong accuracy
- PFCN performs well in terms of NonParity and Differential Fairness, behind NFCF on average, and stays comparatively stable across all four scenarios
- FOCF is consistently mid-pack: never the best, never the worst, ranking third in every single scenario
- FairGo shows the weakest trade-off in every scenario (lose-lose condition)



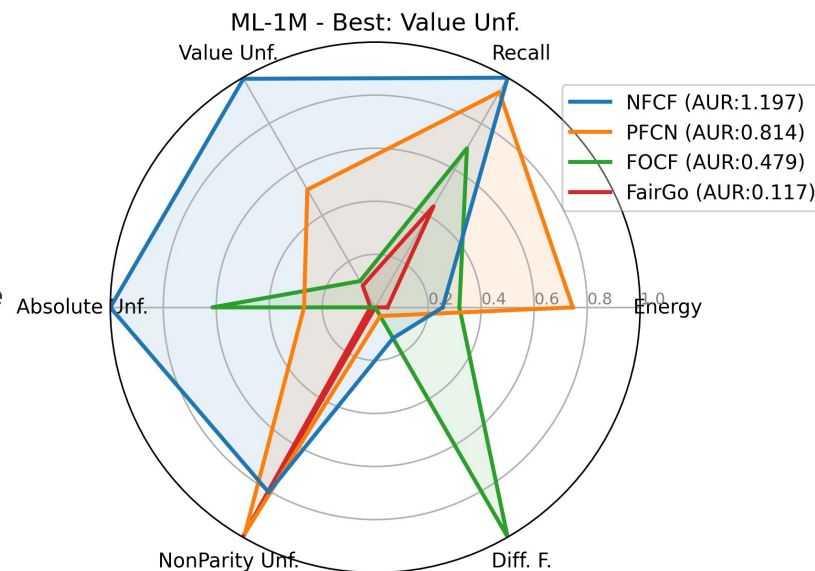
2. Sustainability of RS

Submitted to Journal

Trade-off Among Accuracy, Fairness, and Emissions (ML-1M)

- NFCF offers the best trade-off for Value and Absolute Unfairness, pairing low unfairness with strong accuracy
- PFCN performs well in terms of NonParity and Differential Fairness, behind NFCF on average, and stays comparatively stable across all four scenarios
- FOCF is consistently mid-pack: never the best, never the worst, ranking third in every single scenario
- FairGo shows the weakest trade-off in every scenario (lose-lose condition)

On ML-1M, no single model dominates: NFCF and PFCN split the best trade-off depending on which fairness dimension matters most, while FairGo underperforms across the board.

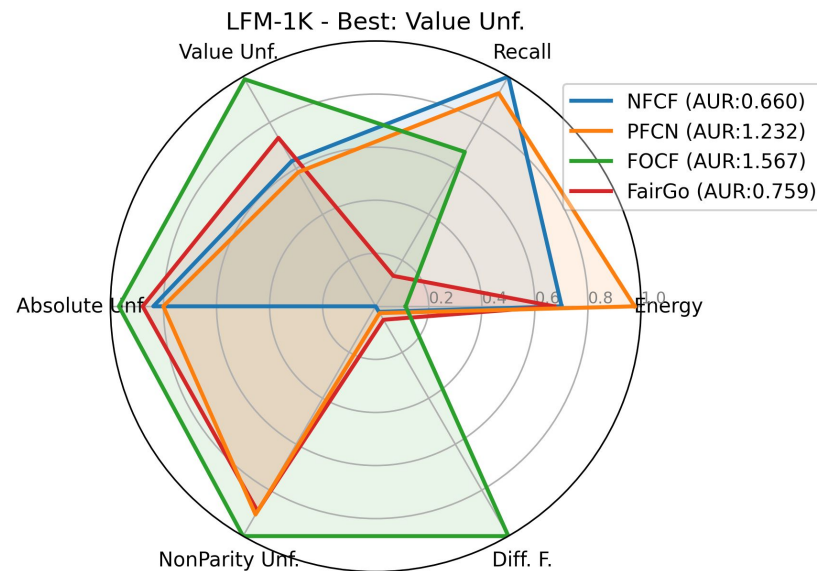


2. Sustainability of RS

Submitted to Journal

Trade-off Among Accuracy, Fairness, and Emissions (LFM-1K)

- FOCF has the highest training energy, but its fairness advantage outweighs the energy cost
- PFCN is the second-best model overall
- NFCF accuracy is lower under LFM-1K's sparsity
- FairGo again shows the weakest trade-off, though here this partly reflects degenerate, near-zero-Recall configurations rather than a genuine fairness strength



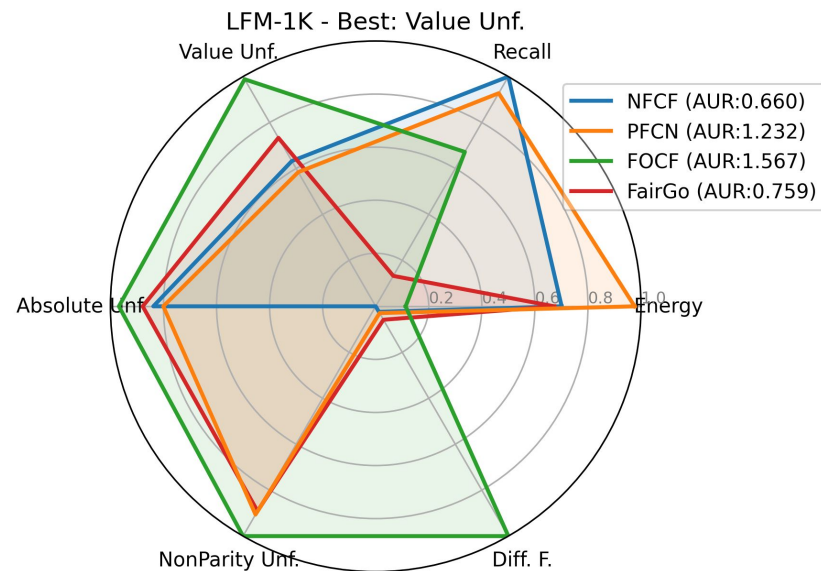
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On LFM-1K, FOCF offers the best trade-off despite its high emissions, since its fairness dominance outweighs the carbon cost; PFCN remains the strongest, most consistent runner-up.



2. Sustainability of RS

Take-home messages

- **Fairness has a green cost**, but only for some models: pursuing fairness raises training energy for FairGo while FOCF and NFCF reach their best fairness at 20-84% lower energy-
The penalty is model-specific, not universal
- **The trade-off flips with the dataset**: which model wins the trade-off changes entirely between ML-1M and LFM-1K, driven by model architecture and dataset density
- **PFCN is the only model ranking among the top two trade-offs on both datasets**.
NFCF's ML-1M dominance does not transfer to the sparser LFM-1K
- **Sustainability needs a multi-pillar, dataset-aware view**: jointly auditing fairness and carbon footprint, while accounting for how dataset density reshapes the trade-off, is what avoids sustainability silos in RS design

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 - then we saw how to improve the energy efficiency
 - now we are moving toward green recommended items

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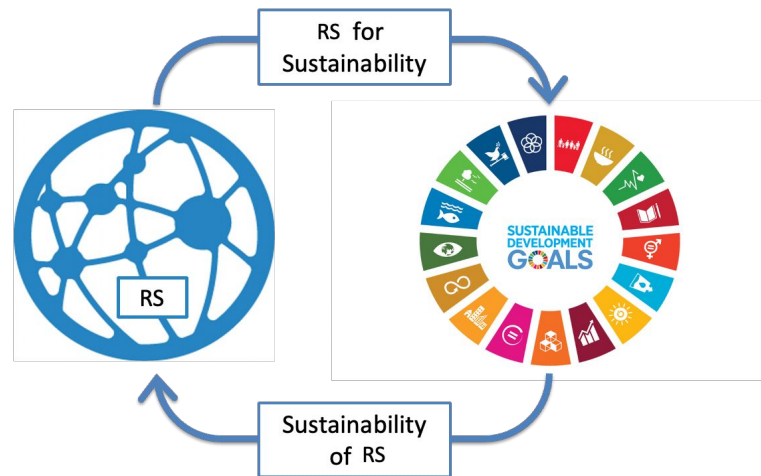
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 - but we need to consider both item- and user-side fairness
- **The economic pillar remains the less investigated**
- Multi-pillar studies are now emerging
 - **This is the right direction for future research!**
 - Can we be fully sustainable, under all perspectives?



3. The Dual Perspective

Can we envision a future where recommender systems **both enable sustainable choices and are themselves sustainable**, without compromising performance or user experience?



Which of the following research directions do you find most promising, are already working on, or would like to explore regarding the integration of Sustainability and RS?



Discussion

Research Direction	SoTA Maturity	Current Practices	Main Gaps
Developing Multi-Pillar RS Applications	Low	Some systems address 2 pillars (e.g., health + environment)	Rarely address all three pillars (environmental, social, economic) simultaneously
Designing Multi-Objective RS	Medium	Growing interest in balancing fairness and accuracy	Lack of frameworks that also balance environmental and economic objectives concurrently
Designing Evaluation Frameworks Beyond Accuracy	Medium	Some systems use fairness metrics or carbon footprint separately	Evaluations are mostly isolated; no integrated multi-pillar evaluation protocols
Establishing Cross Disciplinary Collaborations	Low	Few cases involve stakeholders beyond computer science	Missing practical, interdisciplinary teams including economists, environmentalists, policymaker

RecSoGood 2026



RecSoGood 2026

Third International Workshop on Recommender Systems for Sustainability and Social Good | co-located with RecSys 2026

 The ACM Conference Series on
Recommender Systems



 **When**
October 2nd, 2026

 **Where**
Room Ballroom 4

 **Time**
9:00 – 15:30 (Full day)



Our Vision for RS



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Recommender systems and sustainability: a dual perspective

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ACM RecSys 2026

The 20th ACM Recommender Systems Conference will take place in Minneapolis, Minnesota, USA, from Sep. 27 - Oct. 2, 2026.

Thank you!

Allegra De Filippo, Ludovico Boratto and Giuseppe Spillo



<https://sustainablerecsys.github.io/SustainabilityTutorial26/>